SOCIETY FOR ECOSYSTEM RESTORATION
IN NORTHERN BRITISH COLUMBIA

Restoring Fish Passage in the Fraser Region - 2025

Prepared for
Ministry of Transportation and Infrastructure

and

Society for Ecosystem Restoration in Northern BC (SERNbc)

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on behalf of

Society for Ecosystem Restoration in Northern BC (SERNbc)

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Claude Opus 4.7 (Anthropic) assisted with literature synthesis, drafting, and technical writing. All scientific interpretation, data analysis, and conclusions are the responsibility of the authors.

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Acknowledgement

At New Graph Environment, we understand our well-being as inseparable from the health of the land and waters we work within. When we care for ecosystems, we care for ourselves and for the communities connected to them. This relationship is not metaphorical — it is the foundation of our practice.

Modern civilization has a long journey ahead to acknowledge and address the historic and ongoing impacts of colonialism that have resulted in harm to the cultures and livelihoods living interconnected with our ecosystems for many thousands of years.

This work takes place within the territories of [Nations], who have stewarded these lands and waters since time immemorial. [Project-specific connection to territory, governance, species, or watershed.]

[Funding and partner acknowledgements.]

Executive Summary

Executive Summary (PDF)

This report is available as a PDF and as an online interactive report at https://www.newgraphenvironment.com/fish_passage_fraser_2025_reporting/. We recommend viewing online as the web-hosted HTML version contains more features and is more easily navigable. Please reference the website for the latest version stamped PDF from fish_passage_fraser_2025_reporting.pdf.

Since 2023, the Society for Ecosystem Restoration Northern British Columbia (SERNbc), with support from the Ministry of Transportation and Infrastructure, has been actively involved in planning, coordinating, and conducting fish passage restoration efforts within the Nechako River, Lower Chilako River, Morkhill River, Upper Fraser River, and François Lake watershed groups which are sub-basins of the Upper Fraser River watershed.

The primary objective of this project is to identify and prioritize fish passage barriers within study areas, develop comprehensive restoration plans to address these barriers, and foster momentum for broader ecosystem restoration initiatives. While the primary focus is on fish passage, this work also serves as a lens through which to view the broader ecosystems, leveraging efforts to build capacity for ecosystem restoration and improving our understanding of watershed health. We recognize that the health of life - such as our own - and the health of our surroundings are interconnected, with our overall well-being dependent on the health of our environment.

Although the main purpose of this report is to document field work data and results, it also builds on reporting from past field activities and all reports can be considered living documents that can be updated and improved over time. In addition to the numerous assessments at sites undocumented in past years of the project, field activities in 2024 were also conducted at sites where habitat confirmations were previously documented within the reports linked below. The reports for these sites were edited and updated with data.

- Restoring Fish Passage in the Fraser Region - 2023

Fish passage assessment procedures conducted through SERNbc in the Upper Fraser River Watershed since 2023 are amalgamated online within the Results and Discussion section of the report found here which includes links to project reporting for each site.

In 2025, the project area was expanded to include the Willow river, Lower Salmon River, and Tabor River watershed groups. Fish passage assessments were completed at 12 new sites, focusing on structures with potential barriers to upstream fish movement using standard provincial criteria.

Habitat confirmation assessments were conducted at multiple sites within the Willow river and Tabor River watersheds, covering over XX km of stream length. Detailed habitat metrics were documented alongside eDNA samples to evaluate habitat quality and presence of fish species.

New in 2025, environmental DNA (eDNA) sampling was incorporated into the program at both previously assessed and newly added sites. A total of 38 samples were collected across 16 streams.

Monitoring was conducted at two sites, including post-remediation evaluations at PSCIS crossings 196085 and 196200. Monitoring included eDNA sampling and habitat observations. A custom effectiveness monitoring form tailored to fish passage projects was used and metrics assessed included flow velocity, substrate condition, channel constriction, riparian condition, and cover availability.

A major challenge in advancing fish passage restoration is the complexity of working across jurisdictions and with multiple stakeholders—rail and highway authorities, forestry ministries, licensees, and private landowners. These partners are often being asked to accommodate priorities that originate outside their mandates and budgets. Convincing them to invest in difficult, high-cost interventions—like modifying crossings or relocating infrastructure—requires navigating uncertainty about costs and ecological outcomes, as well as a disconnect between the benefits to watershed health and the internal pressures or performance goals of these agencies. It's a tough ask: to take on massive, uncertain projects when they're already stretched thin with their own responsibilities.

Fish passage restoration across British Columbia is further complicated by the legacy of infrastructure deeply embedded in the landscape. Roads, railways, highways, community infrastructure and private assets often constrain floodplains and disrupt natural hydrological processes. While targeted repairs to individual barriers are essential, they won't resolve the broader systemic issues without rethinking and restructuring how infrastructure interacts with watershed function. Loss of riparian vegetation and intensive beaver management only add to the degradation. Addressing these challenges means making strategic, well-communicated choices—picking battles carefully, building trust, and staying committed to a longer-term transformation.

While preliminary top remediation priorities are provided by watershed group, these rankings are inherently subjective and can depend on the capacity and willingness of infrastructure owners and tenure holders to support implementation—both financially and over the often multi-year project timelines. In practice, we must often act opportunistically, pursuing simpler, lower-cost options to maintain momentum and achieve near-term progress.

NEEDS TO BE UPDATED

To enhance fish passage restoration in the FWCP Peace Region:

- Maintain strong partnerships to support funding, site selection, remediation, and monitoring through adaptive management informed by traditional knowledge and real-time data.
- Prioritize detailed assessments in areas with blockages and high habitat potential, especially near McLeod Lake.
- Use climate modeling to prioritize crossings that enable access to cold, drought-resistant habitats.
- Secure financial commitments for Fern Creek remediation despite uncertainties in harvest planning.
- Continue effectiveness monitoring at key sites using fish sampling, eDNA, PIT tagging, temperature data, and aerial imagery.
- Continue to develop a cost-effective monitoring framework to assess productivity gains from improved passage.
- Collaborate with WLRS, UNC, local fisheries experts, FWCP, and the CEMPRA Project working group.
- Utilize environmental DNA (eDNA) to better understand bull trout and Arctic grayling habitat use at both potential and remediated sites. This will refine prioritization and assess fish passage effectiveness.

1 Introduction

This report is available as a PDF and as an online interactive report at https://www.newgraphenvironment.com/fish_passage_fraser_2025_reporting/. We recommend viewing online as the web-hosted HTML version contains more features and is more easily navigable. Please reference the website for the latest PDF from fish_passage_peace_2024_reporting.pdf.

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This project builds on Society for Ecosystem Restoration Northern BC (SERNbc) work in:

- Restoring Fish Passage in the Fraser Region - 2023 (Irvine and Schick 2024)

The health and viability of freshwater fish populations can depend on access to tributary and off channel areas which provide refuge during high flows, opportunities for foraging, overwintering habitat, spawning habitat and summer rearing habitat (Bramblett et al. 2002; Swales and Levings 1989; Diebel et al. 2015). Culverts can present barriers to fish migration due to low water depth, increased water velocity, turbulence, a vertical drop at the culvert outlet and/or maintenance issues (Slaney et al. 1997; Cote et al. 2005). As road crossing structures are commonly upgraded or removed there are numerous opportunities to restore connectivity by ensuring that fish passage considerations are incorporated into repair, replacement, relocation and deactivation designs.

Although remediation and replacement of stream crossing structures can have benefits to local fish populations, the costs of remedial works can be significant and the impacts of the work often complex to evaluate and quantify. Additionally, allocation of ecosystem restoration funding towards infrastructure upgrades on transportation right of ways are not always considered ethical under all circumstances from all perspectives. When funds are finite and invested groups are engaged in fund raising, cost benefits and the ethics of crossing replacements should be explored collaboratively alongside the cost benefits and ethics of alternative investment activities including transportation corridor relocation/deactivation, land procurement/covenant, cattle exclusion, riparian/floodplain restoration, habitat complexing, water conservation, commercial/recreational fishing management, salt water interventions and research.

Please note that at the time of reporting this document can be considered a living document. See the Open Source Reporting Framework (page 43) section for details on version tracking and project documentation.

2 Background

2.1 Approach

Fish passage assessments in British Columbia follow the provincial protocol (MoE 2011), which scores barrier risk at road-stream crossings from structural parameters — culvert embedment, slope, outlet drop, and culvert diameter relative to channel width. The “Barrier” and “Potential Barrier” classifications represent passability for juvenile salmon or small resident rainbow trout under any flow conditions that may occur throughout the year (Clarkin et al. 2005; Bell 1991; Thompson 2013). The thresholds are deliberately conservative, so the classifications are risk scores rather than determinations of whether fish actually move through. Many structures classified as “Barrier” still pass some species and life stages at some flows, and natural features in adjacent reaches can be as limiting as the structure itself. Passability can be quantified in other ways — fish length, swim speed, and species physiology all affect connectivity outcomes — and finer-grained scoring methods exist for structures already classified as barriers (Bourne et al. 2011; Kemp and O’Hanley 2010; Washington Department of Fish & Wildlife 2009). Whether a remediation changes outcomes for fish therefore depends on the habitat upstream, the species and life stages in question, and the broader condition of the reach the structure connects to — not the classification alone.

The provincial inventory contains thousands of structures on fish-bearing streams. Replacement at a typical forest road culvert is in the six-figure range; structural replacements on highway and rail corridors run into seven and eight figures. Funding for restoration is finite and split across multiple programs and partners. Prioritization across the inventory therefore carries more weight than any individual assessment — the assessment establishes what could be fixed, while prioritization decides what should be.

Authority and funding for these structures are distributed across rail authorities, highway ministries, forestry licensees, and private landowners. Each operates under its own mandate, performance metrics, and budget cycle. Advancing restoration involves working inside organizations whose primary mandate is moving goods, maintaining roads, or harvesting timber — not aquatic ecological outcomes. The work proceeds through shared data, sustained relationships, and incremental progress across multiple budget years and field seasons.

Many road and rail corridors were built without consideration of fish passage or floodplain function. Damage frequently extends beyond the structure at the crossing — channelized streams, dykes that sever lateral connection, drained wetlands, cleared riparian buffers. Replacing a culvert in-kind can restore longitudinal passage while leaving the broader corridor-scale disconnection unaddressed. Understanding where infrastructure constrains watershed function helps distinguish cases where structural replacement is sufficient from cases where remediation needs to address more than the structure itself.

This report compiles the layers used for prioritization in the watershed groups where the work is active: structural assessment data, habitat modelling, climate trajectory, and floodplain extent. Each layer contributes a piece of the picture — what is present, what it could support, how it is changing, and the surrounding watershed context. The work happens over many years across many partners; this report documents one year's contribution to that effort.

2.2 Project Location

The study area spans from Burns Lake to Valemount, British Columbia, and includes the Francois Lake, Lower Chilako River, Lower Salmon River, Morkill River, Nechako River, Tabor River, Upper Fraser River, and Willow River watershed groups. In 2025 the study area was expanded to include the Tabor River, Willow River, and Lower Salmon River watershed groups.

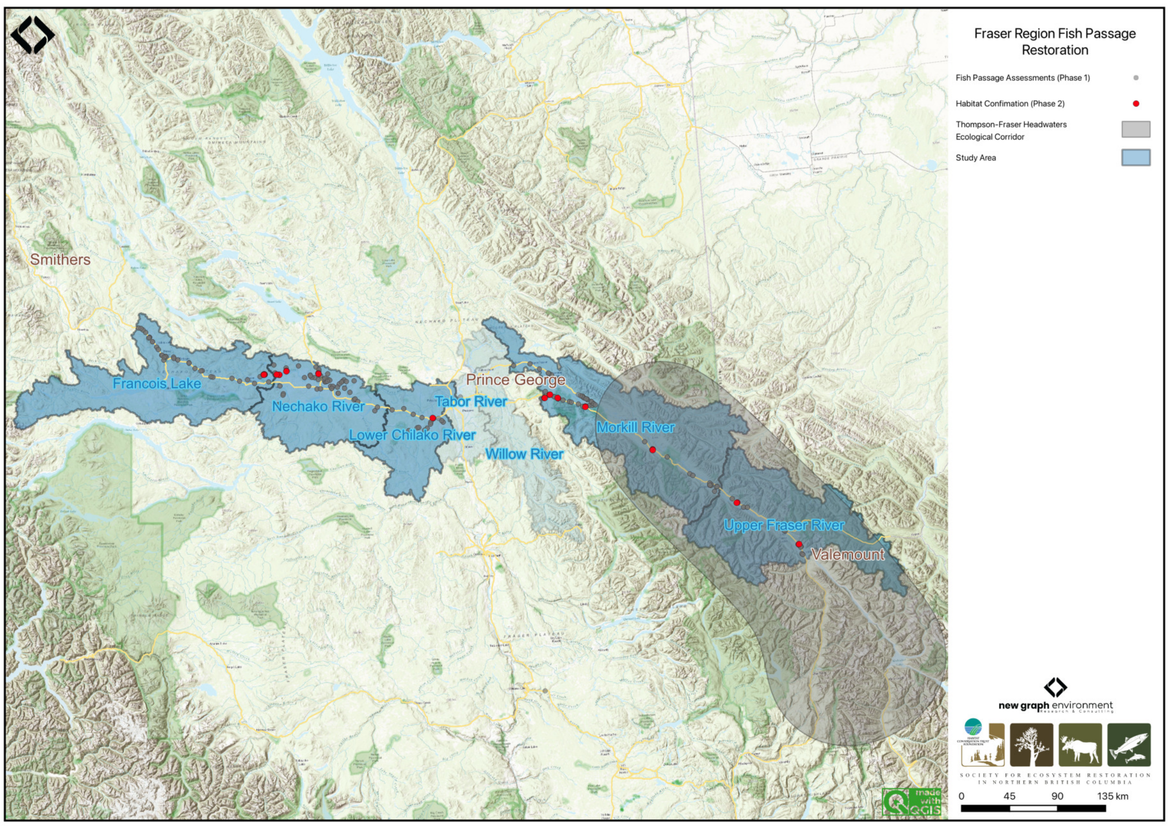

Figure 2.1: Overview map of the Fraser Region study areas

2.3 Lhtakoh

Known as the Lhtakoh, meaning "rivers within one another" to the Dakelh (Carrier) people, the Fraser River stretches nearly 1,400 kilometers from the Rocky Mountains of Mount Robson

Provincial Park to the Strait of Georgia near Vancouver. As the largest salmon-producing river on Canada's west coast (Bradford and Taylor 2023; "Dakleh Placenames," n.d.), it plays a crucial economic role in supporting forestry, agriculture, and hydroelectric power generation. Additionally, the Fraser River is vital for fisheries, especially for salmon populations, which are essential to both the local ecosystem and indigenous communities.

The Upper Fraser River is commonly defined as the section of the mainstem north of Quesnel, flowing through the Cariboo and Fraser Plateau regions. Major tributaries include the Nechako, Quesnel, and McGregor rivers (Shaw and Tuominen, n.d.). This vast expanse supports many indigenous groups who utilize the land for cultural, spiritual, and economic practices.

The Upper Fraser River, an 8th order stream, drains an area of 232,134km^2 upstream of the McGregor River confluence. The seasonal hydrograph has a single broad peak in early summer due to snow and glacial melt from surrounding mountain ranges (Figure 2.2). The mean annual discharge at station 08KA005 in McBride, located roughly 200km southeast of Prince George, is 200.4m^3/s

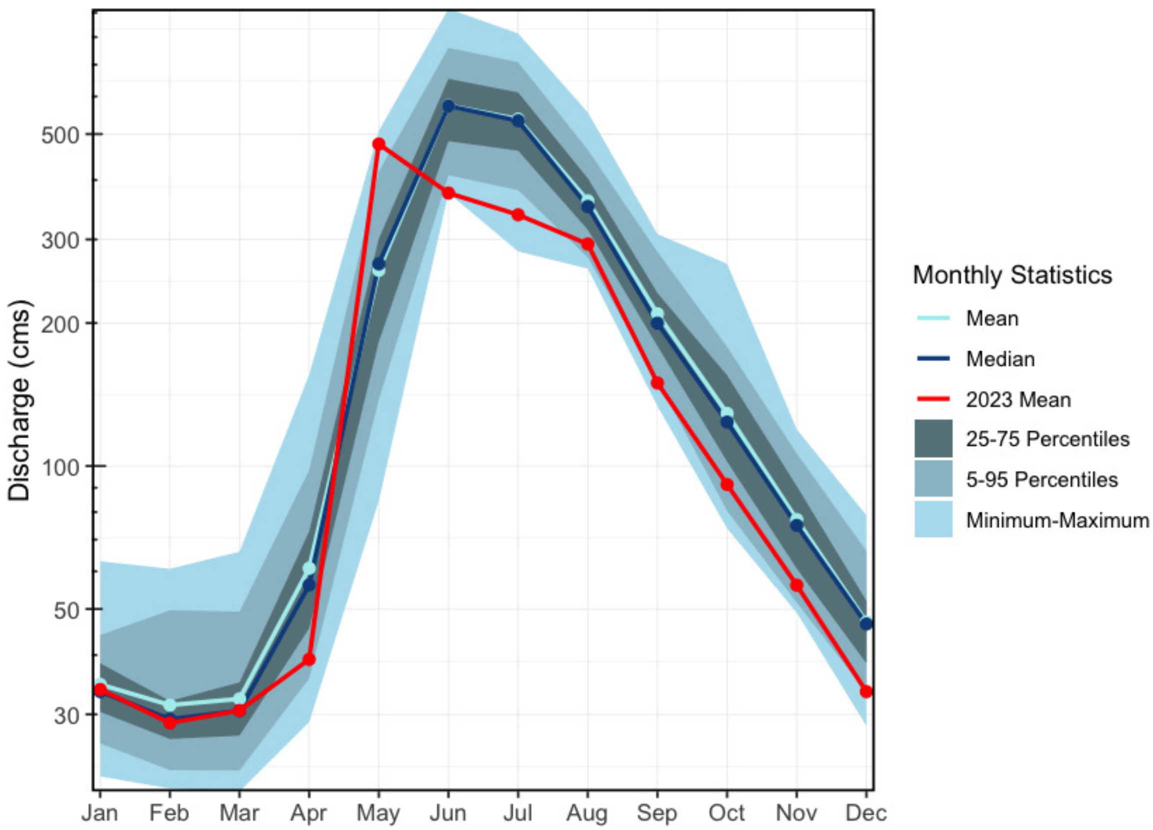

Figure 2.2: Hydrograph for the Fraser River At McBride (Station #08KA005 - Lat 53.30172 Lon -120.14092). Available mean daily discharge data from 1953 to 2023.

2.4 Nechakoh

The Nechako River is an 8th order stream that drains an area of 47,269km^2. Beginning at the Nechako Plateau, it flows north toward Fort Fraser, then east to its confluence with the Fraser River in Prince George. The Nechako River has three main tributaries: the Stuart River, the Endako River, and the Chilako River. It has a mean annual discharge of 278.5m^3/s at station 08JC002, located near Isle Pierre, approximately 25km downstream of the Stuart River confluence. Upstream at station 08JC001 in Vanderhoof, the mean annual discharge is 140.5m^3/s. Flow at Isle Pierre is strongly influenced by inflows from the Stuart River, resulting in higher peak levels and average discharge (Figure 2.3). In contrast, the hydrograph at Vanderhoof shows lower peak levels and mean flows, with peaks occurring in June and August (Figure 2.4).

The Nechako River, meaning “Blackwater people’s river”, is home to the Cheslatta Carrier Nation who are part of the Dakelh people. Traditionally, they lived off the land near Tsetl’adak Bunk’ut “Peak Rock Lake” (Cheslatta Lake), however, in 1952, the construction of the Kenney Dam by the Aluminum Company of Canada (now Rio Tinto Alcan) and the subsequent flooding of the Nechako Reservoir forced the Cheslatta Carrier people to abandon their ancestral lands (“The History of the

Cheslatta Carrier Nation," n.d.; "Dakleh Placenames," n.d.). This relocation was done with little notice or compensation, causing significant disruptions to their community, culture, and way of life. Despite these challenges, the Cheslatta Carrier Nation has worked to preserve their cultural heritage and advocate for their rights and land restoration.

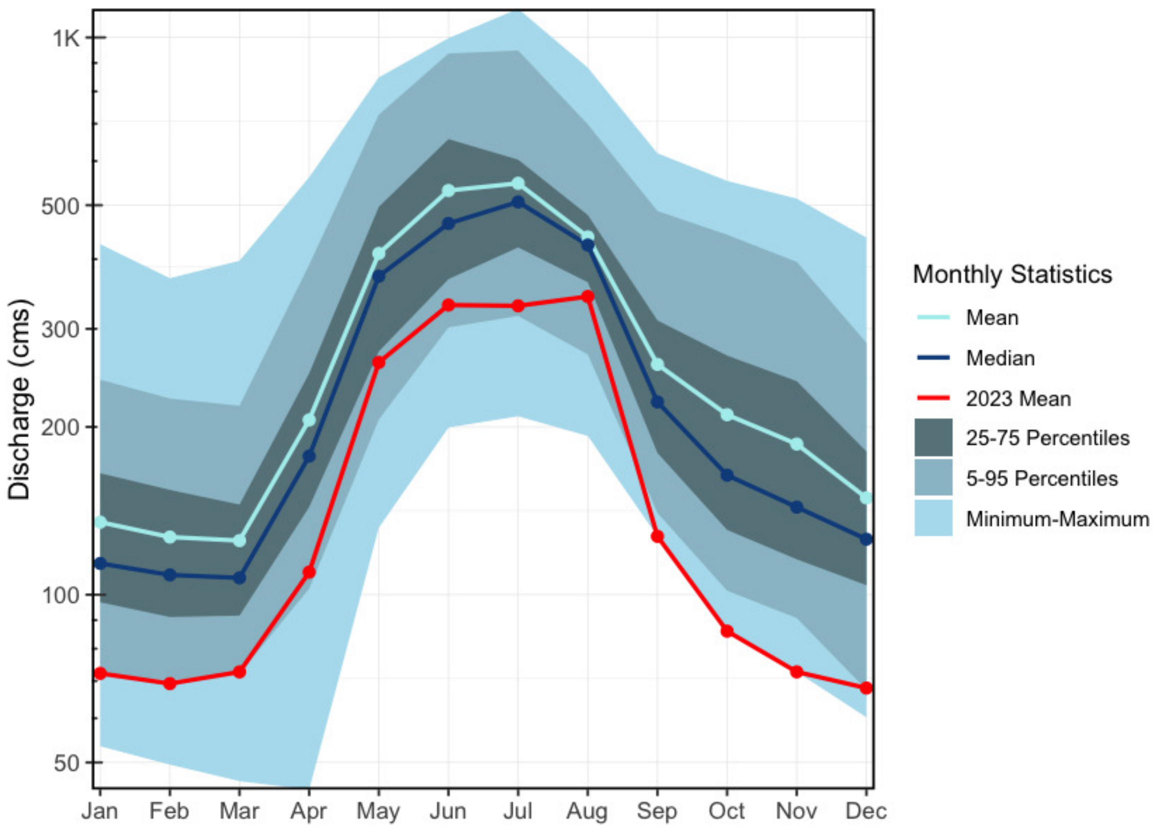

Figure 2.3: Hydrograph for the Nechako River at Isle Pierre, below the confluence of the Stuart River (Station #08JC002). Available mean daily discharge data from 1950 to 2023.

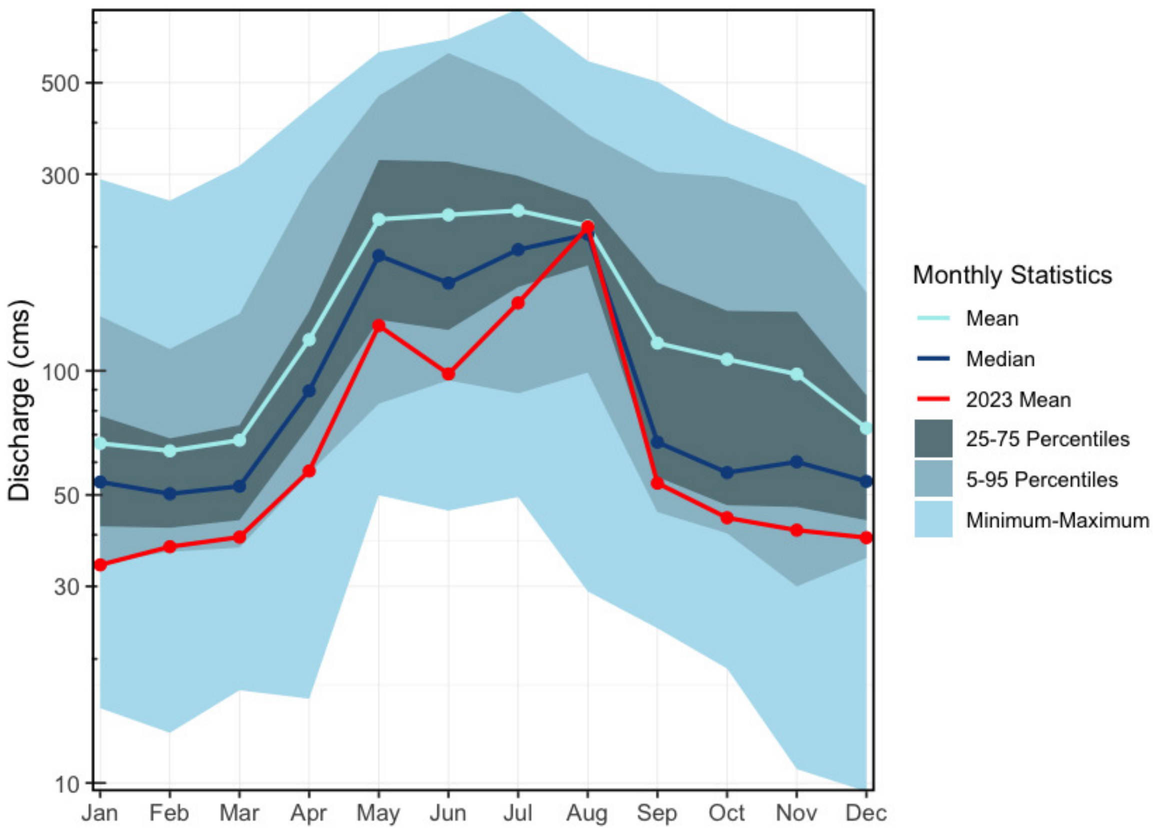

Figure 2.4: Hydrograph for the Nechako River at Vanderhoof (Station #08JC001). Available mean daily discharge data from 1915 to 2023.

2.5 Endako River

The Endako River is a 6th order stream that flows southeast from Burns Lake to Fraser Lake, draining an area of 5,970km^2. One hydrometric station (08JB004) was located in Endako but was only active during 1951. The mean annual discharge for that year was 12.7m^3/s, with the hydrograph peaking in May–June.

2.6 Tsalakoh

The Chilako River, known as Tsalakoh by the Dakelh people, translates to “beaver paw river”. It is a 6th order stream that flows north from the Nechako Plateau to its confluence with the Nechako River, draining an area of 3,634km^2. One hydrometric station (08JC005) is located approximately 10km upstream of the confluence with the Nechako River; however, it was only active from 1960 to 1974. During this period, the mean annual discharge was 13.3m^3/s, with peak flows typically occurring in May–June (Figure 2.5).

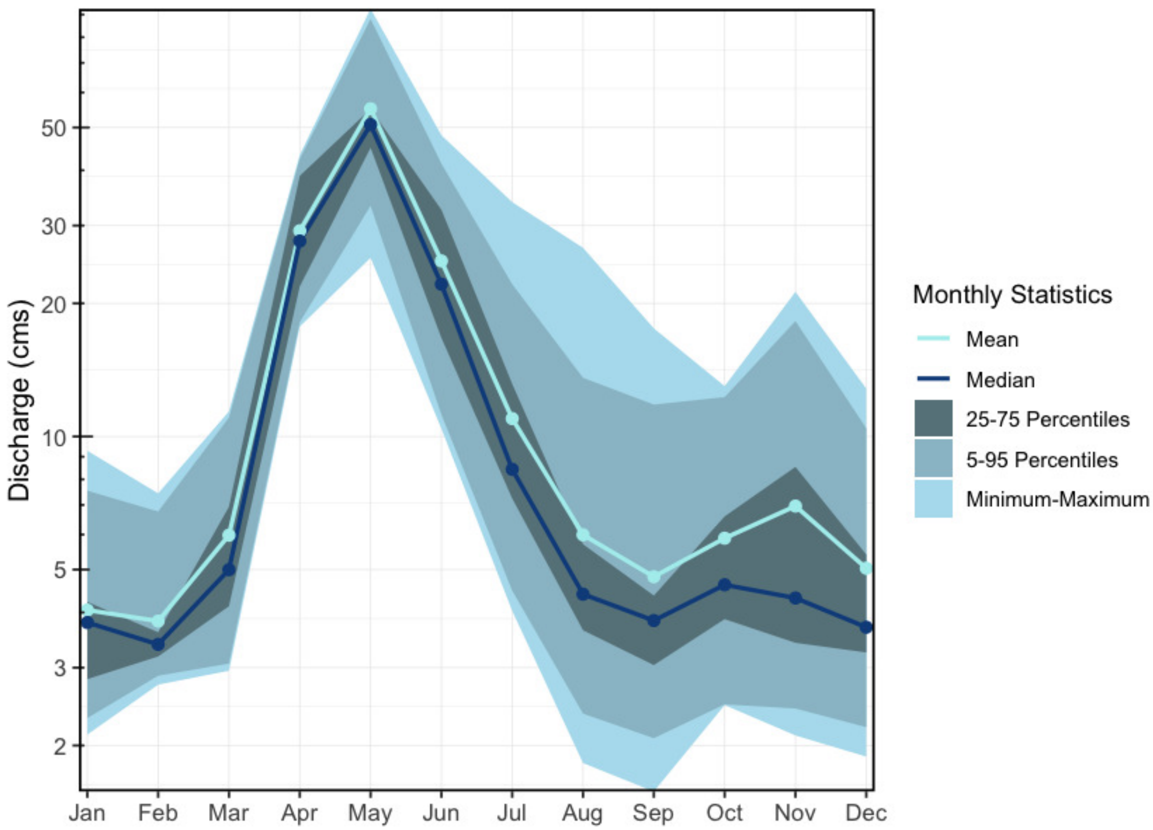

Figure 2.5: Hydrograph for the Chilako River Near Prince George (Station #08JC005 - Lat 53.808891 Lon -122.988892). Available mean daily discharge data from 1960 to 1974.

2.7 Tabor Creek

The Tabor River is a tributary of the Fraser River that drains the eastern portion of the Prince George region. The system originates at Tabor Lake east of Tabor Mountain and flows north–northwest through low-gradient valleys before entering the Fraser River upstream of the Willow River confluence. The watershed is approximately XXXX km^2 and is classified as a 5th order stream. There are two historic hydrometric stations on Tabor Creek. One located where Tabor Creek crossed Highway 97 which was active from 1974 to 1981, and had a mean annual discharge of $0.8 m^3/s$ (Figure 2.6). The second station, active from 1981 to 1999, was located above Swede Creek and had a mean annual discharge of 0.5 (Figure 2.7).

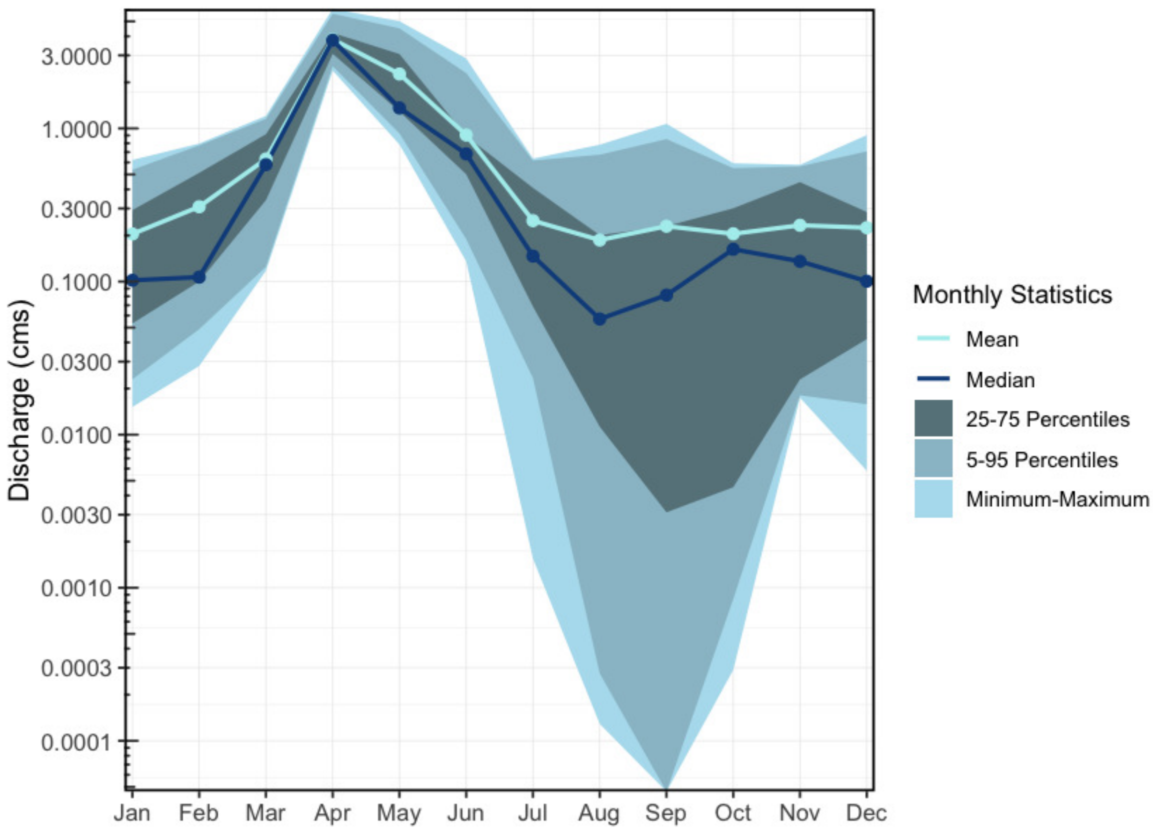

Figure 2.6: Hydrograph for Tabor Creek at Highway 97 Near Prince George (Station #08KE028 - Lat 53.808891 Lon -122.988892). Available mean daily discharge data from 1974 to 1981

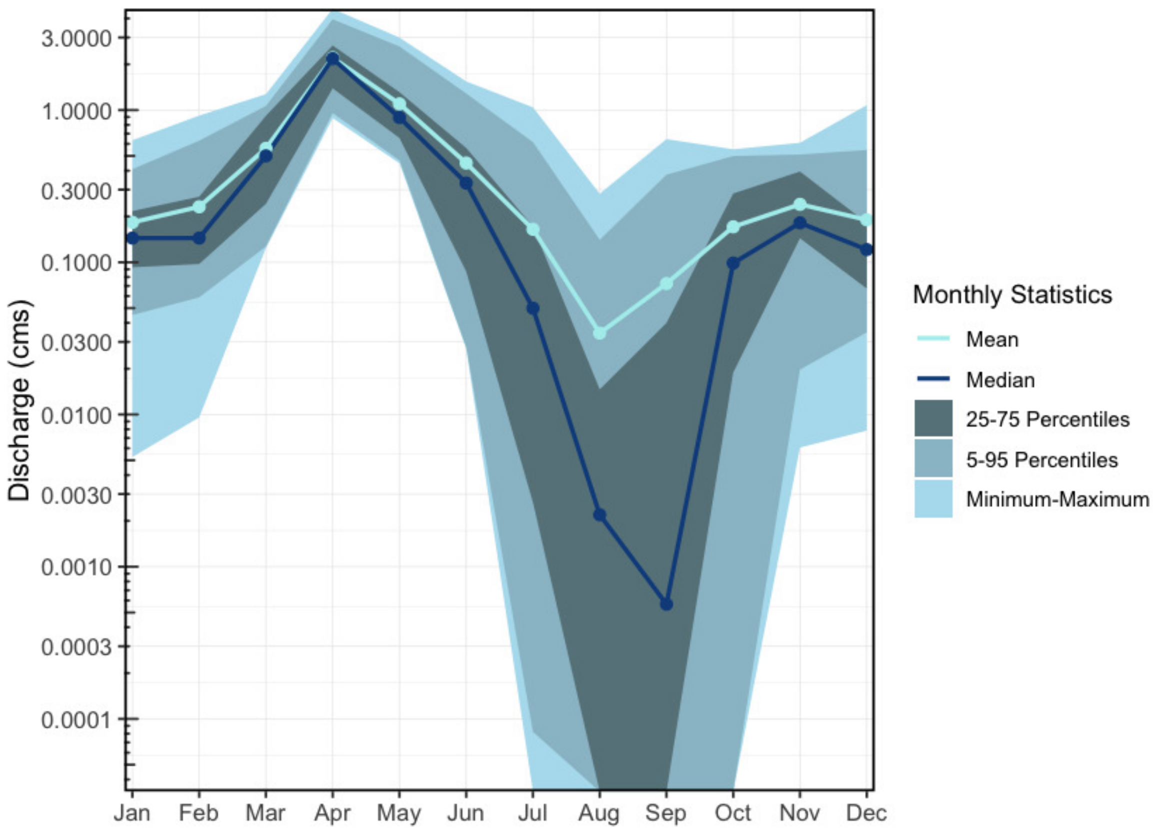

Figure 2.7: Hydrograph for Tabor Creek above Swede Creek near Prince George (Station #08KE032 - Lat 53.808891 Lon -122.988892). Available mean daily discharge data from 1981 to 1999

2.8 Willow River

The Willow River is a tributary of the Fraser River located east of Prince George, British Columbia. The river flows north from its headwaters at Jack of Clubs Lake near the mining community of Wells in the Cariboo Mountains and enters the Fraser River just north of Prince George. At its confluence, the Willow River is classified as a 6th order stream and drains an area of approximately 3,181.3km^2. Several historic hydrometric stations have operated on the Willow River (Stations 08KD006, 08KD003, 08KD002), with the most recent data originating from the station above Hay Creek. This station recorded a mean annual discharge of 36.4m^3/s and was active from 1976 to 2011 (Figure 2.8).

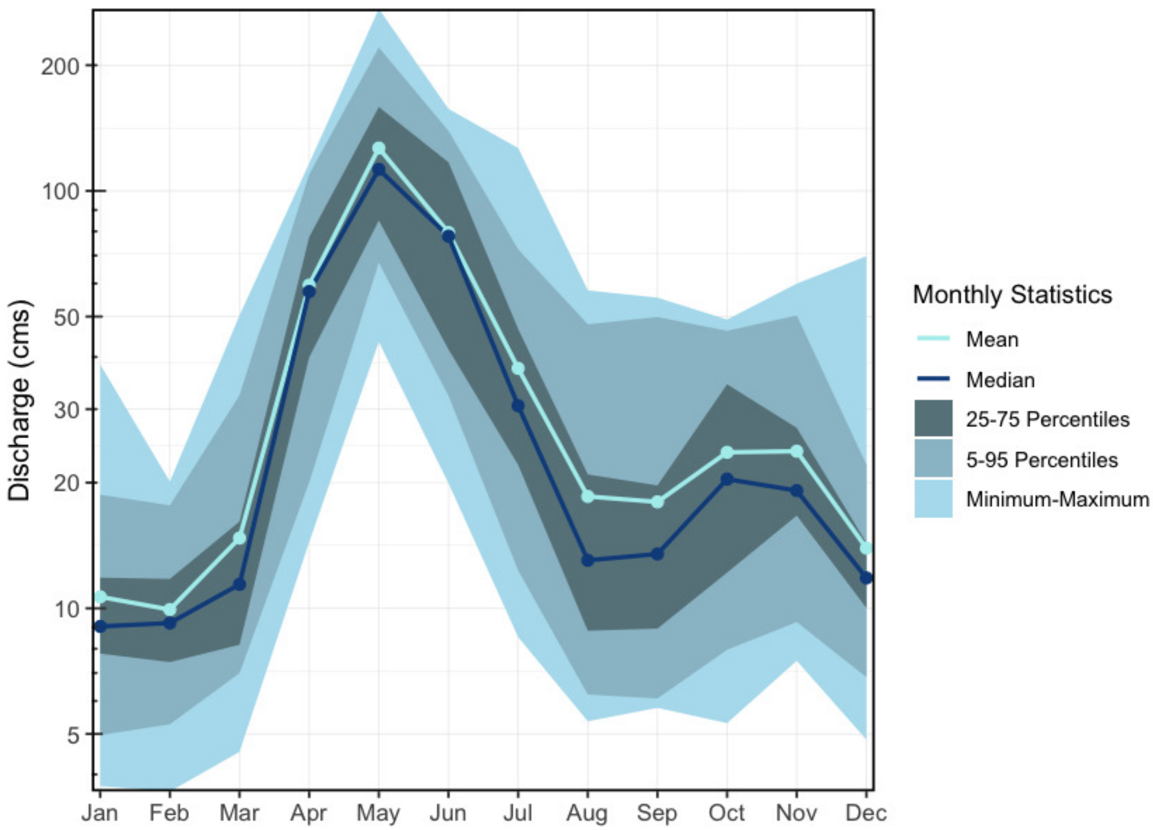

Figure 2.8: Hydrograph for the Willow River above Hay Creek (Station #08KD006 - Lat 54.045556, Lon -122.373056). Available mean daily discharge data from 1976 to 2011

2.9 Lower Salmon River

The Salmon River is a tributary of the Fraser River located north of Prince George, British Columbia. The system drains a low-lying lake district and enters the Fraser River just north of Prince George. The watershed is characterized by predominantly low-relief terrain, with agricultural and residential land use near the confluence. This project focused on the lower Salmon River watershed, defined as the area downstream of the confluence of the Muskeg River and the Salmon River. An active hydrometric station is located approximately 10km upstream of the Fraser River confluence and has a mean annual discharge of 29.3m^3/s (Figure 2.9).

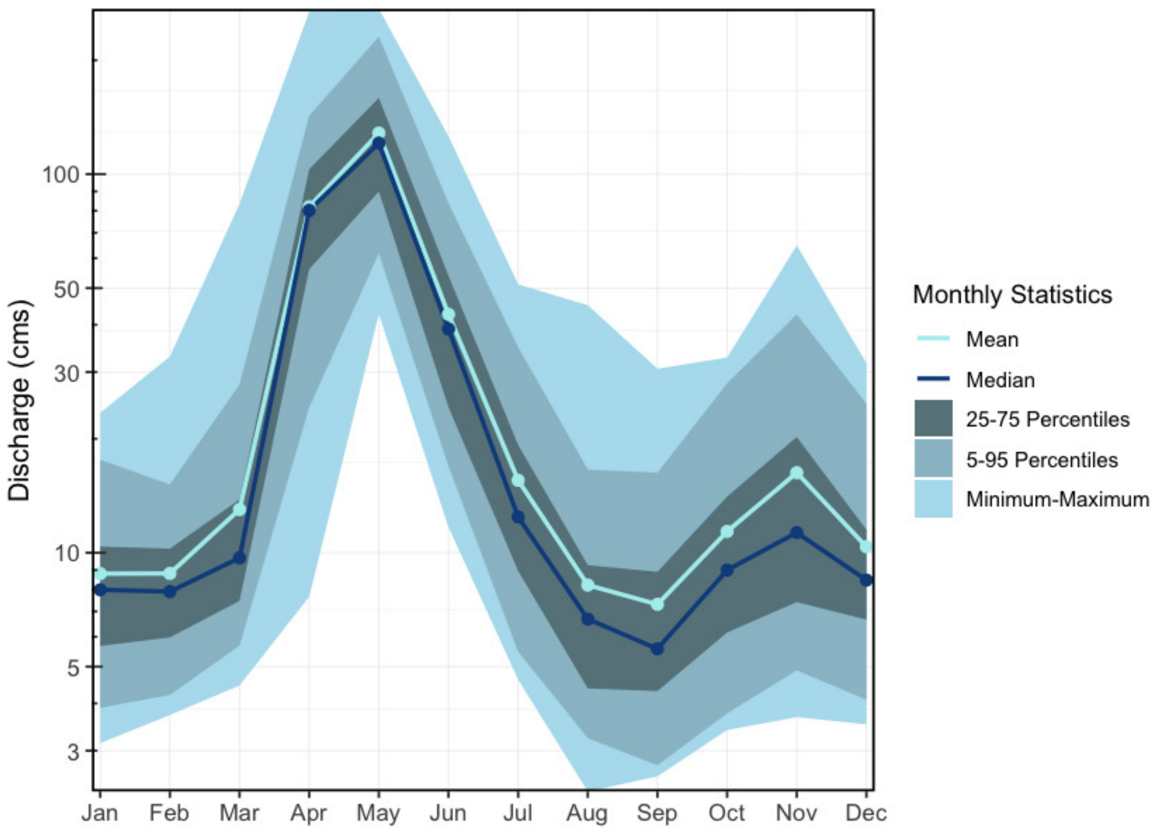

Figure 2.9: Hydrograph for the Salmon River near Prince George (Station #08KC001 - Lat 54.045556, Lon -122.373056). Available mean daily discharge data from 1953 to 2023

2.10 Floodplains and Watershed Function

Floodplains are the low-lying ground adjacent to stream channels that water reclaims during high-flow events. They are not idle margins — they are where rivers do geomorphic work: storing and sorting sediment, exchanging nutrients between terrestrial and aquatic systems, recruiting riparian vegetation, and delivering large woody debris to the channel. For fish, floodplain habitats — side channels, sloughs, beaver complexes, oxbow ponds, and connected wetlands — provide the off-channel conditions that drive freshwater productivity. Juvenile salmonids depend on these areas for low-velocity rearing, overwinter refuge when mainstem conditions become harsh, and high-flow shelter during freshet. In many systems, floodplain habitats are where freshwater survival is won or lost.

Floodplains are often the part of the watershed most compromised by the infrastructure legacy described above. Dykes sever lateral connection; channelization confines the active footprint; drainage eliminates wetland complexes. Mapping floodplain extent at watershed scale therefore complements the linear barrier inventory — prioritization can consider not just whether a structure passes fish but the condition and extent of the off-channel habitat in the reach the structure connects to.

2.11 Fisheries

The Fraser River watershed is the largest salmon-producing river system in British Columbia and one of the most significant on the Pacific Coast. It provides critical habitat for anadromous salmonids, resident freshwater species, and other aquatic organisms. The river supports indigenous, commercial, and recreational fisheries, with Pacific salmon (Oncorhynchus spp.) playing a key ecological and economic role.

The 2019 Big Bar landslide created a significant barrier to salmon migration in the Fraser River, further stressing already vulnerable populations. Extensive mitigation efforts, including rock removal and the construction of a fishway, have helped improve passage, with an estimated 2.9 million salmon successfully navigating the site in 2022. The Upper Fraser Fisheries Conservation Alliance (UFFCA) has played a key role in supporting emergency response and recovery planning, working with federal and provincial agencies to ensure First Nations' interests and traditional knowledge are incorporated into long-term recovery strategies. This includes ongoing monitoring, habitat restoration, and collaborative management to mitigate future risks to Fraser salmon populations (*Upper Fraser Fisheries Conservation Alliance*, n.d.).

2.11.1 Lhtakoh

The Fraser River mainstem serves as the primary migration corridor for all anadromous fish species returning to spawn in its tributaries. Historically, the Fraser supported some of the largest sockeye and chinook salmon runs in North America. However, climate change, habitat degradation, and fisheries pressures have led to declining salmon populations in recent decades.

Upper Fraser sockeye populations have been in decline for several decades, likely due to reduced flows and altered water temperatures, affecting migration and spawning success (Levy and Nicklin 2018). The Early Stuart and Late Stuart sockeye are now classified as “Endangered” by COSEWIC (COSEWIC 2017). Under the Wild Salmon Policy, several Upper Fraser sockeye conservation units have been classified as “red-zoned” (Levy and Nicklin 2018).

In 2005, Nesbit et al. (2005) outlined key fisheries areas in the upper Fraser River watershed as the McLennan River, Swift Creek, and the upper mainstem of the Fraser River near Valemount and Tête Jaune Cache. These systems are known to support chinook salmon, with documented spawning in both Swift Creek and the McLennan River. At the time of reporting by Nesbit et al. (2005), two Wildlife Habitat Areas (WHA 3155 and WHA 6012) were designated in the Tête Jaune Cache area based on high fisheries values. WHA 3155 encompassed the confluence of the McLennan and Fraser Rivers and was recognized as the largest chinook spawning area in the upper Fraser system. WHA 6012 protected an adjacent side channel further downstream, which was identified as important juvenile rearing and refuge habitat during high flow events. Although both WHAs have since been removed, their prior designation underscores the ecological importance of the area.

Swift Creek is the site of ongoing conservation hatchery efforts by the Spruce City Wildlife Association (SCWA), which has operated a volunteer-run hatchery since 1987. Following upgrades in 2020, it is now considered the most technologically advanced hatchery in the province. Focused on restoring upper Fraser River chinook, which are listed as “Endangered” by COSEWIC (COSEWIC 2019), SCWA releases thousands of chinook fry into Swift Creek each spring as part of broader recovery initiatives targeting key spawning systems in the region (Spruce City Wildlife Association 2023).

Three sub-watersheds within the upper Fraser River basin are designated as Fisheries Sensitive Watersheds (FSWs) due to the presence of sensitive fish species, including chinook salmon and bull trout. The Goat River and Milk River watersheds have held FSW status since 2013 (FLNRORD 2004b), and the Walker Creek watershed has been designated since 2004 (FLNRORD 2004a).

The importance of salmon in the upper Fraser is strongly reflected in Valemount’s community identity, education, and ecotourism initiatives. Public engagement with the salmon run is encouraged through accessible viewing areas at George Hicks Regional Park on Swift Creek, the Fraser River at Tête Jaune Cache, and along the McLennan River—locations that serve both as local gathering points and platforms for increasing awareness of salmon life cycles and watershed stewardship (Tourism Valemount, n.d.).

2.11.2 Nechakoh

The Nechako River, a tributary of the Fraser River, supports Chinook salmon populations and serves as a migration corridor for sockeye salmon that spawn in the Stuart and Nadina/Francois Basins. Since the construction of the Kenney Dam in the 1950s, flows in the Nechako River have been regulated by Rio Tinto Alcan for hydroelectric power generation at the Kemano Generating Station on the Pacific Coast. In response to concerns over impacts on Nechako Chinook and Upper Fraser sockeye, flow management practices have been implemented. Since 1987, the Nechako Fisheries Conservation Program (NFCP) has operated an annual monitoring program, focussed on Nechako Chinook and migratory sockeye salmon designed to monitor the effectiveness of conservation measures specified in the 1987 Settlement Agreement between Canada, British Columbia, and Alcan (Levy and Nicklin 2018).

The Nechako River also supports a genetically distinct and endangered population of white sturgeon with low natural recruitment. To address this decline, the Nechako White Sturgeon Recovery Initiative (NWSRI) was established, implementing conservation measures such as the Nechako White Sturgeon Conservation Centre. The facility focuses on conservation aquaculture to maintain genetic diversity and support population recovery. Since 2006, the NWSRI has been releasing tagged juvenile sturgeon annually into the Nechako River to aid in species restoration ("Nechako White Sturgeon Recovery Initiative I Home," n.d.).

Indigenous fisheries organizations, including individual First Nations, the Carrier Sekani Tribal Council, and the Upper Fraser Fisheries Conservation Alliance (UFFCA), are actively involved in fisheries management in the Nechako watershed. These organizations work to support conservation while ensuring the sustainability of Indigenous fisheries, which have cultural and subsistence importance (*Upper Fraser Fisheries Conservation Alliance*, n.d.).

2.11.3 François Lake

François Lake is one of the largest natural lakes in BC and supports landlocked Kokanee (O. nerka), as well as migratory sockeye salmon that access the lake via the Stellako River. François Lake is a key rearing area for juvenile salmon before migrating to the Fraser River.

2.11.4 Morkill River

The Morkill River, a tributary to the upper Fraser, supports populations of bull trout, mountain whitefish, and rainbow trout. While salmon presence in the system is limited, the Morkill provides cold water inputs to the Fraser River, which may offer thermal refuge for migrating salmonids during warm summer months.

2.11.5 Salmon Stock Assessment Data

Fisheries and Oceans Canada stock assessment data was accessed via the NuSEDS-New Salmon Escapement Database System through the Open Government Portal with results presented here (Fisheries and Oceans Canada, n.d.). A brief memo on the data extraction process is available here.

2.11.6 Fish Species

Summary of historical fish observations in the Upper Fraser River, Nechako River, Lower Chilako River, Willow River, Lower Salmon River, Tabor Creek, and François Lake watershed groups (Table 2.1) (MoE 2024).

A review of available fisheries data, for the Upper Fraser River, Nechako River, Lower Chilako River, Willow River, Lower Salmon River, Tabor Creek, and François Lake watershed groups, stratified by different habitat characteristics can provide insight into which habitats may provide the highest intrinsic value for fish species based on the number of fish captured in those habitats in past assessment work (Figures 2.10 - 2.11 - 2.12). It should be noted however that it should not be assumed that all habitat types have been sampled in a non-biased fashion or that particular sites selected do not have a disproportionate influence on the overall dataset composition (ie. fish salvage sites are often located adjacent to construction sites which are more commonly located near lower gradient stream reaches).

Table 2.1: Fish species recorded in the Francois Lake, Lower Chilako River, Lower Salmon River, Morkill River, Nechako River, Tabor River, Upper Fraser River, and Willow River watershed groups.

Scientific Name	Species Name	BC List	COSEWIC	Francois Lake	Lower Chilako	Lower Salmon	Morkill	Nechako	Tabor	Upper Fraser	Willow
Acipenser transmontanus	White Sturgeon	No Status	E/T (Nov 2012)	Yes	Yes	–	Yes	Yes	Yes	–	–
Carassius auratus	Goldfish	Exotic	–	–	Yes	–	–	–	–	–	–
Catostomus bondi	Northern Mountain Sucker	Blue	T (Dec 2022)	–	Yes	–	–	–	–	–	–
Catostomus catostomus	Longnose Sucker	Yellow	–	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Catostomus columbianus	Bridgelip Sucker	Yellow	–	Yes	–	Yes	–	Yes	Yes	–	–
Catostomus commersonii	White Sucker	Yellow	–	Yes	Yes	Yes	Yes	Yes	Yes	–	Yes
Catostomus macrocheilus	Largescale Sucker	Yellow	–	Yes	Yes	Yes	Yes	Yes	Yes	–	Yes
Chrosomus neogaeus	Finescale Dace	Yellow	–	Yes	–	–	–	–	–	–	–
Coregonus clupeaformis	Lake Whitefish	Yellow	–	Yes	Yes	Yes	–	Yes	–	–	Yes
Cottus asper	Prickly Sculpin	Yellow	–	Yes	Yes	Yes	Yes	Yes	Yes	–	Yes
Cottus cognatus	Slimy Sculpin	Yellow	–	Yes	Yes	–	Yes	Yes	Yes	Yes	Yes
Cottus ricei	Spoonhead Sculpin	Yellow	NAR (May 1989)	Yes	–	–	–	–	–	Yes	–
Couesius plumbeus	Lake Chub	Yellow	DD	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Cyprinus carpio	Carp	Exotic	–	Yes	–	–	–	–	–	–	–
Hybognathus hankinsoni	Brassy Minnow	No Status	–	Yes	Yes	Yes	–	Yes	Yes	–	Yes
Lota lota	Burbot	Yellow	–	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Micropterus salmoides	Largemouth Bass	Exotic	–	–	–	–	–	Yes	–	–	–

Scientific Name	Species Name	BC List	COSEWIC	Francois Lake	Lower Chilako	Lower Salmon	Morkill	Nechako	Tabor	Upper Fraser	Willow
Mylocheilus caurinus	Peamouth Chub	Yellow	–	Yes	Yes	Yes	Yes	Yes	Yes	–	Yes
Oncorhynchus clarkii	Cutthroat Trout	No Status	–	Yes	–	–	–	–	–	–	Yes
Oncorhynchus gorbuscha	Pink Salmon	Not Reviewed	–	–	Yes	Yes	–	–	–	–	Yes
Oncorhynchus kisutch	Coho Salmon	Not Reviewed	–	Yes	Yes	–	–	Yes	–	–	Yes
Oncorhynchus mykiss	Rainbow Trout	Yellow	–	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Oncorhynchus mykiss	Steelhead	Yellow	–	–	–	–	Yes	–	–	–	–
Oncorhynchus nerka	Kokanee	Not Reviewed	–	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Oncorhynchus nerka	Sockeye Salmon	Not Reviewed	–	Yes	Yes	–	Yes	Yes	–	Yes	–
Oncorhynchus tshawytscha	Chinook Salmon	Not Reviewed	E/T/SC/DD/NAR (Nov 2020)	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Prosopium coulterii	Pygmy Whitefish	Yellow	NAR (Nov 2016)	–	Yes	–	–	Yes	Yes	Yes	Yes
Prosopium cylindraceum	Round Whitefish	Yellow	–	–	–	–	Yes	–	–	–	–
Prosopium williamsoni	Mountain Whitefish	Yellow	–	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Ptychocheilus oregonensis	Northern Pikeminnow	Yellow	–	Yes	Yes	Yes	Yes	Yes	Yes	–	Yes
Rhinichthys cataractae	Longnose Dace	Yellow	–	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Rhinichthys falcatus	Leopard Dace	Yellow	NAR (May 1990)	Yes	Yes	Yes	–	Yes	Yes	–	Yes
Richardsonius balteatus	Redside Shiner	Yellow	–	Yes	Yes	Yes	Yes	Yes	Yes	–	Yes
Salvelinus confluentus	Anadromous Bull Trout	Blue	SC (Nov 2012)	–	–	–	–	Yes	–	–	–
Salvelinus confluentus	Bull Trout	Blue	SC (Nov 2012)	Yes	Yes	–	Yes	Yes	Yes	Yes	Yes
Salvelinus fontinalis	Brook Trout	Exotic	–	Yes	Yes	Yes	Yes	Yes	Yes	Yes	–
Salvelinus malma	Dolly Varden	Yellow	–	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Salvelinus namaycush	Lake Trout	Yellow	–	Yes	Yes	–	Yes	Yes	Yes	Yes	Yes

Scientific Name	Species Name	BC List	COSEWIC	Francois Lake	Lower Chilako	Lower Salmon	Morkill	Nechako	Tabor	Upper Fraser	Willow
–	All Salmon	–	–	–	Yes	–	–	–	–	–	–
–	Arctic Char	–	–	–	–	–	–	–	–	–	Yes
–	Char, General	–	–	–	–	–	–	–	–	Yes	–
–	Chub (General)	–	–	Yes	Yes	Yes	–	Yes	–	–	Yes
–	Dace (General)	–	–	Yes	Yes	Yes	–	Yes	Yes	–	–
–	Lamprey (General)	–	–	–	–	Yes	–	–	–	–	–
–	Minnow (General)	–	–	Yes	Yes	Yes	Yes	Yes	Yes	–	Yes
–	Northern Pearl Dace	–	–	Yes	–	–	–	–	–	–	–
–	Salmon (General)	–	–	–	Yes	–	Yes	–	–	Yes	Yes
–	Sculpin (General)	–	–	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
–	Stickleback (General)	–	–	–	–	–	–	–	Yes	–	–
–	Sucker (General)	–	–	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
–	Whitefish (General)	–	–	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

* COSEWIC abbreviations :
SC - Special concern
DD - Data deficient
NAR - Not at risk
E - Endangered
T - Threatened

BC List definitions :
Yellow - Species that is apparently secure
Blue - Species that is of special concern
Exotic - Species that have been moved beyond their natural range as a result of human activity

Table 2.2: Summary of historic fish observations vs. stream gradient category for the Francois Lake, Lower Chilako River, Lower Salmon River, Morkill River, Nechako River, Tabor River, Upper Fraser River, and Willow River watershed groups.

species_code	Gradient	Count	total_spp	Percent
BT	0 - 3 %	57	109	52

species_code	Gradient	Count	total_spp	Percent
BT	03 - 5 %	14	109	13
BT	05 - 8 %	19	109	17
BT	08 - 15 %	12	109	11
BT	15 - 22 %	7	109	6
CH	0 - 3 %	168	186	90
CH	03 - 5 %	15	186	8
CH	05 - 8 %	2	186	1
CH	08 - 15 %	1	186	1
CM	0 - 3 %	3	4	75
CM	08 - 15 %	1	4	25
CO	0 - 3 %	393	491	80
CO	03 - 5 %	74	491	15
CO	05 - 8 %	13	491	3
CO	08 - 15 %	7	491	1
CO	15 - 22 %	3	491	1
CO	22+ %	1	491	0
CT	0 - 3 %	391	572	68
CT	03 - 5 %	87	572	15
CT	05 - 8 %	65	572	11
CT	08 - 15 %	25	572	4
CT	15 - 22 %	3	572	1
CT	22+ %	1	572	0
DV	0 - 3 %	414	940	44
DV	03 - 5 %	162	940	17
DV	05 - 8 %	177	940	19
DV	08 - 15 %	149	940	16
DV	15 - 22 %	27	940	3
DV	22+ %	11	940	1
PK	0 - 3 %	78	87	90
PK	03 - 5 %	8	87	9
PK	08 - 15 %	1	87	1
RB	0 - 3 %	813	1139	71
RB	03 - 5 %	138	1139	12

species_code	Gradient	Count	total_spp	Percent
RB	05 - 8 %	77	1139	7
RB	08 - 15 %	95	1139	8
RB	15 - 22 %	11	1139	1
RB	22+ %	5	1139	0
SK	0 - 3 %	44	45	98
SK	08 - 15 %	1	45	2

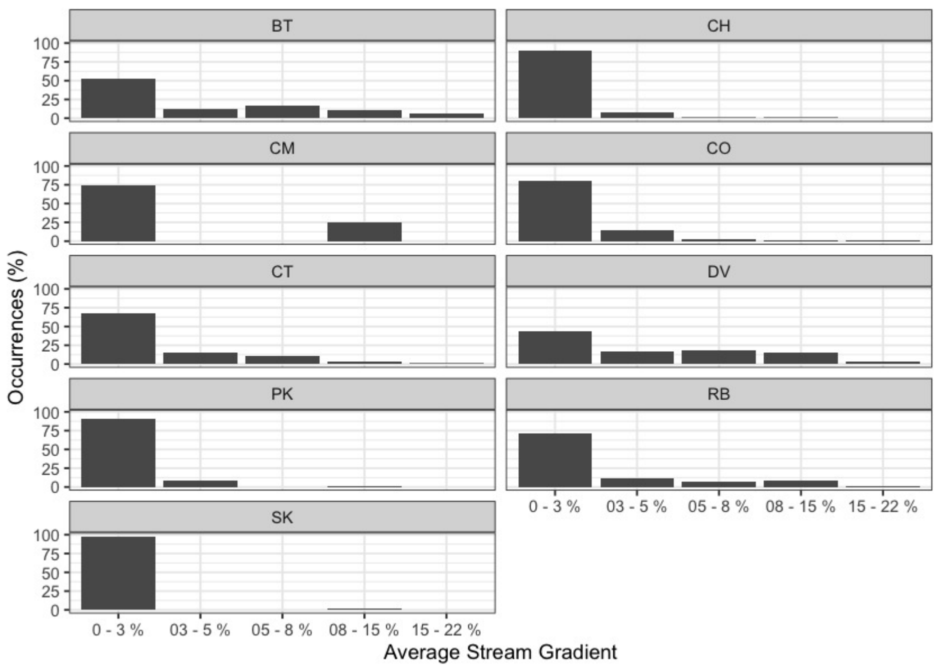

Figure 2.10: Summary of historic fish observations vs. stream gradient category for the Francois Lake, Lower Chilako River, Lower Salmon River, Morkill River, Nechako River, Tabor River, Upper Fraser River, and Willow River watershed groups.

Table 2.3: Summary of historic fish observations vs. channel width category for the Nechako River, Lower Chilako River, François Lake, Morkill River and Upper Fraser River watershed groups.

species_code	Width	Count	total_spp	Percent
BT	0 - 2m	2	109	2
BT	02 - 04m	13	109	12

species_code	Width	Count	total_spp	Percent
BT	04 - 06m	20	109	18
BT	06 - 10m	25	109	23
BT	10 - 15m	18	109	17
BT	15m+	7	109	6
BT	–	24	109	22
CH	0 - 2m	2	186	1
CH	02 - 04m	6	186	3
CH	04 - 06m	9	186	5
CH	06 - 10m	38	186	20
CH	10 - 15m	38	186	20
CH	15m+	79	186	42
CH	–	14	186	8
CM	0 - 2m	2	4	50
CM	04 - 06m	1	4	25
CM	15m+	1	4	25
CO	0 - 2m	8	491	2
CO	02 - 04m	47	491	10
CO	04 - 06m	75	491	15
CO	06 - 10m	110	491	22
CO	10 - 15m	73	491	15
CO	15m+	93	491	19
CO	–	85	491	17
CT	0 - 2m	38	572	7
CT	02 - 04m	97	572	17
CT	04 - 06m	69	572	12
CT	06 - 10m	54	572	9
CT	10 - 15m	26	572	5
CT	15m+	28	572	5
CT	–	260	572	45
DV	0 - 2m	75	940	8
DV	02 - 04m	242	940	26
DV	04 - 06m	134	940	14
DV	06 - 10m	154	940	16

2.11 Fisheries

species_code	Width	Count	total_spp	Percent
DV	10 - 15m	72	940	8
DV	15m+	68	940	7
DV	–	195	940	21
PK	0 - 2m	2	87	2
PK	02 - 04m	2	87	2
PK	04 - 06m	7	87	8
PK	06 - 10m	15	87	17
PK	10 - 15m	10	87	11
PK	15m+	44	87	51
PK	–	7	87	8
RB	0 - 2m	40	1139	4
RB	02 - 04m	158	1139	14
RB	04 - 06m	135	1139	12
RB	06 - 10m	134	1139	12
RB	10 - 15m	74	1139	6
RB	15m+	93	1139	8
RB	–	505	1139	44
SK	0 - 2m	2	45	4
SK	04 - 06m	1	45	2
SK	06 - 10m	2	45	4
SK	10 - 15m	4	45	9
SK	15m+	17	45	38
SK	–	19	45	42

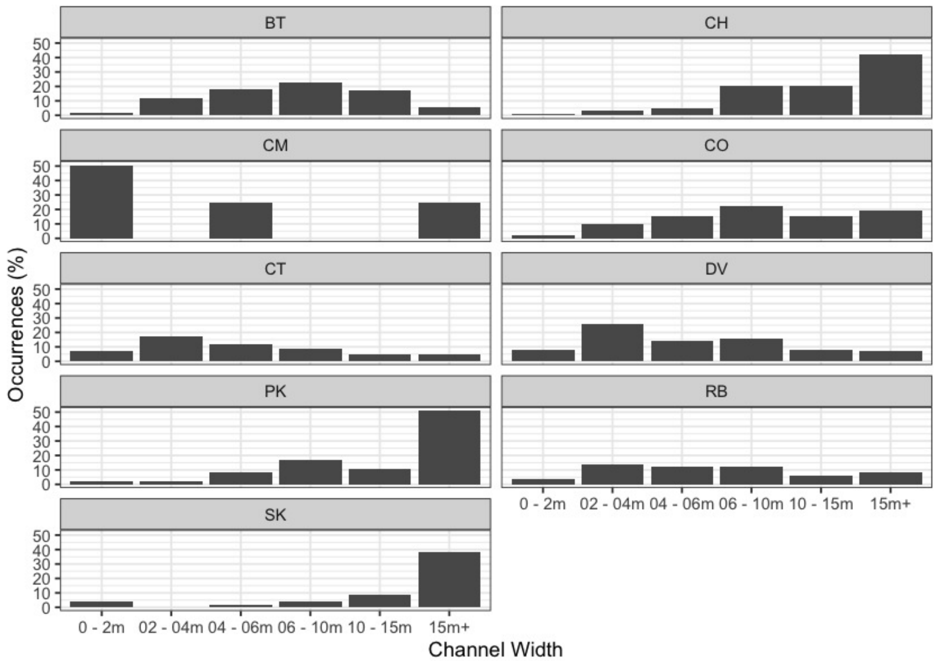

Figure 2.11: Summary of historic fish observations vs. channel width category for the Parsnip River watershed group.

Table 2.4: Summary of historic fish observations vs. watershed size category for the Nechako River, Lower Chilako River, François Lake, Morkill River and Upper Fraser River watershed groups.

species_code	Watershed	count_wshd	total_spp	Percent
BT	0 - 25km2	165	446	37
BT	25 - 50km2	62	446	14
BT	50 - 75km2	12	446	3
BT	75 - 100km2	14	446	3
BT	100km2+	193	446	43
CH	0 - 25km2	76	800	10
CH	25 - 50km2	31	800	4
CH	50 - 75km2	32	800	4
CH	75 - 100km2	4	800	0
CH	100km2+	657	800	82
CM	0 - 25km2	12	49	24
CM	25 - 50km2	6	49	12

2.11 Fisheries

species_code	Watershed	count_wshd	total_spp	Percent
CM	50 - 75km2	4	49	8
CM	75 - 100km2	2	49	4
CM	100km2+	25	49	51
CO	0 - 25km2	35	137	26
CO	25 - 50km2	22	137	16
CO	50 - 75km2	14	137	10
CO	75 - 100km2	11	137	8
CO	100km2+	55	137	40
CT	0 - 25km2	158	284	56
CT	25 - 50km2	15	284	5
CT	50 - 75km2	5	284	2
CT	75 - 100km2	9	284	3
CT	100km2+	97	284	34
DV	0 - 25km2	210	322	65
DV	25 - 50km2	23	322	7
DV	50 - 75km2	8	322	2
DV	75 - 100km2	22	322	7
DV	100km2+	59	322	18
PK	0 - 25km2	15	48	31
PK	25 - 50km2	15	48	31
PK	50 - 75km2	1	48	2
PK	75 - 100km2	1	48	2
PK	100km2+	16	48	33
RB	0 - 25km2	2867	4621	62
RB	25 - 50km2	308	4621	7
RB	50 - 75km2	190	4621	4
RB	75 - 100km2	395	4621	9
RB	100km2+	861	4621	19
SK	0 - 25km2	5	52	10
SK	25 - 50km2	2	52	4
SK	75 - 100km2	2	52	4
SK	100km2+	43	52	83

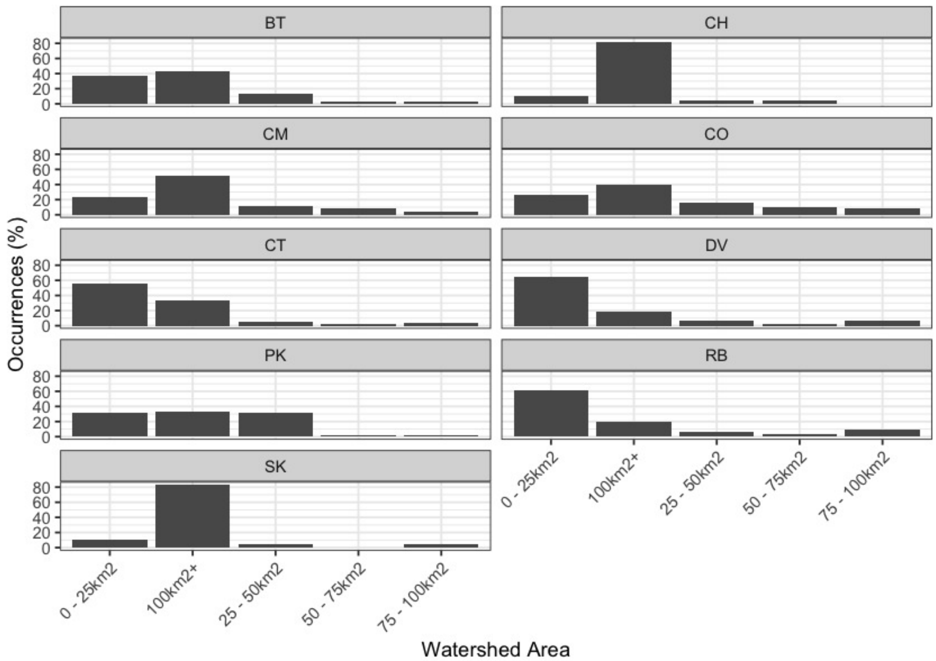

Figure 2.12: Summary of historic fish observations vs. watershed size category for the Nechako River, Lower Chilako River, François Lake, Morkill River and Upper Fraser River watershed groups.

3 Methods

3.1 Climate Departure

Climate departure measures how far recent climate has shifted from a historical baseline — in this case, how much warmer, drier, or snowpack-altered the study area has become relative to the pre-warming period (1951–1980). Where traditional fish-passage assessments focus on the physical structures in and around streams, climate departure adds the atmospheric and hydrologic context those structures sit inside: warming air temperatures that drive warmer water, shifting snowpack that reshapes the freshet, and atmospheric drying that reduces late-summer flows. Understanding these departures at the watershed scale lets us ask whether the habitat above a barrier is gaining or losing value as climate changes — and weight barrier prioritisation accordingly.

We examined 15 climate variables spanning temperature (annual mean, daytime maximum, overnight minimum), moisture (precipitation, vapour pressure deficit, soil moisture, relative humidity), and snowpack (snow water equivalent, snowfall, snowmelt, snow cover, snowfall fraction, peak snow water equivalent, snowmelt timing, and peak melt rate). Each variable was assessed at annual and seasonal scales, across the full study area and within each of eight ecoregions.

Climate departures were computed with the cd R package, which reads ERA5-Land hourly reanalysis (1950–present, ~9 km native grid) aggregated to monthly, seasonal, and annual periods on a single grid covering British Columbia, served as Cloud-Optimised GeoTIFFs from a public S3 catalog. For each variable the package crops the published rasters to the area of interest, takes a yearly spatial mean, and computes anomalies against the 1951–1980 pre-warming reference period — the same base period Hansen et al. (2012) used for global temperature emergence. Long-term trends are tested with Mann-Kendall significance and quantified with Theil-Sen slopes; the recent-decade (2015–2025) versus pre-warming window comparison uses a Welch two-sample t-test on the annual values. The same pipeline runs at any polygon scale — region, ecoregion, or watershed group. The regional and ecoregion-level analyses are summarised in this section; the watershed-group breakdown feeds into the per-watershed barrier prioritisation that follows. Appendix - Climate Departure (page 105) carries the full set of plots and tables.

3.2 Collaborative GIS Environment

Geographical Information Systems are essential for developing and communicating restoration plans as well as the reasons they are required and how they are developed. Without the ability to visualize the landscape and the data that is used to make decisions, it is difficult to conduct and communicate the need for restoration, the details of past and future plans as well as and the potential results of physical works.

To facilitate the planning and implementation of restoration activities, a collaborative GIS environment has been established using QGIS and is served on the cloud using source code stored here. This environment is intended to be a space where project team members can access, view, and contribute to the amalgamation of background spatial data and the development of restoration as well as monitoring for the project. The collaborative GIS environment allows users to view, edit, and analyze shared, up to date spatial data on personal computers in an office setting as well as on phones and tablets in the field. At the time of reporting, the environment was being used to develop and share maps, conduct spatial analyses, communicate restoration plans to stakeholders as well as to provide a central place to store methodologies and tools for conducting field assessments on standardized pre-developed digital forms. The platform can also be used to track the progress of restoration activities and monitor changes in the landscape over time, helping encourage the record keeping of past and future restoration activities in a coordinated manner.

The shared QGIS project was created using scripts currently kept in `dff-2022` with the precise calls to project creation scripts tracked in the `project_creation_and_permissions.txt` document kept in the main QGIS project directory. Information about the scripts used for GIS project creation and updates can be viewed here with outcomes of their use summarized below:

- Download and clip user specified layers from the BC Data Catalogue as well as data layers stored in custom Amazon Web Services buckets for an area of interest defined by a list of watershed groups and load to a geopackage called `background_layers.gpkg` stored in the main directory of the project.
- A project directory is created to hold the spatial data and `QGIS` project information (ie. layer symbology and naming conventions, metadata, etc.).
- Metadata for individual project spatial layers is kept in the `rfp_tracking` table within the `background_layers.gpkg` along with tables related to user supplied stream width/gradient inputs to `bcfishpass` to model potentially high value habitat that is accessible to fish species of interest.

3.2.1 Mapping

The workflows to produce the georeferenced `pdf` maps include using a QGIS layer file defining and symbolizing all layers required and are continuously evolving. At the time of reporting - mapping scripts and associated layer file were kept under version control within `bcfishpass` here. Loading the QGIS layer file within a QGIS project, allows load and representation of all map component layers provided the user points to a postgresql database populated via `bcfishpass` outputs.

3.3 Planning — Habitat, Connectivity and Floodplain Modelling

3.3.1 Habitat Modelling

Habitat modelling used to help guide planning for field assessments is generated by `bcfishpass` (Norris [2020] 2024) which has been designed to prioritize potential fish passage barriers for assessment or remediation by generating a simple model of aquatic habitat connectivity. We utilize the `bcfishpass` access model, linear spawning/rearing habitat model and lateral habitat connectivity for planning purposes. These models provide a valuable starting point, but their results are not definitive and should always be considered with professional judgment. Detailed information regarding model methodology, select parameters and known model limitations are detailed in Norris ([2020] 2024) with key documentation linked below:

- Access model
- Linear spawning/rearing habitat models
- Lateral habitat model

Table 3.1 documents the custom species-specific thresholds for stream gradient and channel width applied to the linear spawning and rearing habitat model for this year's project planning.

Table 3.1: Stream gradient and channel width thresholds used to model potentially highest value fish habitat.

Variable	Bull Trout	Chinook Salmon	Coho Salmon	Sockeye Salmon
Spawning Gradient Max (%)	5.5	4.5	5.5	2.5
Spawning Width Min (m)	2	4	2	2
Rearing Width Min (m)	1.5	1.5	1.5	1.5
Rearing Gradient Max (%)	10.5	5.5	5.5	–

*

3.3.1.1 Statistical Support for `bcfishpass` Fish Habitat Modelling Updates

This project provided the statistical background for updates to `bcfishpass` that facilitated incorporation of channel width (observed or predicted) into species specific linear spawning/rearing habitat models. In early 2021, Bayesian statistical methods were developed to predict channel width in all provincial freshwater atlas stream segments where width measurements had not previously been measured in the field. The model was based on the relationship between watershed area and mean annual precipitation weighted by upstream watershed area (Thorley and Irvine 2021). In December of 2021, Thorley and Irvine (2021) methods were updated using a power model derived by Finnegan et al. (2005) which relates stream discharge to watershed area and mean annual precipitation resulting in Thorley et al. (2021) which was utilized for channel width estimates within `bcfishpass` modelling at the time of reporting. More detailed documentation of the methodology used to facilitate both the data collection and statistical analysis can be sourced in Irvine ([2021] 2022) and Thorley et al. (2021).

In 2024, in collaboration with Poisson Consulting - stream discharge and temperature causal effects pathways were mapped with the intent of focusing aquatic restoration actions in areas of highest potential for positive impacts on fisheries values (ie. elimination of areas from intrinic models where water temperatures are likely too cold to support fish production). The project began with a custom mechanistic model (visually represented here), but the model struggled to converge. The project then shifted to the air2stream model, which offers a middle ground between fully mechanistic models—often data-intensive and reliant on quantities that are difficult to measure or estimate—and purely statistical models, which lack physical justification and perform poorly when extrapolated to new conditions (Toffolon and Piccolroaz (2015)). After several adaptations, the expected stream temperatures were best modeled using the four-parameter version of the air2stream model, with added random effects by site for each of the four parameters (Hill et al. (2024)). The data used for the model were sourced from the following locations, for years 2019-2021:

- Water temperature data collected in the Nechako Watershed were downloaded from Zenodo (Gilbert et al. 2022).
- Hourly air temperature data were obtained from the ERA-5-Land dataset via the Copernicus Climate Change Service (Muñoz Sabater (2019))
- Daily baseflow and surface runoff data were sourced from the Pacific Climate Impacts Consortium's Gridded Hydrologic Model Output using the ACCESS1-0_rcp85 scenario (Pacific Climate Impacts Consortium (n.d.)).

3.3.2 Floodplain Delineation

To complement the linear barrier inventory, we mapped functional floodplain extent at watershed scale for the Nechako River Watershed Group. Overlaying floodplain footprint on the crossing inventory lets us ask not just whether fish can pass a structure, but how much off-channel habitat sits within the affected corridor and whether infrastructure has compromised the lateral connectivity that sustains it.

We used the `flooded` R package (Nagel et al. 2014), which models the area a river actively inundates during regular high-flow events — the **functional floodplain**. This is distinct from the narrow active channel (the wetted perimeter at low flow) and the broader geologic valley bottom (which includes terraces untouched by water for centuries). The functional floodplain is the scale where off-channel habitat is created and maintained: where side channels form, seasonal flooding sustains wetlands, and riparian forests recruit. Bankfull depth is estimated from the regional regression of Hall et al. (2007), calibrated against field-mapped floodplain sites in Pacific Northwest streams. Inputs were the MRDEM-30 30 m DEM, the chinook accessible stream network (stream order ≥ 3) from bcfishpass, and BC Freshwater Atlas lakes and wetlands on that network. Full methodology and watershed-scale mapping are in Appendix - Floodplain Delineation (page 101).

3.4 Fish Passage Assessments

3.4.1 Natural Barriers to Fish Passage

Our assessments may include natural features such as waterfalls that could limit fish passage. This informs whether upstream culvert upgrades would restore access for anadromous species (e.g., salmon) or primarily benefit resident fish already upstream. We document these features by measuring height, gradient, and pool depth, recording field observations, capturing site photographs, and reviewing background sources for context.

3.4.2 Road Stream Crossings

In the field, crossings prioritized for follow-up were first assessed for fish passage following the procedures outlined in “Field Assessment for Determining Fish Passage Status of Closed Bottomed Structures” (MoE 2011). The reader is referred to (MoE 2011) for detailed methodology. Crossings surveyed included closed bottom structures (CBS), open bottom structures (OBS) and crossings considered “other” (i.e. fords). Photos were taken at surveyed crossings and when possible included images of the road, crossing inlet, crossing outlet, crossing barrel, channel downstream and channel upstream of the crossing and any other relevant features. The following information was recorded for all surveyed crossings: date of inspection, crossing reference, crew member initials, Universal Transverse Mercator (UTM) coordinates, stream name, road name and kilometer, road tenure information, crossing type, crossing subtype, culvert diameter or span for OBS, culvert length or width for OBS. A more detailed “full assessment” was completed for all closed bottom structures and included the following parameters: presence/absence of continuous culvert embedment (yes/no), average depth of embedment, whether or not the culvert bed resembled the native stream bed, presence of and percentage backwatering, road fill depth, outlet drop, outlet pool depth, inlet drop, culvert slope, average downstream channel width, stream slope, presence/absence of beaver activity, presence/absence of fish at time of survey, type of valley fill, and a habitat value rating. Habitat value ratings were based on channel morphology, flow characteristics (perennial, intermittent, ephemeral), fish migration patterns, the presence/absence of deep pools, un-embedded boulders, substrate, woody debris, undercut banks, aquatic vegetation and overhanging riparian vegetation (Table 3.2).

Table 3.2: Habitat value criteria (Fish Passage Technical Working Group, 2011).

Habitat Value	Fish Habitat Criteria
High	The presence of high value spawning or rearing habitat (e.g., locations with abundance of suitably sized gravels, deep pools, undercut banks, or stable debris) which are critical to the fish population.
Medium	Important migration corridor. Presence of suitable spawning habitat. Habitat with moderate rearing potential for the fish species present.
Low	No suitable spawning habitat, and habitat with low rearing potential (e.g., locations without deep pools, undercut banks, or stable debris, and with little or no suitably sized spawning gravels for the fish species present).

Fish passage potential was determined for each stream crossing identified as a closed bottom structure as per MoE (2011). The combined scores from five criteria: depth and degree to which the structure is embedded, outlet drop, stream width ratio, culvert slope, and culvert length were used to screen whether each culvert was a likely barrier to some fish species and life stages (Table 3.3, Table 3.4). These criteria were developed based on data obtained from various studies and reflect an estimation for the passage of a juvenile salmon or small resident rainbow trout (Clarkin et al. 2005; Bell 1991; Thompson 2013). For crossings determined to be potential barriers or barriers based on the data, a culvert fix and recommended diameter/span was proposed.

Table 3.3: Fish Barrier Risk Assessment (MoE 2011).

Risk	LOW	MOD	HIGH
Embedded	>30cm or >20% of diameter and continuous	<30cm or 20% of diameter but continuous	No embedment or discontinuous
Value	0	5	10
Outlet Drop (cm)	<15	15-30	>30
Value	0	5	10
SWR	<1.0	1.0-1.3	>1.3
Value	0	3	6
Slope (%)	<1	1-3	>3
Value	0	5	10
Length (m)	<15	15-30	>30
Value	0	3	6

Table 3.4: Fish Barrier Scoring Results (MoE 2011).

Cumlative Score	Result
0-14	passable
15-19	potential barrier
>20	barrier

The habitat gain index is the quantity of modelled habitat upstream of the subject crossing and represents an estimate of habitat gained with remediation of fish passage at the crossing. For this project, a gradient threshold between accessible and non-accessible habitat was set at 25% (for a minimimum length of 100m) intended to represent the maximum gradient of which the strongest swimmers of anadromous species (bull trout) are likely to be able to migrate upstream.

For reporting of Phase 1 - fish passage assessments within the body of this report (Table 3.3), a “total” value of habitat <20% output from `bcfishpass` was used to estimate the amount of habitat upstream of each crossing less than 25% gradient before a falls of height >5m - as recorded in MoE (2020) or documented in other `bcfishpass` online documentation. For Phase 2 - habitat confirmation sites, conservative estimates of the linear quantity of habitat to be potentially gained by fish passage restoration, bull trout rearing maximum gradient threshold (10.5%) was used. To generate estimates for area of habitat upstream (m^2), the estimated linear length was multiplied by half the downstream channel width measured (overall triangular channel shape) as part of the fish passage assessment protocol. Although these estimates are not generally conservative, have low accuracy and do not account for upstream stream crossing structures they allow a rough idea of the best candidates for follow up.

Potential options to remediate fish passage were selected from MoE (2011) and included:

- Removal (RM) - Complete removal of the structure and deactivation of the road.
- Open Bottom Structure (OBS) - Replacement of the culvert with a bridge or other open bottom structure. Based on consultation with FLNR road crossing engineering experts, for this project we considered bridges as the only viable option for OBS type .
- Streambed Simulation (SS) - Replacement of the structure with a streambed simulation design culvert. Often achieved by embedding the culvert by 40% or more. Based on consultation with FLNR engineering experts, we considered crossings on streams with a channel width of <2m and a stream gradient of <8% as candidates for replacement with streambed simulations.
- Additional Substrate Material (EM) - Add additional substrate to the culvert and/or downstream weir to embed culvert and reduce overall velocity/turbulence. This option was considered only when outlet drop = 0, culvert slope <1.0% and stream width ratio < 1.0.
- Backwater (BW) - Backwatering of the structure to reduce velocity and turbulence. This option was considered only when outlet drop < 0.3m, culvert slope <2.0%, stream width ratio < 1.2 and stream profiling indicates it would be effective..

3.4.3 Cost Estimates

Cost estimates for structure replacement with bridges and embedded culverts were generated based on the channel width, slope of the culvert, depth of fill, road class and road surface type. Road details were sourced from FLNRORD (2020b) and FLNRORD (2020a) through `bcfishpass`. Interviews with Phil MacDonald, Engineering Specialist FLNR - Kootenay, Steve Page, Area Engineer - FLNR - Northern Engineering Group and Matt Hawkins - MoTi - Design Supervisor for Highway Design and Survey - Nelson were utilized to help refine estimates which have since been adjusted for inflation in 2020 and based on past experience.

Base costs for installation of bridges on forest service roads and permit roads with surfaces specified in provincial GIS road layers as rough and loose was estimated at $30000/linear m and assumed that the road could be closed during construction and a minimum bridge span of 15m. For streams with channel widths <2m, embedded culverts were reported as an effective solution with total installation costs estimated at $100k/crossing (pers. comm. Phil MacDonald, Steve Page then adjusted for inflation in 2020). For larger streams (>6m), estimated span width increased proportionally to the size of the stream. For crossings with large amounts of fill (>3m), the replacement bridge span was increased by an additional 3m for each 1m of fill >3m to account for cutslopes to the stream at a 1.5:1 ratio. To account for road type, a multiplier table was generated to estimate incremental cost increases with costs estimated for structure replacement on paved surfaces, railways and arterial/highways costing up to 15 times more than forest service roads due to expenses associate with design/engineering requirements, traffic control and paving. The cost multiplier table (Table 3.5) should be considered very approximate with refinement recommended for future projects.

Table 3.5: Cost multiplier table based on road class and surface type.

Class	Surface	Class Multiplier	Surface Multiplier	Bridge $/15m	Streambed Simulation $
FSR	Rough	1	1	450,000	100,000
FSR	Loose	1	1	450,000	100,000
Resource	Loose	1	1	450,000	100,000
Collector	Loose	1	1	450,000	100,000
Recreation	Loose	1	1	450,000	100,000
Resource	Rough	1	1	450,000	100,000
Permit	Unknown	1	1	450,000	100,000
Permit	Loose	1	1	450,000	100,000
Permit	Rough	1	1	450,000	100,000
Unclassified	Loose	1	1	450,000	100,000
Unclassified	Rough	1	1	450,000	100,000
Unclassified	Paved	1	2	750,000	150,000
Unclassified	Unknown	1	2	750,000	150,000
Local	Loose	4	1	1,500,000	200,000
Local	Paved	4	2	3,000,000	400,000
Collector	Paved	4	2	3,000,000	400,000
Arterial	Paved	15	2	11,250,000	1,500,000
Highway	Paved	15	2	11,250,000	1,500,000
Rail	Rail	15	2	11,250,000	1,500,000

3.4.4 Climate Change Risk Assessment

In collaboration with the Ministry of Transportation and Infrastructure (MoTi), a new climate change replacement program aims to prioritize vulnerable culverts for replacement (pers. comm Sean Wong, 2022) based on data collected and ranked related to three categories - culvert condition, vulnerability and priority. Within the “condition” risk category - data was collected and crossings were ranked based on erosion, embankment and blockage issues. The “climate” risk category included ranked assessments of the likelihood of both a flood event affecting the culvert as well as the consequence of a flood event affecting the culvert. Within the “priority” category the following factors were ranked - traffic volume, community access, cost, constructability, fish bearing status and environmental impacts (Table 3.6). This project is still in its early stages with methodology changes going forward.

Table 3.6: Climate change data collected at MoTi culvert sites

Parameter	Description
erosion_issues	Erosion (scale 1 low - 5 high)
embankment_fill_issues	Embankment fill issues 1 (low) 2 (medium) 3 (high)
blockage_issues	Blockage Issues 1 (0-30%) 2 (>30-75%) 3 (>75%)
condition_rank	Condition Rank = embankment + blockage + erosion
condition_notes	Describe details and rational for condition rankings
likelihood_flood_event_affecting_culvert	Likelihood Flood Event Affecting Culvert (scale 1 low - 5 high)
consequence_flood_event_affecting_culvert	Consequence Flood Event Affecting Culvert (scale 1 low - 5 high)
climate_change_flood_risk	Climate Change Flood Risk (likelihood x consequence) 1-6 (low) 6-12 (medium) 10-25 (high)
vulnerability_rank	Vulnerability Rank = Condition Rank + Climate Rank
climate_notes	Describe details and rational for climate risk rankings
traffic_volume	Traffic Volume 1 (low) 5 (medium) 10 (high)
community_access	Community Access - Scale - 1 (high - multiple road access) 5 (medium - some road access) 10 (low - one road access)
cost	Cost (scale: 1 high - 10 low)
constructability	Constructibility (scale: 1 difficult -10 easy)
fish_bearing	Fish Bearing 10 (Yes) 0 (No) - see maps for fish points
environmental_impacts	Environmental Impacts (scale: 1 high -10 low)
priority_rank	Priority Rank = traffic volume + community access + cost + constructability + fish bearing + environmental impacts
overall_rank	Overall Rank = Vulnerability Rank + Priority Rank
priority_notes	Describe details and rational for priority rankings

3.5 Habitat Confirmation Assessments

Following fish passage assessments, habitat confirmations were completed in accordance with procedures outlined in the document “A Checklist for Fish Habitat Confirmation Prior to the Rehabilitation of a Stream Crossing” (Fish Passage Technical Working Group 2011). The main objective of the field surveys was to document upstream habitat quantity and quality and to determine if any other obstructions exist above or below the crossing. Habitat value was assessed based on channel morphology, flow characteristics (perennial, intermittent, ephemeral), the presence/absence of deep pools, un-embedded boulders, substrate, woody debris, undercut banks, aquatic vegetation and overhanging riparian vegetation. Criteria used to rank habitat value was based on guidelines in Fish Passage Technical Working Group (2011) (Table 3.2).

During habitat confirmations, to standardize data collected and facilitate submission of the data to provincial databases, information was collected on digital field forms adapted from provincial “Site Cards”. Habitat characteristics recorded included channel widths, wetted widths, residual pool depths, gradients, bankfull depths, stage, temperature, conductivity, pH, cover by type, substrate and channel morphology (among others). When possible, the crew surveyed downstream of the crossing to a minimum distance 300m and upstream to a minimum distance of 500 - 600m. Any potential obstacles to fish passage were inventoried with photos, physical descriptions and locations recorded on site cards. Surveyed routes were recorded with time-signatures on handheld GPS units.

3.5.1 Fish Sampling

3.5.1.1 Electrofishing

Fish sampling was conducted on a subset of sites when biological data was considered to add significant value to the physical habitat assessment information. Electrofishing was utilized for fish sampling according to stream inventory standards and procedures found in the Reconnaissance (1:20 000) Fish and Fish Habitat Inventory Manual (Resources Inventory Committee 2001). A Haltech 2000 backpack electrofisher was used within discrete site units both upstream and downstream of the subject crossing with electrofisher settings and seconds, water quality parameters (i.e. conductivity, temperature and ph), start and end locations, length of site and wetted widths (average of a minimum of three) recorded.

3.5.1.2 Fish Handling and Processing

Captured fish were held in buckets with sufficient water to minimize stress until processing, and multiple buckets were used when catch numbers were high. For each fish captured, fork length, weight and species was recorded with results documented in the fish data submission spreadsheet.

3.5.1.3 Pit Tagging

Fish with a fork length greater than 60 mm and belonging to species approved under the scientific fish collection permit PG25-983997 and WL25-993485 were tagged with Passive Integrated Transponders (PIT tags) using the Abdominal Cavity method outlined by Biomark. To anesthetize fish prior to pit tagging, we used a solution of approximately 0.1 mL of clove oil per 1 L of water (1:10,000). This concentration was selected for its efficiency in providing effective sedation with minimal residual effects, making it ideal for studies in which fish are released back into their natural habitats (Fernandes et al. 2017). The clove oil solution was prepared in advance by dissolving pure clove oil in ethyl alcohol in a 1:9 ratio (clove oil: ethyl alcohol) to enhance solubility, then mixed into the water bucket (Fernandes et al. 2017). Fish were immersed in this solution until they reached an appropriate level of anesthesia for handling and then were tagged. To maintain needle sharpness and minimize injury risk, needles were replaced approximately every 10 fish. Each tagged fish was scanned with the PIT reader, and both the PIT tag ID and row ID were recorded. Once tagged, fish were placed into a bucket of fresh water and allowed to recover before being released back into the stream. Fish information and habitat data will be submitted to the province under scientific fish collection permit PG25-983997 and WL25-993485.

3.5.2 Environmental DNA (eDNA) sampling

3.5.2.1 eDNA Sampling and Filtration

Environmental DNA (eDNA) samples were collected using two approaches: in-stream filtration with a Haltech EMOS eDNA sampler and a grab-and-go bottle collection method following Hobbs et al. (2025). For both approaches, all water collected at a site (typically 1–3L) was treated as a single sample and filtered onto one MCE filter, resulting in one eDNA replicate per site. Typically, one sample was collected upstream and one downstream of the road–stream crossing, although in some cases only a downstream sample was collected. Field blanks were collected at the start of each sampling day using distilled water.

In-stream filtration was completed using Haltech reusable aluminum housings pre-loaded with mixed cellulose ester (MCE) filters (0.45μm, 1μm, or 1.5μm pore sizes). When turbidity was high, a Haltech pre-filter equipped with a 100μm nitex screen was installed upstream of the MCE filter to remove coarse material and maintain filtration efficiency. After filtration, aluminum housings were returned to their labelled bags and stored in a cooler with ice until the end of the day. Each evening, filters were removed from housings in a clean environment, transferred into coin envelopes placed inside labelled ziplock bags with silica desiccant, and stored in a freezer until shipment. Aluminum housings were sanitized with a bleach solution, reloaded with new filters, and stored in clean ziplock bags for future use.

At sites where the grab-and-go method was used, stream water was collected in sanitized 1L HDPE Nalgene bottles. For detailed procedures, see Hobbs et al. (2025). Bottles were stored in a cooler with ice until the end of the day and then filtered through MCE filters identical to those used for the

EMOS sampler. Post-filtration handling followed the same procedures as the in-stream method, with filters transferred to labelled bags with silica desiccant and frozen until shipment. Nalgene bottles were sanitized with a bleach solution after use.

All eDNA samples were shipped frozen to the University of Northern British Columbia (UNBC) laboratory for analysis.

3.5.2.2 eDNA Extraction

eDNA was extracted in the dedicated eDNA lab at UNBC and analysed with the protocols developed by Murray and Booth (2023).

3.5.3 Open Source Reporting Framework

This report is produced using open-source tools (R, bookdown) with full version control via git, enabling iterative updates. All code, data, and revision history are publicly accessible in the project repository, with ongoing tasks tracked via GitHub issues and version history maintained in the project changelog.

3.5.4 Aerial Imagery

Scripted processing and serving of UAV imagery collected during the project is available at https://github.com/NewGraphEnvironment/stac_uav_bc/ (Irvine [2025] 2025). `OpenDroneMap` was utilized to produce orthomosaics, digital surface models (DSMs), and digital terrain models (DTMs) (OpenDroneMap Authors [2014] 2025). To support efficient web-based access - imagery products were converted to cloud-optimized GeoTIFFs (COGs) using `rio-cogeo`, then collated accordiong to the SpatioTemporal Asset Catalog (STAC) specification with `pystac` and uploaded to S3 storage Amazon Web Services (2025). A `titiler` tile server was set up to facilitate interactive viewing of the orthoimagery and an Application Program Interface (API) leveraging `stac-fastapi-pgstac` is served at https://images.a11s.one to enable linking of collection images through QGIS as well as remote spatial and temporal querying using open source software such as `rstac` (Development Seed [2019] 2025; stac-utils 2025; Simoes et al. 2021).

3.6 Engineering Design

Engineering designs were signed and sealed by professional engineers. When possible - completed designs are loaded to the PSCIS data portal.

3.7 Remediations

Structure replacement was conducted by project specific contractors. If not already completed, as-built drawings will be loaded to the PSCIS data portal.

3.8 Monitoring

Monitoring of fish passage restoration sites — both proposed and completed - is essential to ensure restoration investments lead to meaningful ecological outcomes. Monitoring enables evaluation of whether remediation actions improve connectivity for fish and provides critical feedback to refine future prioritization and restoration strategies.

Baseline data collection, including fish sampling and aerial surveys via drone, are core components of this monitoring. While detailed methods for these activities are included in the previous habitat confirmation section, they are also fundamental to effectiveness monitoring, as they provide context to not only facilitate prioritization and communications but also for detecting change following restoration.

To support consistent and targeted assessment, a custom field form was developed for routine effectiveness monitoring based loosely on Forest Investment Account (2003) but tailored specifically for fish passage projects. Table 3.7) outlines the monitoring metrics used.

Table 3.7: Description of monitoring metrics used for effectiveness monitoring.

Parameter	Description
Dewatering	Have the remediation works led to dewatering of the channel due to substrate aggradation or other factors?
Velocity	Are flow velocities similar to those within the natural channel? Are they expected to exceed swim speeds of particular fish species/life stages of interest?
Constriction	Have the remediation works led to constriction of the channel. Compare channel width underneath structure and within construction footprint to average channel widths upstream and downstream?
Substrate	Is the substrate within/under and adjacent to the remediated structure generally equivalent to that found upstream and downstream where natural channel conditions exist?
Riparian	What is the condition of the riparian area within the construction footprint?
UAV Flight	Was a flight conducted with unmmanned aerial vehicle to document conditions at time of monitoring?
Flow_depth	What are the flow depths at the time of assessment within project footprint. Are depths expected to be sufficient to facilitat upstream passage for specific species/life stages of interest?
Stability	Does the structure appear to be stable or is there evidence of erosion/shifting?
Revegetation	How were riparian areas rehabillitated and are they improving fish habitat value?
Cover	Is cover available for fish within the construction footprint in the form of overhanging vegetation, large/small woody debris, boulders, undercut banks, etc?
Maintenance	If required, provide maintenance recommendations.
Recommendations	General recommendations for follow up. Could include revegetation, addition of substrate, fish sampling, etc.

4 Results and Discussion

4.1 Site Assessment Data

Fish passage assessment procedures conducted through SERNbc since 2023 are amalgamated online within the Results and Discussion section of the report found here.

Since 2023, orthoimagery and elevation model rasters have been generated and stored as Cloud Optimized Geotiffs on a cloud service provider (AWS) with select imagery linked to in the collaborative GIS project. Additionally - a tile service has been set up to facilitate viewing and downloading of individual images, provided athttps://www.newgraphenvironment.com/fish_passage_fraser_2025_reporting//results-and-discussion.html.

Table 4.1: Summary of fish passage assessment procedures conducted in northern British Columbia through SERNbc. **NOTE: To view all columns in the table - please click on one of the sort arrows within column headers before scrolling to the right.**

Show 5 entries

Watershed Group	Assessment	Reassessment	Habitat Confirmation	Design	Rem
All	All	All	All	All	All
Francois Lake	1	0	4	0	0
Lower Chilako River	1	0	1	0	0
Morkill River	1	0	5	0	0
Nechako River	2	0	4	0	0
Tabor River	0	1	0	0	0
Upper Fraser River	2	2	3	0	0
Total	7	3	17	0	0

Showing 1 to 7 of 7 entries Excel CSV Previous 1 Next

Table 4.2: Details of fish passage assessment procedures conducted in northern British Columbia through SERNbc.

Show 5 entries

4.1 Site Assessment Data

Stream Crossing Id	My Crossing Reference	Watershed Group	Stream Name	Road Name	Assessment
All	All	All	All	All	All
4931		Upper Fraser River	Crooked Creek	Tinsley Pit Road	yes
7620		Upper Fraser River	Tributary of McLennan River	CN Rail access road	yes
7622		Francois Lake	UNN Flows into Fraser Lake	Stella Rd	
196200		Tabor River	Bittner Creek	Forman Road	yes
199171		Francois Lake	Tributary to Fraser Lake	Gala Bay Road	
199172		Francois Lake	Scotch Creek	Stella Road	
199173		Nechako River	Tributary to Nechako River	Dog Creek Road	
199174		Nechako River	Tributary to Nechako River	Sutherland FSR	
199190		Nechako River	Clear Creek	Highway 27 S	
199204		Nechako River	Nine Mile Creek	Settlement Rd	

Showing 1 to 10 of 26 entries Excel CSV

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4.2 Climate Departure

Four findings from the regional climate-departure analysis carry to fish-passage prioritisation.

Warming is broad and significant region-wide. All eight ecoregions of the study area warmed about 1.6 °C since 1951 — annual mean (+1.6 °C), daytime maximum (+1.5 °C), and overnight minimum (+1.8 °C) all tell the same story (Mann-Kendall $p < 0.001$ in every ecoregion). The rate has been steady rather than accelerating — the 1981–present slope is shallower than the 1951–present slope. A west-to-east gradient layers on top, with the interior plateau ecoregions warming roughly 0.5 °C more than the eastern Rocky Mountain ecoregions — see Appendix - Climate Departure (page 105) Figure 5.20.

The atmosphere is drying despite flat precipitation. Vapour pressure deficit rose 0.34 hPa region-wide, significant in every ecoregion. Annual precipitation rose +3 % regionally but the long-term trend is not statistically significant in any of the eight ecoregions; soil moisture is essentially flat because the warmer atmosphere is drinking the modest precipitation increase back through evaporation and transpiration.

The snowpack signal is about timing, not quantity. Total annual snowfall is essentially unchanged (-6 %); annual snow water equivalent is down 10 %, neither significant. The unambiguous signals are on the summer side and at the midpoint: summer SWE collapsed 52 % and the snowmelt midpoint moved 12 days earlier ($p < 0.001$ regionally and in every ecoregion). Counterbalancing rises in winter snowmelt (+45 %) and spring snowmelt (+18 %) point in the same direction but are noisier season-by-season. The freshet is arriving weeks earlier and the high-elevation snowpack that historically lingered into summer no longer does.

Day-night asymmetry is present. Overnight minimums warmed 1.8 °C while daytime maximums warmed 1.5 °C — overnight outpacing daytime is the textbook fingerprint of greenhouse warming (Karl et al. 1993). Mean diurnal range has narrowed by 0.3 °C across the record.

For barrier prioritisation these signals cut both ways. Where upstream reaches are cold-limited, added growing-degree-days can raise the intrinsic potential of habitat above the barrier and *increase* the value of restoring passage; where upstream reaches sit closer to the upper edge of the cold-water thermal niche, the same warming reduces usable habitat and lowers the payoff. The watershed-group breakdown in Appendix - Climate Departure (page 105) maps the eight-ecoregion signal onto the seven watershed groups via a percentage crosswalk.

4.3 Collaborative GIS Environment

In addition to numerous layers documenting fieldwork activities since , a summary of background information spatial layers and tables loaded to the collaborative GIS project (sern_fraser_2024) at the time of writing (2026-07-31) are included online here

4.4 Planning — Habitat, Connectivity and Floodplain Modelling

4.4.1 Habitat Modelling

Habitat modelling from `bcfishpass` including access model, linear spawning/rearing habitat model and lateral habitat connectivity models for watershed groups within our study area were updated for the spring of 2025 and are included spatially in the collaborative GIS project. A snapshot of these outputs related to each modeled and PSCIS stream crossing structure are also included within an `sqlite` database within this year's project reporting/code repository here.

4.4.1.1 Statistical Support for `bcfishpass` Fish Habitat Modelling Updates

Initial mapping of stream discharge and temperature causal effects pathways for the future purpose of focusing aquatic restoration actions in areas of highest potential for positive impacts on fisheries values (ie. elimination of areas from intrinsic models where water temperatures are likely too cold to support fish production) are detailed in Hill et al. (2024) which is included as Attachment - Water Temperature Modelling (page 149).

4.4.2 Floodplain Delineation

Modelled functional floodplain in the Nechako River Watershed Group covers 54,306 ha — 13.1 % of the 4,148 km² watershed group — attached to 1,312 km of chinook accessible order-3+ streams. Within that footprint sit 439 lakes and 702 mapped wetland polygons — the off-channel inventory that crossing remediations in these reaches stand to reconnect or leave severed, depending on whether design accounts for the floodplain context around the structure. The full metric table, watershed-scale map, and Murray Creek confluence detail map are in Appendix - Floodplain Delineation (page 101).

4.5 Fish Passage Assessemnts

Field assessments were conducted between September 15, 2025 and October 05, 2025 by Allan Irvine, R.P.Bio. and Lucy Schick, B.Sc. A total of 12 Fish Passage Assessments were completed, including 12 Phase 1 assessments and 0 reassessments.

Of the 12 sites where fish passage assessments were completed, 12 were not yet inventoried in the PSCIS system. This included 3 crossings considered "passable", 2 crossings considered a "potential" barrier, and 7 crossings were considered "barriers" according to threshold values based on culvert embedment, outlet drop, slope, diameter (relative to channel size) and length (MoE 2011).

A summary of crossings assessed, a rough cost estimate for remediation, and a priority ranking for follow-up for Phase 1 sites is presented in Table 4.3. Detailed data with photos are presented in Attachment - Phase 1 Data and Photos (page 145).

Table 4.3: Upstream habitat estimates and cost benefit analysis for Phase 1 assessments ranked as a 'barrier' or 'potential' barrier. Bull trout network model used for habitat estimates (total length of stream network <25% gradient).

PSCIS ID	External ID	Priority	Stream	Road	Barrier Result	Habitat value	Habitat Upstream (km)	Stream Width (m)	Fix	Cost Est ($K)
203577	5400047	low	Van Lear Creek	Unnamed	Barrier	Low	20.1	2.5	OBS	450
203578	24403467	moderate	Tributary To Willow River	Willow FSR	Barrier	Medium	20.1	3.2	OBS	450
203579	24403486	low	Tributary To Willow River	Willow FSR	Potential	Medium	7.8	2.0	OBS	450
203580	5400321	moderate	Cordella Creek	Colleymount Creek	Potential	High	23.9	3.2	OBS	450
203581	19703295	low	Tributary To Fraser River	Railway	Barrier	Medium	30.3	4.0	OBS	26625
203582	19703257	moderate	Tabor Creek	Railway	Barrier	Medium	196.4	10.0	OBS	27000
203583	5405348	low	Tributary To Nadina River	Tahtsa FSR	Barrier	Low	29.3	1.5	SS-CBS	100
203584	5405311	moderate	Tributary To Nadina River	Tahtsa FSR	Barrier	Medium	12.5	2.3	OBS	450
203585	5400078	low	Van Lear Creek	Colleymount Road	Barrier	Low	20.1	2.5	OBS	450

4.6 Habitat Confirmation Assessments

During the field assessments, habitat confirmation assessments were conducted at 6 sites within the Francois Lake, Lower Chilako River, Lower Salmon River, Morkill River, Nechako River, Tabor River, Upper Fraser River, and Willow River watershed groups. A total of approximately 4 km of stream was assessed.

As collaborative decision-making was ongoing at the time of reporting, site prioritization can be considered preliminary. Results are summarized in Table 4.4 with raw habitat data included in

Attachment - Data (page 147). A summary of preliminary modelling results illustrating quantities of bull trout spawning and rearing habitat potentially available upstream of each crossing as estimated by measured/modelled channel width and upstream accessible stream length are presented in Figure 4.1. Detailed information for each site assessed with Phase 2 assessments (including maps) are presented within site-specific appendices to this document.

Table 4.4: Overview of habitat confirmation sites. Bull trout rearing model used for habitat estimates (total length of stream network <10.5% gradient).

PSCIS ID	Stream	Road	Tenure	UTM	UTM zone	Fish Species	Habitat Gain (km)	Habitat Value	Priority	Comments
203581	Tributary To Fraser River	Railway	CN Rail	526469 5985767	10	–	11.6	Medium	Low	Landslide ~300m upstream of railway tracks - Wp208. Low gradient stream with abundant gravels and numerous fish to 60mm observed. Entrenched steep wall valley with several sections where clay banks have collapsed into the stream. Surveyed upstream from railway culvert for 650m. eDNA samples collected upstream of the crossing.
196076	Tributary To Fraser River	Beaver FSR	McLean R02924 A	527373 5985002	10	–	10.3	Medium	Moderate	Moderate-sized, low-gradient stream with abundant gravels and habitat suitable for fry and juvenile salmonid rearing, including shallow pools, small and large woody debris, and undercut banks. No fish or barriers were observed at the time of assessment. eDNA samples collected upstream and downstream of the crossing.
196085	Tabor Creek	Willow-Cale FSR	MoTi	518498 5962011	10	–	75.3	Medium	Moderate	Channel was dewatered at the time of assessment with shallow pools present at the downstream end of the site. Gravels were abundant throughout and suitable for chinook and sockeye spawning under adequate flow conditions. Deep pools were likely present when the stream was flowing. Chinook were documented downstream of the Willow FSR crossing in 2004. eDNA sample collected at the bottom end of the survey (upstream of the crossing).
203582	Tabor Creek	Railway	CN Rail	518844 5961981	10	–	74.8	Medium	Moderate	Stream was dewatered until approximately 360m upstream of the railway culverts. The farthest downstream pool contained numerous

	PSCIS ID	Stream	Road	Tenure	UTM	UTM zone	Fish Species	Habitat Gain (km)	Habitat Value	Priority	Comments
fry and was where eDNA sampling was conducted. Upstream, occasional pools transitioned to increasing flow toward the upper end of the site where a continually flowing channel reemerged. The system was extremely large with high flows during freshet based on the size of the substrates. Gravels were abundant throughout, with limited woody debris or instream cover at low flow, restricted mainly to interstitial spaces between cobbles and boulders.											
	126158	Tributary To Stony Lake	Willow FSR	MoF	577622 5918587	10	–	4.4	Medium	High	Low-gradient stream with abundant gravels and diverse cover types including boulders, small and large woody debris, and undercut banks. Good flow with no natural barriers observed within the surveyed area. eDNA sample was collected at the downstream end of the site. Bull trout documented downstream of the crossing in 2014.
	196332	South Yuzkli Creek	24 Y Road	W Fraser R14273 100S	578271 5895125	10	–	2.8	Medium	High	Steep stream with good flow and numerous step-pool features ranging from 30–50cm in height. Shallow pools were 30–40cm deep with pockets of gravel suitable for resident Bull trout spawning. Large woody debris was abundant throughout. eDNA sample was collected at the downstream end of the site.

Table 4.5: Summary of Phase 2 fish passage reassessments.

PSCIS ID	Embedded	Outlet Drop (m)	Diameter (m)	SWR	Slope (%)	Length (m)	Final score	Barrier Result
126158	No	1.60	1.5	2.80	2.5	37	37	Barrier
196076	No	0.55	1.2	3.33	2.0	50	37	Barrier
196085	No	0.00	4.0	2.50	2.5	32	27	Barrier
196332	No	0.60	1.2	2.83	6.5	23	39	Barrier
203581	No	2.50	2.1	1.90	2.5	50	37	Barrier
203582	No	0.30	5.2	1.92	0.5	50	32	Barrier

Table 4.6: Cost benefit analysis for Phase 2 assessments. Bull trout rearing model used for habitat estimates (total length of stream network <10.5% gradient).

PSCIS ID	Stream	Road	Barrier Result	Habitat value	Stream Width (m)	Fix	Cost Est ($K)	Habitat Upstream (m)	Cost Benefit (m / $K)	Cost Benefit (m2 / $K)
126158	Tributary To Stony Lake	Willow FSR	Barrier	Medium	3.5	OBS	900	4447	4941.1	10376.3
196076	Tributary To Fraser River	Beaver FSR	Barrier	Medium	3.5	OBS	585	10278	17569.2	35138.5
196085	Tabor Creek	Willow-Cale FSR	Barrier	Medium	11.0	OBS	3600	75254	20903.9	104519.4
196332	South Yuzkli Creek	24 Y Road	Barrier	Medium	3.0	OBS	450	2774	6164.4	10479.6
203581	Tributary To Fraser River	Railway	Barrier	Medium	4.5	OBS	26625	11634	437.0	873.9
203582	Tabor Creek	Railway	Barrier	Medium	14.6	OBS	27000	74788	2769.9	13849.6

Table 4.7: Summary of Phase 2 habitat confirmation details.

PSCIS ID	Length surveyed upstream (m)	Average Channel Width (m)	Average Wetted Width (m)	Average Pool Depth (m)	Average Gradient (%)	Total Cover	Habitat Value
–	–	–	–	–	–	–	–
−−−:	−−−−−−−−−−:	−−−−−−−−−:	−−−−−−−−−:	−−−−−−−−:	−−−−−−−:	:−−−−	:−−−−-

Table 4.8: Summary of watershed area statistics upstream of Phase 2 crossings.

Site	Area Km	Elev Site	Elev Min	Elev Max	Elev Median	Elev P60	Aspect
126158	10.4	956	929	1870	1376	1297	SSW
196076	14.9	629	572	784	696	691	SSW
196085	145.6	592	558	1254	703	683	SSW
196332	5.1	1218	1200	1636	1417	1398	SW
203581	17.5	592	572	784	688	680	SW
203582	145.6	601	558	1254	703	683	SSW

* Elev P60 = Elevation at which 60% of the watershed area is above

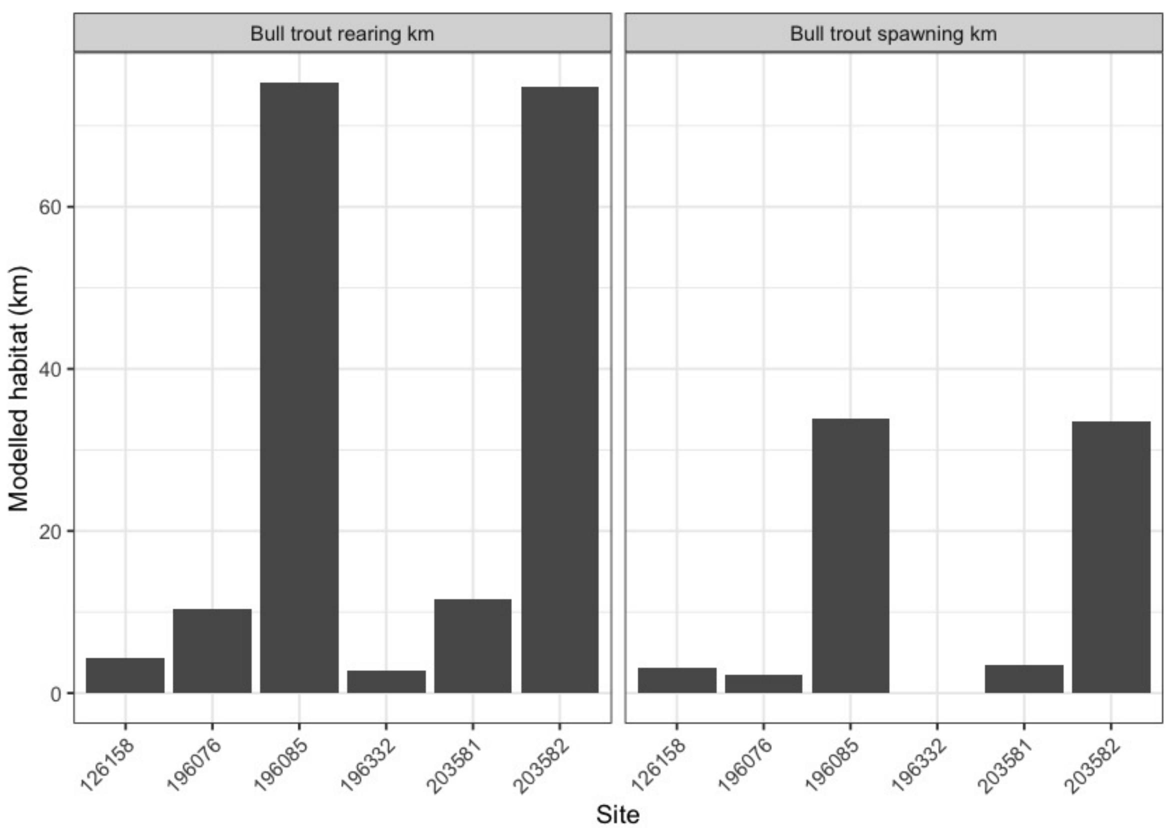


Figure 4.1: Summary of potential rearing and spawning habitat upstream of habitat confirmation assessment sites. Bull trout rearing and spawning models used for habitat estimates (total length of stream network <10.5% and <5.5% gradient, respectively).

4.6.1 Environmental DNA (eDNA) sampling

New in 2025, environmental DNA (eDNA) sampling was incorporated into the program at both previously assessed and newly added sites. A total of 38 collection events were undertaken across 16 streams, comprising 33 real environmental samples, 2 field blanks, and 3 office blanks collected for quality assurance.

Per-species detection results across the 33 real environmental samples are summarized in Table 4.9. Detections use the standard ddPCR call threshold of ≥4 positive droplets; sub-threshold signals (1-3 droplets) are tracked separately because they are not interpretable as species presence under ddPCR convention. Site-level detail — the per-site detection table, field blanks, and retests — is provided in Appendix - Environmental DNA Results (page 129), and the interactive Fraser eDNA map shows toggleable per-species layers.

Table 4.9: Per-species summary of eDNA results across real environmental samples in the Fraser project area (excludes field and office blanks).

Species	Code	Sites tested	Detected (≥4 droplets)	Sub-threshold only (1-3 droplets)	No detection (0 droplets)
Rainbow Trout	RAIN	23	20	1	2
Chinook Salmon	CHIN	30	9	3	18
Bull Trout	BULT	33	5	5	23
Burbot	BURB	5	1	1	3
Sockeye Salmon	SOCK	12	1	1	10
Coho Salmon	COHO	21	0	1	20
Rainbow Trout (confirmation primer)	RAIN2	2	0	0	2

4.7 Engineering Design

MIGHT NEED TO UPDATE

No new designs were commissioned in due to uncertainty related to forest harvesting activities and lack of funding for the 50% costs of replacing structures. All sites with a design for remediation of fish passage can be found online within the Results and Discussion section of the report found here.

4.8 Monitoring

In , baseline or follow up monitoring data was gathered through completion of an effectivness montoring form (sites where remediation has been completed) and/or through eDNA sampling and aquistion of aerial imagery at the following sites (Table 4.10:

- Effectiveness monitoring was conducted on Tabor Creek (PSCIS crossing 196085) on Willow-Cale FRS which was replaced in XXXX.
- Effectiveness monitoring was conducted on Bittner Creek (PSCIS crossing 196200) on Foreman Road which was replaced in XXXX.

Table 4.10: Summary of monitoring sites.

PSCIS ID	Stream	Road	Crossing type	Crossing subtype	Diameter or span (m)	Length or width (m)	Assessment comments
196085	Tabor Creek	Willow-Cale FRS	CBS	OVAL	4	32	eDNA samples collected upstream and downstream of the crossing during the monitoring assessment.
196200	Bittner Creek	Foreman Road	OBS	BRIDGE	–	–	–

5 Recommendations

A major challenge in advancing fish passage restoration is the complexity of working across jurisdictions and with multiple stakeholders—rail and highway authorities, forestry ministries, licensees, and private landowners. These partners are often being asked to accommodate priorities that originate outside their mandates and budgets. Convincing them to invest in difficult, high-cost interventions—like modifying crossings or relocating infrastructure—requires navigating uncertainty about costs and ecological outcomes, as well as a disconnect between the benefits to watershed health and the internal pressures or performance goals of these agencies. It's a tough ask: to take on massive, uncertain projects when they're already stretched thin with their own responsibilities.

Fish passage restoration across British Columbia is further complicated by the legacy of infrastructure deeply embedded in the landscape. Roads, railways, highways, community infrastructure and private assets often constrain floodplains and disrupt natural hydrological processes. While targeted repairs to individual barriers are essential, they won't resolve the broader systemic issues without rethinking and restructuring how infrastructure interacts with watershed function. Loss of riparian vegetation and intensive beaver management only add to the degradation. Addressing these challenges means making strategic, well-communicated choices—picking battles carefully, building trust, and staying committed to a longer-term transformation.

While preliminary top remediation priorities are provided by watershed group, these rankings are inherently subjective and can depend on the capacity and willingness of infrastructure owners and tenure holders to support implementation—both financially and over the often multi-year project timelines. In practice, we must often act opportunistically, pursuing simpler, lower-cost options to maintain momentum and achieve near-term progress.

Government, community groups, landowners, non-profits, industry and other stakeholders should work collaboratively to address high and moderate priority barriers identified online within the Results and Discussion section of the report found here. Although the table presents many options - currently - linked reports specify whether each site is a low, moderate, or high priority. Progress on any front is meaningful, and aiming to remediate at least one high-priority site per year per watershed group—regardless of its overall rank—is a practical and effective approach.

To enhance fish passage restoration in the FWCP Peace Region:

- Maintain strong partnerships to support funding, site selection, remediation, and monitoring through adaptive management informed by traditional knowledge and real-time data.
- Prioritize detailed assessments in areas with blockages and high habitat potential, especially near McLeod Lake.
- Use climate modeling to prioritize crossings that enable access to cold, drought-resistant habitats.
- Secure financial commitments for Fern Creek remediation despite uncertainties in harvest planning.

- Continue effectiveness monitoring at key sites using fish sampling, eDNA, PIT tagging, temperature data, and aerial imagery.
- Continue to develop a cost-effective monitoring framework to assess productivity gains from improved passage.
- Collaborate with WLRS, UNC, local fisheries experts, FWCP, and the CEMPRA Project working group.
- Utilize environmental DNA (eDNA) to better understand bull trout and Arctic grayling habitat use at both potential and remediated sites. This will refine prioritization and assess fish passage effectiveness.

Tributary To Stony Lake - 126158 - Appendix

Site Location

PSCIS crossing 126158 is located on Tributary To Stony Lake, on the eastern side of Stony Lake, approximately 100km southeast of Prince George, BC, in the Willow River watershed group (Figure ??). The crossing is located 0.8km upstream of Stony Lake, on Willow FSR, and is the responsibility of the Ministry of Forest.

Background

At PSCIS crossing 126158, Tributary To Stony Lake is a third order stream and drains a watershed of approximately 10.4km^2. The watershed ranges in elevation from a maximum of 1870m to 956m near the crossing (Table 5.1).

PSCIS crossing 126158 was first assessed with a fish passage assessment in 2014 by Hooft (2015) and ranked as a barrier to fish passage with medium value habitat present. The crossing was reassessed in 2025 and a habitat confirmation assessment was conducted due to the presence of high value habitat and a significant amount of modelled habitat upstream. No fisheries information is documented for the stream; however, downstream in Stony Lake, bull trout and other species have been previously recorded (Norris [2018] 2024; MoE 2024).

A summary of habitat modelling outputs for the crossing are presented in Table 5.2.

Table 5.1: Summary of derived upstream watershed statistics for PSCIS crossing 126158.

Site	Area Km	Elev Site	Elev Min	Elev Max	Elev Median	Elev P60	Aspect
126158	10.4	956	929	1870	1376	1297	SSW

* Elev P60 = Elevation at which 60% of the watershed area is above

Table 5.2: Summary of fish habitat modelling for PSCIS crossing 126158.

Habitat	Potential	Remediation Gain	Remediation Gain (%)
BT Rearing (km)	4.4	4.4	100

Habitat	Potential	Remediation Gain	Remediation Gain (%)
BT Spawning (km)	3.2	3.2	100
BT Network (km)	11.7	7.5	64
BT Stream (km)	10.2	6.0	59
BT Lake Reservoir (ha)	7.7	7.7	100
BT Wetland (ha)	11.4	11.4	100
BT Slopeclass03 (km)	1.4	1.4	100
BT Slopeclass05 (km)	2.9	2.9	100
BT Slopeclass08 (km)	1.6	0.7	44
BT Slopeclass15 (km)	3.7	0.8	22

* Model data is preliminary and subject to adjustments.

Stream Characteristics at Crossing 126158

At the time of the 2025 assessment, PSCIS crossing 126158 on Willow FSR was un-embedded, non-backwatered and ranked as barrier to upstream fish passage according to the provincial protocol (MoE 2011) (Table 5.3). The culvert had a 1.6m outlet drop and a 1.2m deep outlet pool.

The water temperature was 3°C, pH was 7.4 and conductivity was 79 uS/cm.

Table 5.3: Summary of fish passage assessment for PSCIS crossing 126158.

Location and Stream Data	•	Crossing Characteristics	–
Date	2025-10-05	Crossing Sub Type	Round Culvert
PSCIS ID	126158	Diameter (m)	1.5
External ID	–	Length (m)	37
Crew	AI	Embedded	No
UTM Zone	10	Depth Embedded (m)	–
Easting	577622	Resemble Channel	No
Northing	5918587	Backwatered	No
Stream	Tributary To Stony Lake	Percent Backwatered	–
Road	Willow FSR	Fill Depth (m)	8
Road Tenure	MoF	Outlet Drop (m)	1.6
Channel Width (m)	4.2	Outlet Pool Depth (m)	1.2
Stream Slope (%)	3	Inlet Drop	No
Beaver Activity	No	Slope (%)	2.5
Habitat Value	Medium	Valley Fill	Deep Fill
Final score	37	Barrier Result	Barrier
Fix type	Replace with New Open Bottom Structure	Fix Span / Diameter	30

Comments: Huge outlet drop. Habitat confirmation assessments and EDNA samples conducted upstream and downstream. of the crossing. Originally prioritized by HOOFT in 2015.

Photos: From top left clockwise: Road/Site Card, Barrel, Outlet, Downstream, Upstream, Inlet.

Location and Stream Data	Crossing Characteristics





Stream Characteristics Downstream of Crossing 126158

The stream was surveyed downstream from crossing 126158 for 240m (Figure 5.1). The habitat was rated as medium value for salmonid spawning and rearing. Abundant gravels were present throughout with no barriers observed between the road crossing and Stony Lake. The stream was relatively large with good flow, containing occasional shallow pools and sparse large and small woody debris. The dominant substrate was gravels with cobbles sub-dominant.The average channel width was 4.3m, the average wetted width was 2.2m, and the average gradient was 2.7%.Total cover amount was rated as moderate with undercut banks dominant. Cover was also present as small woody debris, deep pools, and overhanging vegetation.

Stream Characteristics Upstream of Crossing 126158

The stream was surveyed upstream from crossing 126158 for 330m (Figure 5.1). The habitat was rated as medium value. Total cover amount was rated as moderate with undercut banks dominant. Cover was also present as small woody debris, large woody debris, boulders, and overhanging vegetation.The average channel width was 3.5m, the average wetted width was 2.7m, and the average gradient was 4.8%.The dominant substrate was gravels with cobbles sub-dominant. Low-gradient stream with abundant gravels and diverse cover types including boulders, small and large woody debris, and undercut banks. Good flow with no natural barriers observed within the surveyed area.

Environmental DNA Sampling

eDNA samples were collected both upstream and downstream of crossing 126158 on Willow FSR, with sampling techniques summarized in Table 5.4.

Table 5.4: Summary of eDNA samples collected at site 126158.

Site	Stream	Habitat Type	Sample Method	Filter Type	Filter Size (um)	Volume filtered (L)	Site Description
126158_ds_ed1	Tributary To Stony Lake	riffle	grab	mce	1	1.9	Sampled collected 240m downstream of the culvert on the Willow FSR on the right bank. Walked down from the road on the left of the stream.
126158_us_ed1	Tributary To Stony Lake	riffle	grab	mce	1	1.9	Sample collected 3m upstream of the culvert inlet on the left bank.

Detection results are presented in Table 5.5. Format is `CODE(max_droplets)`, with $*$ flagging samples UNBC reran. Detected means at least 4 positive droplets, the standard ddPCR call threshold; sub-threshold signals of 1-3 droplets are not interpretable as species presence. Site-level detail for the whole project, along with field blanks and retests, is in Appendix - Environmental DNA Results (page 129).

Table 5.5: eDNA detection results at PSCIS crossing 126158.

Site ID	UNBC ID	Detected	Sub-threshold	Not detected
126158_ds_ed1	F61121	RAIN(77)	—	BULT(0), CHIN(0), COHO(0)
126158_us_ed1	F61122	RAIN(21)	—	BULT(0), CHIN(0), COHO(0)

Structure Remediation and Cost Estimate

Should restoration/maintenance activities proceed, replacement of the Willow FSR crossing (126158) with a bridge (30m span) is recommended. At the time of reporting in 2025, the cost of the work is estimated at $900,000.

Conclusion

The habitat upstream of PSCIS crossing 126158 on Willow FSR was documented as medium value for salmonid spawning and rearing, and the crossing is rated as a high priority for replacement due to the large outlet drop.

Table 5.6: Summary of habitat details for PSCIS crossing 126158.

Site	Location	Length Surveyed (m)	Average Channel Width (m)	Average Wetted Width (m)	Average Pool Depth (m)	Average Gradient (%)	Total Cover	Habitat Value
126158	Downstream	240	4.3	2.2	0.5	2.7	moderate	medium
126158	Upstream	330	3.5	2.7	0.3	4.8	moderate	medium

Figure 5.1: Left: Typical habitat downstream of PSCIS crossing 126158. Right: Typical habitat upstream of PSCIS crossing 126158.

Tabor Creek - 196085 & 203582 - Appendix

Site Location

PSCIS crossings 196085 and 203582 are located on Tabor Creek , approximately 8km south of Prince George, BC, in the Tabor River watershed group (Figure 5.2). Crossing 196085 is located 1.4km upstream Tabor Creek’s confluence with the Fraser River. The crossing is located on Willow-Cale FSR and is the responsibility of the Ministry of Transportation and Infrastructure (`chris_culvert_id`: 31046). Approximately 400m upstream, crossing 203582 is located on and the responsibility of the CN Railway.

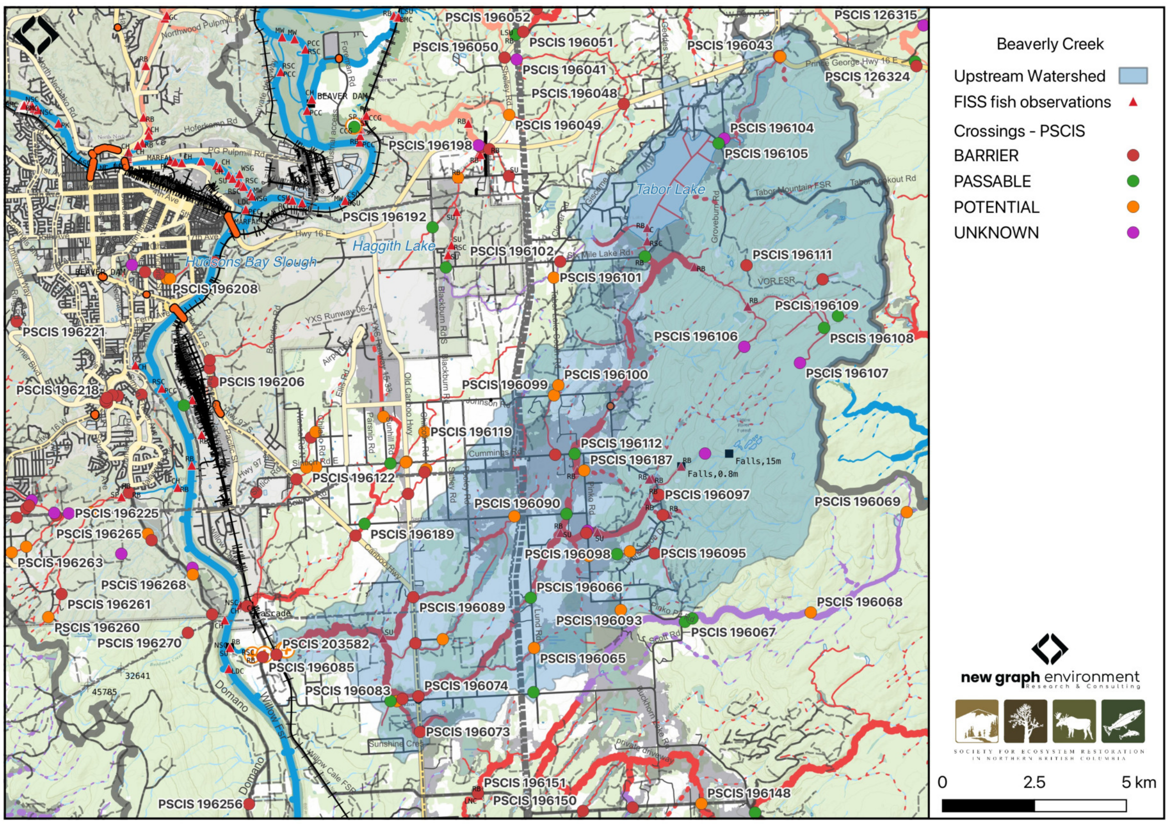


Figure 5.2: Map of Tabor Creek

Background

At the location of these crossings, Tabor Creek is a fifth order stream and drains a watershed of approximately 145.6km^2. The watershed ranges in elevation from a maximum of 1254m to 592m near the crossing (Table 5.7).

Crossing 196085 on Willow-Cale FSR was previously assessed in 2014 by Hooft (2015) and ranked as a barrier to fish passage, with medium value habitat documented and fish observed at the time of assessment. Chinook were recorded downstream of the crossing in 2004, and upstream of the crossing rainbow trout and dolly varden have been previously documented (Norris [2018] 2024; MoE 2024). Crossing 203582 on the CN Railway had not been assessed prior to this project. Barrier assessments were completed at both crossings, and a habitat confirmation encompassing both sites was conducted.

A summary of habitat modelling outputs for the crossing are presented in Table 5.8.

Table 5.7: Summary of derived upstream watershed statistics for PSCIS crossing 196085.

Site	Area Km	Elev Site	Elev Min	Elev Max	Elev Median	Elev P60	Aspect
196085	145.6	592	558	1254	703	683	SSW

* Elev P60 = Elevation at which 60% of the watershed area is above

Table 5.8: Summary of fish habitat modelling for PSCIS crossing 196085.

Habitat	Potential	Remediation Gain	Remediation Gain (%)
BT Rearing (km)	75.3	0.5	1
BT Spawning (km)	33.9	0.5	1
BT Network (km)	196.9	0.5	0
BT Stream (km)	166.1	0.5	0
BT Lake Reservoir (ha)	408.8	0.0	0
BT Wetland (ha)	385.0	0.0	0
BT Slopeclass03 (km)	74.4	0.5	1
BT Slopeclass05 (km)	25.0	0.0	0
BT Slopeclass08 (km)	42.9	0.0	0
BT Slopeclass15 (km)	21.9	0.0	0

* Model data is preliminary and subject to adjustments.

Stream Characteristics at Crossings 196085 and 203582

At the time of the 2025 assessment, both PSCIS crossing 196085 and 203582 on Willow-Cale FSR were un-embedded, non-backwatered and ranked as a barrier to upstream fish passage according to the provincial protocol (MoE 2011) (Tables 5.9 - 5.10). In 2019, work was completed at crossing 196085 on Willow-Cale FSR to backwater the outlet and remove the outlet drop as well as to put baffles within the culvert.

Water temperature was 9°C, pH was 7.6 and conductivity was 497 uS/cm.

Table 5.9: Summary of fish passage assessment for PSCIS crossing 196085.

Location and Stream Data	•	Crossing Characteristics	–
Date	2025-09-25	Crossing Sub Type	Oval Culvert
PSCIS ID	196085	Diameter (m)	4
External ID	–	Length (m)	32
Crew	AI	Embedded	No
UTM Zone	10	Depth Embedded (m)	–
Easting	518498	Resemble Channel	No
Northing	5962011	Backwatered	No
Stream	Tabor Creek	Percent Backwatered	–
Road	Willow-Cale FSR	Fill Depth (m)	4
Road Tenure	MoTi	Outlet Drop (m)	0
Channel Width (m)	10	Outlet Pool Depth (m)	1
Stream Slope (%)	2.5	Inlet Drop	No
Beaver Activity	No	Slope (%)	2.5
Habitat Value	Medium	Valley Fill	Deep Fill
Final score	27	Barrier Result	Barrier
Fix type	Replace with New Open Bottom Structure	Fix Span / Diameter	18

Comments: Work was completed in 2019 to backwater the outlet and remove the outlet drop as well as to put baffles within the culvert. Routine monitoring form also filled out a can be found in accompanying report. eDNA collected in 2025 upstream and downstream of this crossing as well as above the railway crossing upstream, see report for results.

Photos: From top left clockwise: Road/Site Card, Barrel, Outlet, Downstream, Upstream, Inlet.

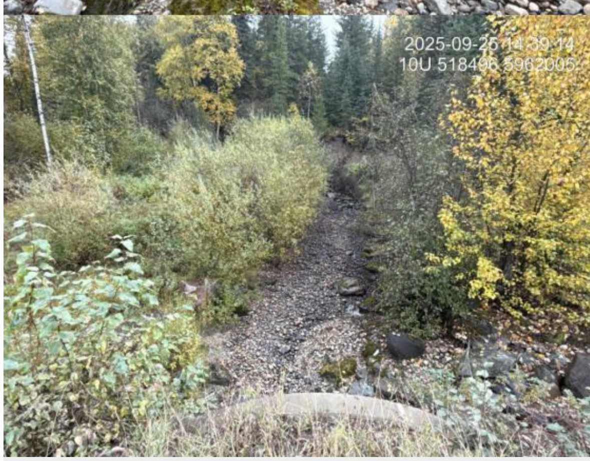
Location and Stream Data •	Crossing Characteristics –
	
	
	

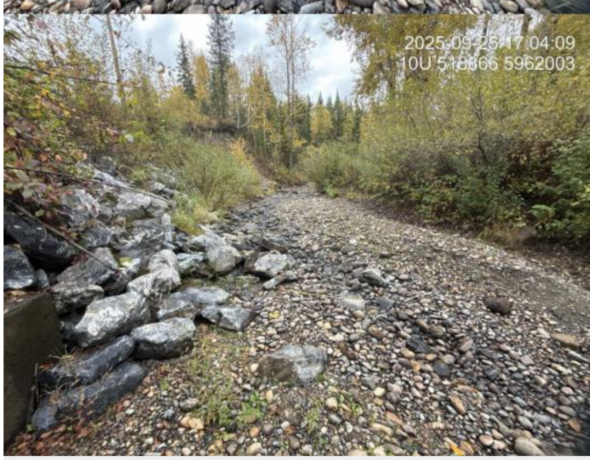
Table 5.10: Summary of fish passage assessment for PSCIS crossing 203582.

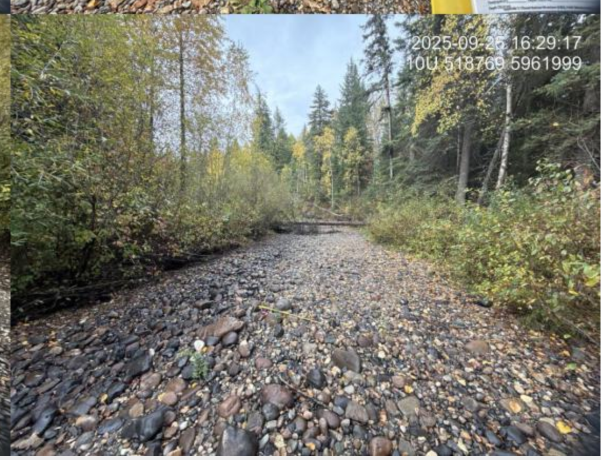
Location and Stream Data	•	**Crossing Characteristics**	–
Location and Stream Data	•	**Crossing Characteristics**	–
Date	2025-09-25	Crossing Sub Type	Round Culvert
PSCIS ID	203582	Diameter (m)	5.2
External ID	19703257	Length (m)	50
Crew	AI	Embedded	No
UTM Zone	10	Depth Embedded (m)	–
Easting	518844	Resemble Channel	–
Northing	5961981	Backwatered	No
Stream	Tabor Creek	Percent Backwatered	–
Road	Railway	Fill Depth (m)	10
Road Tenure	CN Rail	Outlet Drop (m)	0.3
Channel Width (m)	10	Outlet Pool Depth (m)	1
Stream Slope (%)	1.5	Inlet Drop	No
Beaver Activity	No	Slope (%)	0.5
Habitat Value	Medium	Valley Fill	Deep Fill
Final score	32	Barrier Result	Barrier
Fix type	Replace with New Open Bottom Structure	Fix Span / Diameter	36

Comments: Two pipes at 2.6 m each. Pipes are in decent shape although it appears they had metal skirt sections (previously attached to bottom of pipes) at the outlets that have ripped off. Difficult to determine the size of the outlet drop as the stream was dry at the time of assessment. Culvert also crosses underneath a road and powerline on the east side of the railway

Photos: From top left clockwise: Road/Site Card, Barrel, Outlet, Downstream, Upstream, Inlet.

Location and Stream Data | **Crossing Characteristics**





Stream Characteristics Downstream of Crossing 196085

The stream was surveyed downstream from crossing 196085 for 350m (Figure 5.3). The stream was flowing, but discharge appeared minimal relative to channel-forming flows. Extensive algae growth near the highway suggested potential nutrient inputs, possibly from the upstream mill site; algae presence decreased farther downstream (Figure 5.3).. Large areas of gravels were present and suitable for chinook and sockeye spawning, contingent on adequate flows during spawning periods. Total cover amount was rated as trace with boulders dominant. Cover was also present as undercut banks and instream vegetation.The average channel width was 9.9m, the average wetted width was 3.4m, and the average gradient was 2%.The dominant substrate was boulders with cobbles sub-dominant. The habitat was rated as medium value.

Stream Characteristics Upstream of Crossing 196085 and Downstream of Crossing 203582

The stream was surveyed upstream from crossing 196085 for 410m , up to crossing 203582 (Figure 5.4). The habitat was rated as medium value. Total cover amount was rated as trace with boulders dominant. Cover was also present as undercut banks.The average channel width was 11m, the average wetted width was 1.9m, and the average gradient was 1.5%.The dominant substrate was gravels with cobbles sub-dominant. The channel was dewatered at the time of assessment with shallow pools present at the downstream end of the site. Gravels were abundant throughout and suitable for chinook and sockeye spawning under adequate flow conditions. Deep pools were likely present when the stream was flowing.

Stream Characteristics Upstream of Crossing 203582

The stream was surveyed upstream from crossing 203582 for 650m (Figure 5.4). The habitat was rated as medium value. The stream was dewatered until approximately 360m upstream of the railway culverts, and the farthest downstream pool contained numerous fry. Upstream, occasional pools transitioned to increasing flow toward the upper end of the site where a continually flowing channel reemerged. The system was extremely large with high flows during freshet based on the size of the substrates. Gravels were abundant throughout, with limited woody debris or in stream cover at low flow, restricted mainly to interstitial spaces between cobbles and boulders. The average channel width was 14.6m, the average wetted width was 2.5m, and the average gradient was 0.8%.Total cover amount was rated as trace with boulders dominant. Cover was also present as undercut banks.The dominant substrate was gravels with cobbles sub-dominant.

Environmental DNA Sampling

One eDNA sample was collected downstream of crossing 196085 on Willow-Cale FSR, and two samples were collected upstream of the crossing in a pool where fish were observed, using different pore sizes. Upstream of the CN railway crossing, two samples were collected using different pore sizes, and the pool sampled contained numerous fry. Sampling techniques are summarized in Table 5.11.

Table 5.11: Summary of eDNA samples collected at sites 196085 and 203582.

Site	Stream	Habitat Type	Sample Method	Filter Type	Filter Size (um)	Volume filtered (L)	Site Description
196085_ds_ed1	Tabor Creek	riffle	grab	mce	1.0	–	Sample collected approximately 400m downstream with culvert on left bank at pool Inlet.
196085_us_ed1a	Tabor Creek	pool	instream	mce	1.0	1.0	Sample collected 30m upstream of FSR/culvert inlet on the right bank.
196085_us_ed1b	Tabor Creek	pool	grab	mce	1.5	1.0	Sample collected 30m upstream of FSR/culvert inlet on the right bank.
203582_us_ed1a	Tabor Creek	pool	grab	mce	1.0	0.5	Sample collected 360m upstream of the railway culverts. Sampled from right bank. Site is adjacent to large slumping clay bank. There are numerous fish in this first pool since extensive dewatering which started just above the Forest service road.
203582_us_ed1b	Tabor Creek	pool	grab	mce	1.5	0.8	Sample collected 360m upstream of the railway culverts. Sampled from right bank. Site is adjacent to large slumping clay bank. There are numerous fish in this first pool since extensive dewatering which started just above the Forest service road.

Detection results are presented in Table 5.12. Format is `CODE(max_droplets)`, with $*$ flagging samples UNBC reran. Detected means at least 4 positive droplets, the standard ddPCR call threshold; sub-threshold signals of 1-3 droplets are not interpretable as species presence. Site-level detail for the whole project, along with field blanks and retests, is in Appendix - Environmental DNA Results (page 129).

Table 5.12: eDNA detection results at PSCIS crossings 196085 and 203582.

Site ID	UNBC ID	Detected	Sub-threshold	Not detected
196085_ds_ed1	F40915	BURB(8), CHIN(181)	BULT(1)	SOCK(0)
196085_us_ed1a	F41019	CHIN(8), RAIN(6)	—	BULT(0), BURB(0)
196085_us_ed1b	F41020	CHIN(144), RAIN(128)	—	BULT(0), BURB(0)
203582_us_ed1a	F60506	CHIN(56)*, RAIN(1281)	SOCK(1)	BULT(0), COHO(0)
203582_us_ed1b	F60507	CHIN(37)*, RAIN(1871)	—	BULT(0), COHO(0), SOCK(0)

Structure Remediation and Cost Estimate

Should restoration/maintenance activities proceed, replacement of the Willow-Cale FSR crossing (196085) with a bridge (18m span) is recommended. At the time of reporting in 2025, the cost of the work is estimated at $3,600,000.

Should restoration/maintenance activities proceed at the CN Railway crossing (203582) replacement with a bridge (36 m span) is recommended. At the time of reporting in 2025, the cost of the work is estimated at $27,000,000.

Conclusion

Tabor Creek provided medium value habitat for salmonid spawning and rearing, with chinook documented downstream of the Willow-Cale FSR crossing. Due to previous remediation work completed at this site, the crossing is ranked as a moderate priority for replacement. The upstream crossing on the CN Railway is ranked as a moderate priority for replacement due to a 0.3m outlet drop. Replacement would be costly because the structure is located on the railway, and the stream exhibited sections of dewatering above Willow-Cale FSR; however, flow was regained upstream and evidence of high flows was observed. Both crossings are situated near the Fraser River and are in close proximity to the City of Prince George, offering opportunities for community engagement.

Table 5.13: Summary of habitat details for PSCIS crossings 196085 and 203582.

Site	Location	Length Surveyed (m)	Average Channel Width (m)	Average Wetted Width (m)	Average Pool Depth (m)	Average Gradient (%)	Total Cover	Habitat Value
196085	Downstream	350	9.9	3.4	0.2	2.0	trace	medium
196085	Upstream	410	11.0	1.9	–	1.5	trace	medium
203582	Upstream	650	14.6	2.5	–	0.8	trace	medium

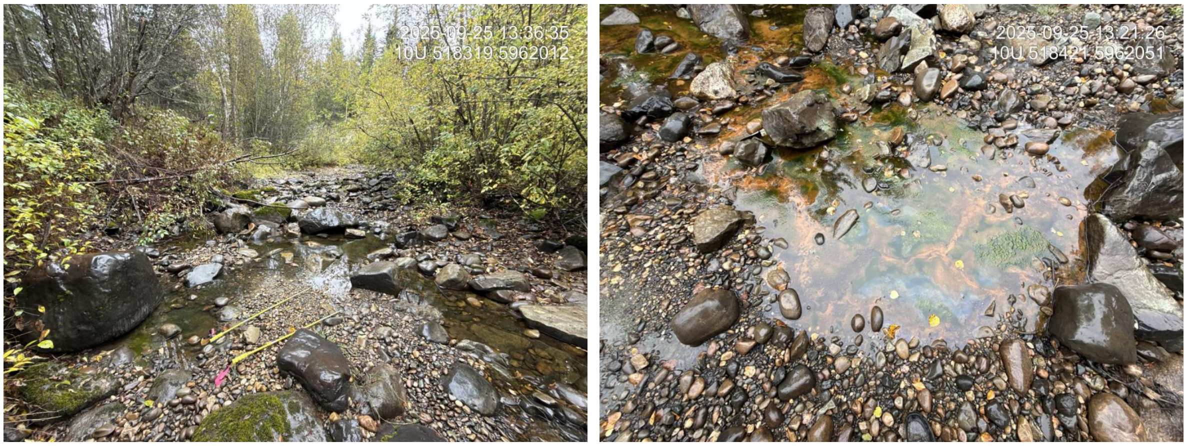


Figure 5.3: Left: Typical habitat downstream of PSCIS crossing 196085. Right: Algae observed dowstream of crossing 196085.

Figure 5.4: Left: Typical habitat upstream of PSCIS crossing 196085 and downstream of PSCIS crossing 203582. Right: Typical habitat upstream of PSCIS crossing 203582.

South Yuzkli Creek - 196332 - Appendix

Site Location

PSCIS crossing 196332 is located on South Yuzkli Creek , approximately 50km northeast of Quesnel, BC, in the Willow River watershed group (Figure **??**). The crossing is located 1.1km upstream of the stream's confluence with Yuzkli Creek, which flows into the Willow River. The crossing is located on 24 Y Road and is situated under West Fraser Mills Ltd. road tenure R14273 100S.

Background

At PSCIS crossing 196332, South Yuzkli Creek is a second order stream and drains a watershed of approximately 5.1km^2. The watershed ranges in elevation from a maximum of 1636m to 1218m near the crossing (Table 5.14).

PSCIS crossing 196332 was first assessed with a fish passage assessment in 2014 by Hooft (2015) and ranked as a barrier to fish passage with medium value habitat documented. The crossing was reassessed in 2025 and a habitat confirmation assessment was conducted due to the presence of medium value habitat and bull trout documented just downstream of the crossing in 2014 (Norris [2018] 2024; MoE 2024).

A summary of habitat modelling outputs for the crossing are presented in Table 5.15.

Table 5.14: Summary of derived upstream watershed statistics for PSCIS crossing 196332.

Site	Area Km	Elev Site	Elev Min	Elev Max	Elev Median	Elev P60	Aspect
196332	5.1	1218	1200	1636	1417	1398	SW

* Elev P60 = Elevation at which 60% of the watershed area is above

Table 5.15: Summary of fish habitat modelling for PSCIS crossing 196332.

Habitat	Potential	Remediation Gain	Remediation Gain (%)
BT Rearing (km)	2.8	2.8	100

Habitat	Potential	Remediation Gain	Remediation Gain (%)
BT Spawning (km)	0.0	0.0	–
BT Network (km)	6.1	4.7	77
BT Stream (km)	5.9	4.6	78
BT Lake Reservoir (ha)	0.0	0.0	–
BT Wetland (ha)	0.9	0.1	11
BT Slopeclass03 (km)	0.6	0.6	100
BT Slopeclass05 (km)	1.6	1.6	100
BT Slopeclass08 (km)	0.5	0.0	0
BT Slopeclass15 (km)	2.9	2.4	83

* Model data is preliminary and subject to adjustments.

Stream Characteristics at Crossing 196332

At the time of the 2025 assessment, PSCIS crossing 196332 on 24 Y Road was un-embedded, non-backwatered and ranked as barrier to upstream fish passage according to the provincial protocol (MoE 2011) (Table 5.16). The culvert had a 0.6m outlet drop and a 0.7m deep outlet pool.

The water temperature was 2°C, pH was 7.2 and conductivity was 116 uS/cm.

Table 5.16: Summary of fish passage assessment for PSCIS crossing 196332.

Location and Stream Data	•	Crossing Characteristics	–
Date	2025-10-05	Crossing Sub Type	Round Culvert
PSCIS ID	196332	Diameter (m)	1.2
External ID	–	Length (m)	23
Crew	AI	Embedded	No
UTM Zone	10	Depth Embedded (m)	–
Easting	578271	Resemble Channel	–
Northing	5895125	Backwatered	No
Stream	South Yuzkli Creek	Percent Backwatered	–
Road	24 Y Road	Fill Depth (m)	3
Road Tenure	W Fraser R14273 100S	Outlet Drop (m)	0.6
Channel Width (m)	3.4	Outlet Pool Depth (m)	0.7
Stream Slope (%)	8	Inlet Drop	Yes
Beaver Activity	No	Slope (%)	6.5
Habitat Value	Medium	Valley Fill	Deep Fill
Final score	39	Barrier Result	Barrier
Fix type	Replace with New Open Bottom Structure	Fix Span / Diameter	15

Comments: Steep pipe with large outlet drop. Prioritized in Hooft 2015. Bull Trout documented just downstream of the crossing in 2014.

Photos: From top left clockwise: Road/Site Card, Barrel, Outlet, Downstream, Upstream, Inlet.

Location and Stream Data	•	Crossing Characteristics	–

Stream Characteristics Downstream of Crossing 196332

The stream was surveyed downstream from crossing 196332 for 220m (Figure 5.5). The habitat was rated as medium value for salmonid spawning and rearing. The stream was steep with numerous step-pool features formed by large woody debris ranging from 40–60cm in height. Shallow pools were present, typically 30–40cm deep, with boulder and overhead cover distributed throughout. The dominant substrate was cobbles with boulders sub-dominant.Total cover amount was rated as moderate with boulders dominant. Cover was also present as small woody debris, large woody debris, undercut banks, deep pools, and overhanging vegetation.The average channel width was 3.4m, the average wetted width was 2.6m, and the average gradient was 7.8%.

Stream Characteristics Upstream of Crossing 196332

The stream was surveyed upstream from crossing 196332 for 400m (Figure 5.5). The habitat was rated as medium value. Total cover amount was rated as moderate with boulders dominant. Cover was also present as small woody debris, large woody debris, undercut banks, and overhanging vegetation.The dominant substrate was cobbles with boulders sub-dominant.The average channel width was 3m, the average wetted width was 2.3m, and the average gradient was 8.8%. The stream was steep with good flow and numerous step-pool features ranging from 30–50cm in height. Shallow pools were 30–40cm deep with pockets of gravel suitable for resident bull trout spawning. Large woody debris was abundant throughout.

Environmental DNA Sampling

eDNA samples were collected both upstream and downstream of crossing 196332 on 24 Y Road, with sampling techniques summarized in Table 5.17.

Table 5.17: Summary of eDNA samples collected at site 196332.

Site	Stream	Habitat Type	Sample Method	Filter Type	Filter Size (um)	Volume filtered (L)	Site Description
196332_ds_ed1	S. Yuzkli Creek	–	grab	mce	1	1.9	220m downstream of culvert on 24 Y Road. Left bank.
196332_us_ed1	S. Yuzkli Creek	riffle	grab	mce	1	1.9	Immediately upstream of culvert on 24 Y Road. Left bank.

Detection results are presented in Table 5.18. Format is `CODE(max_droplets)`, with * flagging samples UNBC reran. Detected means at least 4 positive droplets, the standard ddPCR call threshold; sub-threshold signals of 1-3 droplets are not interpretable as species presence. Site-level detail for the whole project, along with field blanks and retests, is in Appendix - Environmental DNA Results (page 129).

Table 5.18: eDNA detection results at PSCIS crossing 196332.

Site ID	UNBC ID	Detected	Sub-threshold	Not detected
196332_ds_ed1	F21237	—	—	BULT(0), RAIN2(0)
196332_us_ed1	F21238	—	BULT(2)*	RAIN(0)*, RAIN2(0)

Structure Remediation and Cost Estimate

Should restoration/maintenance activities proceed, replacement of the 24 Y Road crossing (196332) with a bridge (15m span) is recommended. At the time of reporting in 2025, the cost of the work is estimated at $450,000.

Conclusion

The habitat upstream of PSCIS crossing 196332 on 24 Y Road was documented as medium value for salmonid spawning and rearing, and the crossing is rated as a high priority for replacement due to the large outlet drop and presence of bull trout documented downstream of the crossing.

Table 5.19: Summary of habitat details for PSCIS crossing 196332.

Site	Location	Length Surveyed (m)	Average Channel Width (m)	Average Wetted Width (m)	Average Pool Depth (m)	Average Gradient (%)	Total Cover	Habitat Value
196332	Downstream	220	3.4	2.6	0.3	7.8	moderate	medium
196332	Upstream	400	3.0	2.3	0.3	8.8	moderate	medium

Figure 5.5: Left: Typical habitat downstream of PSCIS crossing 196332. Right: Typical habitat upstream of PSCIS crossing 196332.

Tributary To Fraser River - 203581 & 196076 - Appendix

Site Location

PSCIS crossings 203581 and 196076 are located on Tributary To Fraser River , approximately 13km east of Prince George, BC, in the Tabor River watershed group (Figure **??**). Crossing 203581 is located 80m upstream of the Fraser River, and is located on and the responsibility of the CN Railway. Approximately 1.3km further upstream, PSCIS crossing 196076 is located on Beaver FSR, and is situated under McLean road tenure R02924 A.

Background

At the location of these crossings, Tributary To Fraser River is a fourth order stream and drains a watershed of approximately 17.5km^2. The watershed ranges in elevation from a maximum of 784m to 592m near the crossing (Table 5.20).

The uppermost crossing, crossing 196076 on Beaver FSR, was previously assessed in 2014 by Hooft (2015) and ranked as a barrier to fish passage with medium value habitat documented. Due to the large amount of chinook and bull trout habitat modelled upstream, the site was reassessed in 2025 along with the previously unassessed downstream CN railway crossing (crossing 203581). Habitat confirmation assessments were completed at both crossings in 2025. No fisheries information is documented for the stream; however, the crossings are near the confluence with the Fraser River, where chinook and bull trout have been recorded.

A summary of habitat modelling outputs for the crossing are presented in Table 5.21.

Table 5.20: Summary of derived upstream watershed statistics for PSCIS crossing 203581.

Site	Area Km	Elev Site	Elev Min	Elev Max	Elev Median	Elev P60	Aspect
203581	17.5	592	572	784	688	680	SW

* Elev P60 = Elevation at which 60% of the watershed area is above

Habitat	Potential	Remediation Gain	Remediation Gain (%)
BT Rearing (km)	11.6	1.4	12
BT Spawning (km)	3.4	1.1	32
BT Network (km)	30.3	2.4	8
BT Stream (km)	24.1	2.1	9
BT Lake Reservoir (ha)	22.2	2.6	12
BT Wetland (ha)	40.7	0.0	0
BT Slopeclass03 (km)	15.3	0.9	6
BT Slopeclass05 (km)	6.8	0.6	9
BT Slopeclass08 (km)	1.1	0.0	0
BT Slopeclass15 (km)	0.9	0.6	67

* Model data is preliminary and subject to adjustments.

Stream Characteristics at Crossings 203581 and 196076

At the time of the 2025 assessment, both PSCIS crossing 203581 and 196076 on Railway were un-embedded, non-backwatered and ranked as a barrier to upstream fish passage according to the provincial protocol (MoE 2011) (Tables 5.22 - 5.23). The downstream crossing on the CN railway (crossing 203581) had a 2.5m outlet drop, however the outlet is directly on the floodplain of the Fraser River and when the river is 2.5 - 3.5m higher the structure would backwater.

Water temperature was 8°C, pH was 7.6 and conductivity was 489 uS/cm.

Table 5.22: Summary of fish passage assessment for PSCIS crossing 203581.

Location and Stream Data	•	Crossing Characteristics	–
Date	2025-09-24	Crossing Sub Type	Concrete Box
PSCIS ID	203581	Diameter (m)	2.1
External ID	19703295	Length (m)	50
Crew	AI	Embedded	No
UTM Zone	10	Depth Embedded (m)	–
Easting	526469	Resemble Channel	No
Northing	5985767	Backwatered	No
Stream	Tributary To Fraser River	Percent Backwatered	–
Road	Railway	Fill Depth (m)	9.9
Road Tenure	CN Rail	Outlet Drop (m)	2.5
Channel Width (m)	4	Outlet Pool Depth (m)	0.1
Stream Slope (%)	2.5	Inlet Drop	No
Beaver Activity	No	Slope (%)	2.5
Habitat Value	Medium	Valley Fill	Deep Fill
Final score	37	Barrier Result	Barrier
Fix type	Replace with New Open Bottom Structure	Fix Span / Diameter	35.5

Comments: Large concrete box structure with the outlet directly on the floodolain of the Fraser River. When the river is 2.5 - 3.5m higher the structure would backwater. Appears that the high watermark is near level with the outlet of the culvert.

Photos: From top left clockwise: Road/Site Card, Barrel, Outlet, Downstream, Upstream, Inlet.

•

Location and Stream Data | **Crossing Characteristics** –

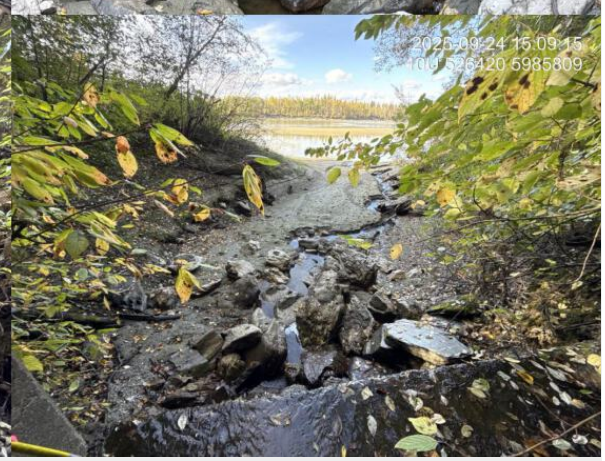


Table 5.23: Summary of fish passage assessment for PSCIS crossing 196076.

Location and Stream Data	•	**Crossing Characteristics**	**–**
Location and Stream Data	•	**Crossing Characteristics**	**–**
Date	2025-09-24	Crossing Sub Type	Round Culvert
PSCIS ID	196076	Diameter (m)	1.2
External ID	–	Length (m)	50
Crew	AI	Embedded	No
UTM Zone	10	Depth Embedded (m)	–
Easting	527373	Resemble Channel	No
Northing	5985002	Backwatered	No
Stream	Tributary To Fraser River	Percent Backwatered	–
Road	Beaver FSR	Fill Depth (m)	4.5
Road Tenure	McLean R02924 A	Outlet Drop (m)	0.55
Channel Width (m)	4	Outlet Pool Depth (m)	0.1
Stream Slope (%)	3	Inlet Drop	No
Beaver Activity	Yes	Slope (%)	2
Habitat Value	Medium	Valley Fill	Deep Fill
Final score	37	Barrier Result	Barrier
Fix type	Replace with New Open Bottom Structure	Fix Span / Diameter	19.5

Comments: Very long culvert in poor condition, with water flowing out through the eroded pipe bottom rather than the outlet, which landed on a dry boulder riprap cascade. Large beaver dams began approximately 50–100m downstream. Habitat confirmation was conducted upstream and downstream of the crossing, as well as at the railway crossing approximately 1.5km downstream near the Fraser River.

Photos: From top left clockwise: Road/Site Card, Barrel, Outlet, Downstream, Upstream, Inlet.

Location and Stream Data	•	Crossing Characteristics	–

Stream Characteristics Downstream of Crossing 203581

The stream was surveyed downstream from crossing 203581 for 80m (Figure 5.6). The habitat was rated as low value for salmonid spawning and rearing. The outlet of the culvert was directly on the floodplain of the Fraser River. The average channel width was 8.3m, the average wetted width was 1m, and the average gradient was 2.5%.

Stream Characteristics Upstream of Crossing 203581

The stream was surveyed upstream from crossing 203581 for 650m (Figure 5.6). The habitat was rated as medium value. The dominant substrate was gravels with fines sub-dominant.Total cover amount was rated as moderate with small woody debris dominant. Cover was also present as large woody debris, boulders, undercut banks, deep pools, and overhanging vegetation.The average channel width was 4.5m, the average wetted width was 1.9m, and the average gradient was 2.5%. A landslide was located approximately 300m upstream of the railway crossing. The stream was low-gradient with abundant gravels and numerous fish to 60mm observed. The channel flowed through an entrenched, steep-walled valley with several sections where clay banks had collapsed into the stream.

Stream Characteristics Downstream of Crossing 196076

The stream was surveyed downstream from crossing 196076 for 250m (Figure 5.7). The habitat was rated as medium value. Beaver dams ranging from 0.6–1m in height began approximately 150m downstream of the culvert within an incised gully. Upstream of the dams, abundant gravels, shallow pools, and small woody debris provided habitat suitable for parr and juvenile rearing. The dominant substrate was NA with NA sub-dominant.The average channel width was 7m, the average wetted width was 5.7m, and the average gradient was 3%.Total cover amount was rated as moderate with deep pools dominant. Cover was also present as small woody debris, undercut banks, and overhanging vegetation.

Stream Characteristics Upstream of Crossing 196076

The stream was surveyed upstream from crossing 196076 for 700m (Figure 5.7). The habitat was rated as medium value. Total cover amount was rated as moderate with undercut banks dominant. Cover was also present as small woody debris, deep pools, and overhanging vegetation.The average channel width was 3.5m, the average wetted width was 2.1m, and the average gradient was 2.8%.The dominant substrate was NA with NA sub-dominant. The stream was moderate-sized, low-gradient stream with abundant gravels and habitat suitable for fry and juvenile salmonid rearing, including shallow pools, small and large woody debris, and undercut banks. No fish or barriers were observed at the time of assessment.

Environmental DNA Sampling

Two eDNA samples were collected upstream of crossing 203581 on the CN railway using different pore sizes. Two samples were also collected downstream of crossing 196076 on Beaver FSR, positioned upstream of the lowest beaver dam. Fish were observed downstream of the lowest beaver dam ~500m from the Fraser River; samples were therefore collected to assess whether fish could be detected upstream of the dams but downstream of the FSR. Filtering rates were extremely slow, requiring a second filter. One additional sample was collected upstream of 196076 on Beaver FSR. Sampling techniques are summarized in Table 5.24.

Table 5.24: Summary of eDNA samples collected at sites 203581 and 196076.

Site	Stream	Habitat Type	Sample Method	Filter Type	Filter Size (um)	Volume filtered (L)	Site Description
196076_ds_ed2a	Tributary To Fraser River	pool	instream	mce	1.00	1.3	150m downstream of culvert on Beaver FSR upstream of first beaver dam (from culvert downstream).
196076_ds_ed2b	Tributary To Fraser River	pool	instream	mce	1.00	1.6	150m downstream of culvert on Beaver FSR upstream of first beaver dam (from culvert downstream).
196076_us_ed1	Tributary To Fraser River	pool	instream	mce	1.00	2.0	10 m upstream of culvert inlet in pool. Too shallow to sample riffle soul sample top end of shallow pool. Right bank
203581_us_ed1a	Tributary To Fraser River	–	grab	mce	0.45	1.0	20m upstream of railway culvert
203581_us_ed1b	Tributary To Fraser River	–	grab	mce	1.50	1.0	20m upstream of railway culvert

Detection results are presented in Table 5.25. Format is `CODE(max_droplets)`, with $*$ flagging samples UNBC reran. Detected means at least 4 positive droplets, the standard ddPCR call threshold; sub-threshold signals of 1-3 droplets are not interpretable as species presence. Site-level detail for the whole project, along with field blanks and retests, is in Appendix - Environmental DNA Results (page 129).

Table 5.25: eDNA detection results at PSCIS crossings 203581 and 196076.

Site ID	UNBC ID	Detected	Sub-threshold	Not detected
196076_ds_ed2a	F41016	RAIN(23)	BURB(1)	BULT(0), CHIN(0)
196076_ds_ed2b	F41017	RAIN(5)	—	BULT(0), BURB(0), CHIN(0)
196076_us_ed1	F61125	RAIN(44)	—	BULT(0), CHIN(0), COHO(0)
203581_us_ed1a	F61123	RAIN(54)	BULT(2), *CHIN(2)*	COHO(0)

Site ID	UNBC ID	Detected	Sub-threshold	Not detected
203581_us_ed1b	F61124	RAIN(51)	CHIN(3)*	BULT(0), COHO(0)

Structure Remediation and Cost Estimate

Should restoration/maintenance activities proceed, replacement of the CN Railway crossing (203581) with a bridge (35.5m span) is recommended. At the time of reporting in 2025, the cost of the work is estimated at $26,625,000.

Should restoration/maintenance activities proceed at the Beaver FSR crossing (196076) replacement with a bridge (19.5 m span) is recommended. At the time of reporting in 2025, the cost of the work is estimated at $585,000.

Conclusion

Tributary To Fraser River provided medium value habitat for salmonid spawning and rearing. The downstream railway crossing is ranked as a low priority for replacement because the structure is likely backwatered when the Fraser River is higher. The upstream crossing on Beaver FSR had a significant outlet drop and is ranked as a moderate priority for replacement. Crossing 196076 on Beaver FSR could be a strong candidate for replacement because the site is close to the Fraser River, is in close proximity to the City of Prince George (supporting community engagement opportunities), and is located on an active logging road, providing potential for collaborative remediation with the road tenure holder McLean.

Table 5.26: Summary of habitat details for PSCIS crossings 203581 and 196076.

Site	Location	Length Surveyed (m)	Average Channel Width (m)	Average Wetted Width (m)	Average Pool Depth (m)	Average Gradient (%)	Total Cover	Habitat Value
196076	Downstream	250	7.0	5.7	0.4	3.0	moderate	medium
196076	Upstream	700	3.5	2.1	0.4	2.8	moderate	medium
203581	Downstream	80	8.3	1.0	–	2.5	–	–
203581	Upstream	650	4.5	1.9	0.3	2.5	moderate	medium

Figure 5.6: Left: Typical habitat downstream of PSCIS crossing 203581. Right: Typical habitat upstream of PSCIS crossing 203581.

Figure 5.7: Left: Beaver dam complex located downstream of PSCIS crossing 196076. Right: Typical habitat upstream of PSCIS crossing 196076.

Appendix - Floodplain Delineation

This appendix details the floodplain delineation for the Nechako River Watershed Group. The rationale for mapping floodplain extent alongside the fish passage barrier inventory is described in the report background — in brief, the off-channel habitats that floodplains sustain are often as important to fish productivity as the mainstem passage that crossing remediations restore, and infrastructure corridors can compromise floodplain function in ways that structure replacement alone does not address.

Floodplain footprints were delineated using the `flooded` R package, which implements the Valley Confinement Algorithm of Nagel et al. (2014). The algorithm combines a digital elevation model with the stream network and a modelled bankfull flood surface to distinguish valley bottoms from confined canyon reaches. Bankfull depth — the flow depth at which a stream begins to spill onto its floodplain — is estimated from upstream watershed area and mean annual precipitation using the regional regression of Hall et al. (2007), calibrated against field-mapped floodplain sites in Pacific Northwest streams.

We mapped the **functional floodplain**: the area a river actively inundates and reworks during regular high-flow events. This sits between two other scales — the narrow active channel (the wetted perimeter at low flow) and the broader geologic valley bottom (which includes terraces untouched by water for centuries). The functional floodplain matters most for habitat planning because it is the ground where side channels form, seasonal flooding sustains wetlands, and riparian forests recruit. Inputs were the MRDEM-30 30 m DEM, the chinook accessible stream network (stream order ≥ 3) from bcfishpass, and BC Freshwater Atlas lakes and wetlands on that network. Waterbodies were included so the valley confinement model fills cells that gradient and cost-distance masks would otherwise exclude.

Table 5.27: Nechako River Watershed Group — modelled functional floodplain and the off-channel habitat units within it.

Metric	Value
Watershed group area (km²)	4,148
Floodplain area (ha)	54,306
Floodplain as % of watershed group	13.1
Chinook accessible streams, order 3+ (km)	1,451
Streams within floodplain (km)	1,312
Lakes within floodplain (count)	439
Lakes within floodplain (ha)	8,331
Wetlands within floodplain (count)	702
Wetlands within floodplain (ha)	5,599

The Nechako River Watershed Group covers roughly 4,148 km². Modelled functional floodplain (Table 5.27) covers about 54,306 ha, or 13.1 % of the watershed group, attached to 1,312 km of the 1,451 km chinook accessible order-3+ network. Within that footprint sit 439 lakes and 702 mapped wetland polygons — the off-channel habitat units that lateral floodplain connectivity supports.

The watershed-wide footprint is shown in Figure 5.8; a detail view of the Murray Creek confluence area near Vanderhoof is shown in Figure 5.9.

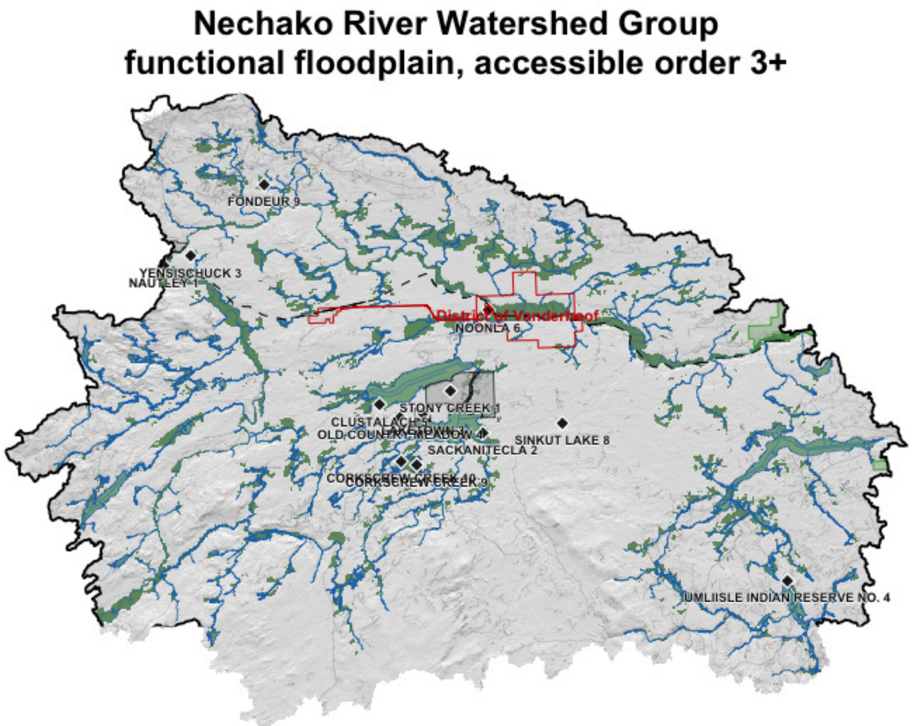


Figure 5.8: Nechako River Watershed Group functional floodplain (green) over hillshaded terrain. Chinook accessible order-3+ streams in blue; lakes and wetlands in light blue; parks light green; First Nations reserves light grey with diamond markers and labels; municipalities outlined in red; forest service / resource roads grey; railways black dashed; watershed boundary heavy black.

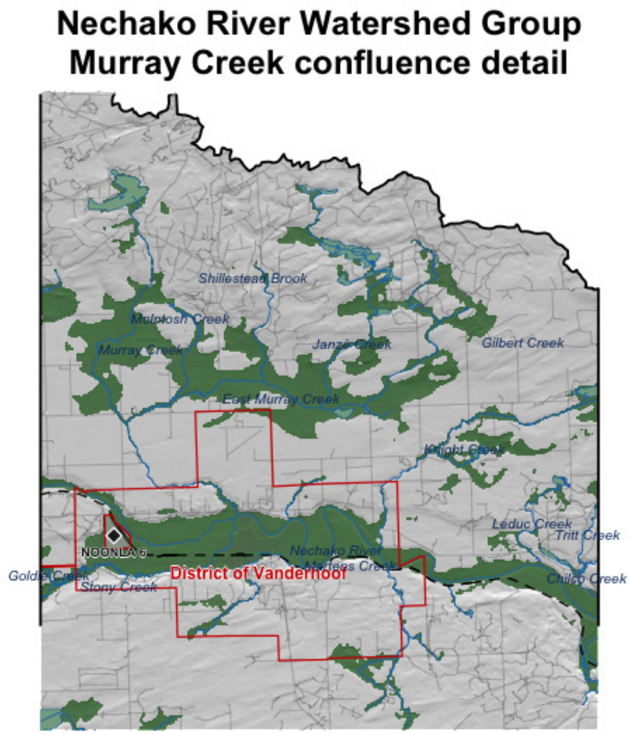


Figure 5.9: Detail view of the Murray Creek confluence with the Nechako River. Symbology as Figure 5.8; named streams labelled in italic blue; municipalities outlined and labelled in red.

The watershed-wide view shows floodplain tracking the trunk Nechako River through broad valley reaches and tributary confluences. The detail view near Vanderhoof resolves how road corridors and settlement intersect the floodplain footprint at reach scale — context that informs whether crossing remediation at a given site should account for restoring lateral connectivity beyond the structure itself. We plan to extend this mapping to additional watershed groups in coming years so that floodplain context can inform barrier prioritisation and restoration scoping across the region.

Appendix - Climate Departure

Climate departure and fish passage

Stream crossings are the visible structures in a fish-passage inventory: culverts, bridges, weirs, fords. The streams they cross sit inside a climate that is changing — warmer air, shifting snowpack, longer shoulder seasons — and those climate changes set the hydrologic and thermal context the structures pass fish *through*. A culvert installed in the 1980s is now passing a different hydrograph: earlier freshet, lower late-summer flows, warmer water. The barrier inventory does not see that on its own.

Warmer air drives warmer streams. For the cold-water salmonids the study area supports — bull trout, chinook, mountain whitefish, rainbow trout — that can move habitat in either direction: cold-limited headwaters gain growing-degree-days (larger fish, better overwinter survival, stronger out-migration), while reaches near the upper edge of the thermal niche lose usable habitat. The recovery value of removing a downstream barrier scales with which side of that line each upstream reach sits on. Climate departure tells you which side that is, watershed by watershed.

Beyond temperature, hydrologic timing matters. Earlier and faster snowmelt shifts the freshet weeks earlier in the calendar and compresses the high-flow pulse. That changes design assumptions for crossing structures, and it changes the timing of migration, gravel mobilisation, and off-channel rearing — the in-stream processes barrier removal is meant to restore.

Quantifying departure at the watershed scale lets us carry climate context into barrier prioritisation alongside the structural and biological inputs.

Area of interest

The study area for this climate-departure analysis is the union of seven Freshwater Atlas watershed groups in the upper and central Fraser drainage: Lower Chilako (LCHL), Nechako (NECR), Francois Lake (FRAN), Morkill River (MORK), Upper Fraser (UFRA), Tabor River (TABR), and Willow River (WILL). Together they cover 34,020 km² and span 8 ecoregions (Figure 5.10) — broad climate-physiography zones based on latitude, elevation, and dominant vegetation. We carry that ecoregion partition through the analysis because climate departure can vary across them in ways the regional average hides, then map the ecoregion signal back onto the seven watershed groups at the end of the appendix.

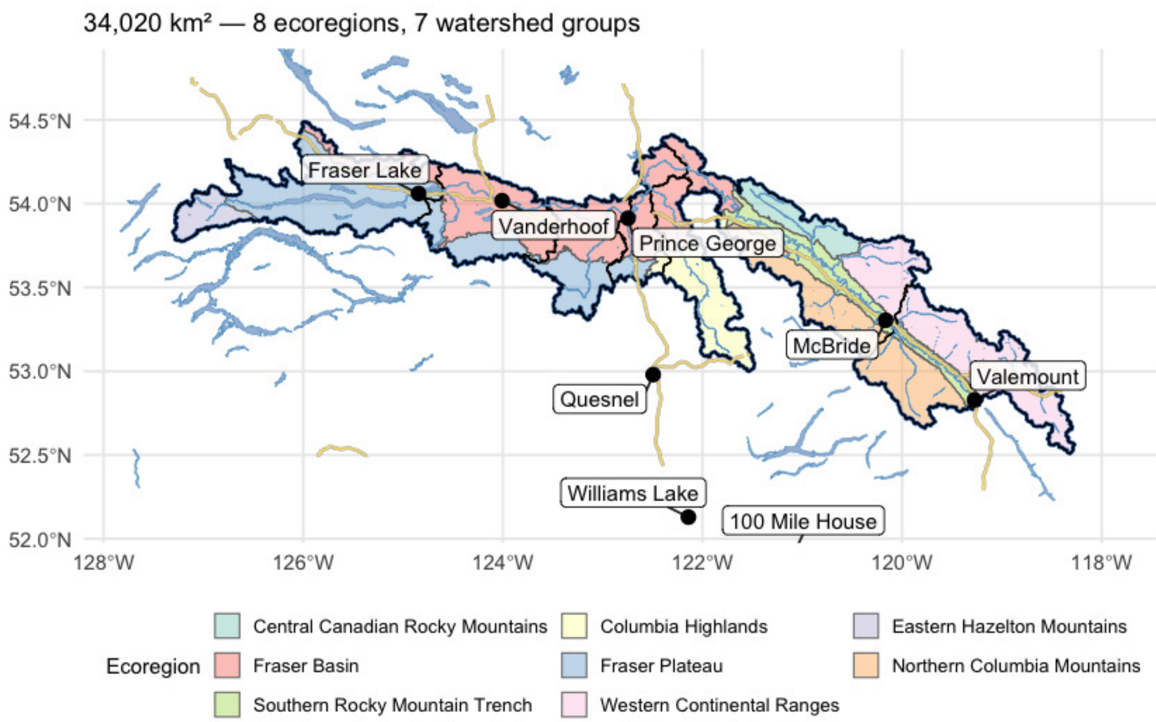


Figure 5.10: Study area in central British Columbia, coloured by ecoregion, with the seven watershed groups (heavy black outlines) that define the analysis extent. The Fraser flows south-east through the interior plateau (Fraser Basin / Fraser Plateau ecoregions) before turning south near Prince George; the eastern WSGs sit in the Western Continental Ranges and adjacent Rocky Mountain ecoregions.

Recent decade versus pre-warming reference

The table below compares two windows directly — the recent decade (2015–2025) against the pre-warming reference (1951–1980). The difference is reported in the variable's native units, alongside percent change where it is meaningful (precipitation, soil moisture, vapour pressure deficit, relative humidity). Two p-values appear, answering different questions. **Δ p (windows)** is the Welch t-test on the annual values within each window — this tells us if the recent decade differs from the pre-warming reference. **Trend p (75-yr)** is the Mann-Kendall test on the full 1951–present series — this tells us if there is a steady year-on-year ramp. Step changes show up as significant Δ p with non-significant trend p; gradual ramps show up as both significant.

Table 5.28: Recent decade (2015-2025) versus pre-warming reference (1951-1980) for all 15 climate variables and all available periods. Two p-values answer different questions — see text.

Variable	Period	Recent (2015–2025)	Pre-warming (1951–1980)	Δ absolute	Δ %	Δ p (windows)	Trend p (75-yr)
prcp	annual	970.08	942.14	27.95	3.0	0.464	0.784
prcp	fall	294.17	269.19	24.98	9.3	0.318	0.118
prcp	spring	182.92	183.84	-0.93	-0.5	0.940	0.540
prcp	summer	255.27	243.80	11.47	4.7	0.597	0.756
prcp	winter	237.73	245.30	-7.57	-3.1	0.642	0.310
rh	annual	71.68	71.95	-0.28	-0.4	0.561	0.191
rh	fall	77.09	76.48	0.61	0.8	0.448	0.749
rh	spring	65.92	67.72	-1.80	-2.7	0.003	0.028
rh	summer	64.04	65.01	-0.97	-1.5	0.521	0.242

The recent decade ran 1.6 °C warmer than the pre-warming reference for annual mean temperature, with both window and trend p-values below 0.001. Vapour pressure deficit rose 0.34 hPa, significant on both tests. Annual precipitation rose about +3 % regionally but neither test confirms a real shift (window p ≈ 0.5, trend p ≈ 0.8). Soil moisture and relative humidity show no significant change. The snowpack rows (swe summer, snowmelt spring, snowmelt winter, snowmelt_doy_50) carry the seasonal-redistribution and freshet-timing signals described below.

Trends over the long record

Anomalies are computed against the 1951–1980 pre-warming reference. Saying a year is “+1.5 °C” means it was 1.5 °C warmer than the average year between 1951 and 1980. The trend table below has two rows per variable — trends computed from two different start years:

- **1951–present (75 years)** — the long view. Captures the full magnitude of warming since the pre-warming reference.
- **1981–present (45 years)** — starts at the beginning of the current 30-year climate normal (1981–2010), the reference used in most published climate products.

Comparing the two slopes is informative. If the 45-year slope is steeper than the 75-year slope, warming has accelerated; if shallower, warming has slowed (though “slower” almost never means “stopped”). Similar slopes mean a roughly steady rate of change across the full record. Total change is slope multiplied by the number of years — the cumulative shift over the trend window.

Table 5.29: Trend statistics for all variables and periods — Theil-Sen slope, Mann-Kendall p-value, and cumulative change over each trend window.

Parameter	Period	Slope	Years	Total Change	Unit	p-value
Precipitation	Annual	0.021	75	1.6	%	0.7837
Relative humidity	Annual	-0.009	75	-0.7	%	0.1908
Snow cover	Annual	-0.059	75	-4.4	%	0.0026
Snowfall	Annual	-0.102	75	-7.7	%	0.2723
Snowfall fraction	Annual	-0.057	75	-4.3	%	0.0536
Snowmelt	Annual	-0.077	75	-5.8	%	0.3848
Day of 50% melt	Annual	-0.172	75	-12.9	day	0.0009
Peak weekly melt rate	Annual	-0.036	75	-2.7	mm/wk	0.6940

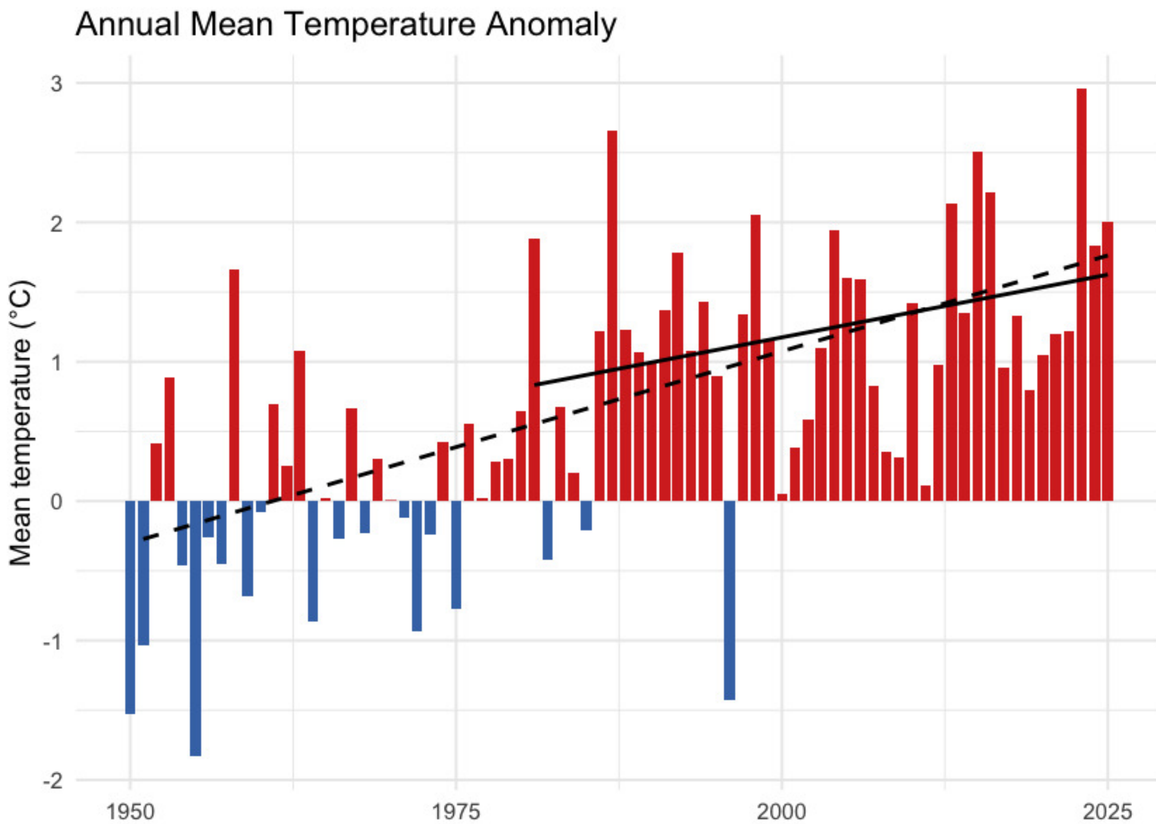


Figure 5.11: Annual mean temperature anomaly relative to the 1951-1980 baseline. Bars are degrees above (red) or below (blue) the pre-warming average; the two trend lines are the 75-year (1951-present, dashed) and 45-year (1981-present, solid) Theil-Sen slopes.

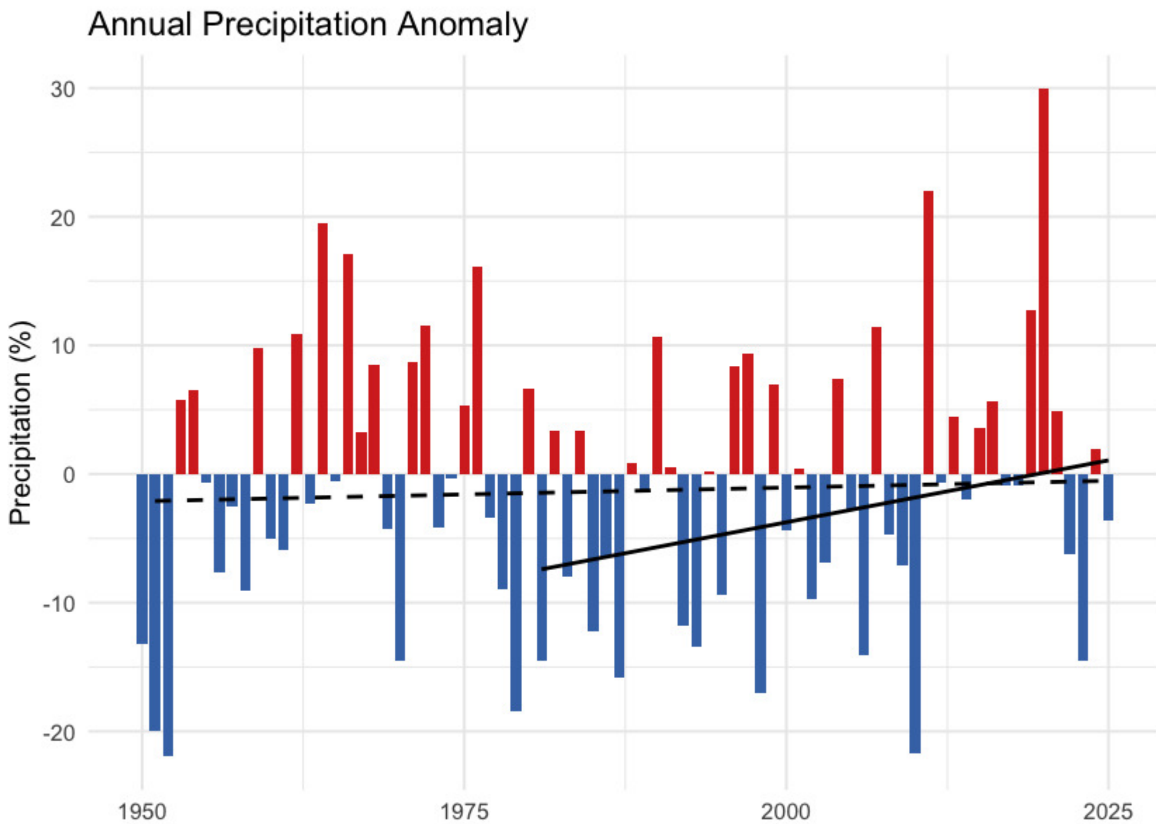


Figure 5.12: Annual precipitation anomaly (% of the 1951-1980 baseline). The long-term trend is essentially flat — regional precipitation has not shifted meaningfully against the natural year-to-year variability.

One pattern in the trend table is worth noting: the 45-year (since 1981) tmean slope is **shallower** than the 75-year slope — 0.18 versus 0.28 °C per decade. The same direction holds for tmax, tmin, and VPD. Warming since 1981 has not been faster than the longer-term average rate; the cumulative shift since 1951 reflects steady, significant warming rather than a recent acceleration.

Daytime highs and overnight lows

Daytime maximum (tmax) and overnight minimum (tmin) temperatures carry distinct information. Overnight minimums warming faster than daytime maximums — the "day-night asymmetry" — is one of the textbook fingerprints of greenhouse warming. Karl et al. (1993) documented this first at the global scale, finding overnight minimums rose at roughly three times the rate of daytime maximums between 1951 and 1990.

In this study area, the asymmetry is present. Daytime maximums warmed 0.24 °C per decade since 1951 (+1.5 °C cumulative); overnight minimums warmed 0.30 °C per decade (+1.8 °C cumulative). The overnight side warmed about 0.3 °C more than the daytime side over the recent decade, narrowing the diurnal temperature range by the same amount (Figure 5.15). The asymmetry is present but weaker than the canonical signal — the local tmin/tmax warming ratio is about 1.2:1 versus the ~3:1 ratio Karl et al. reported at the global scale between 1951 and 1990.

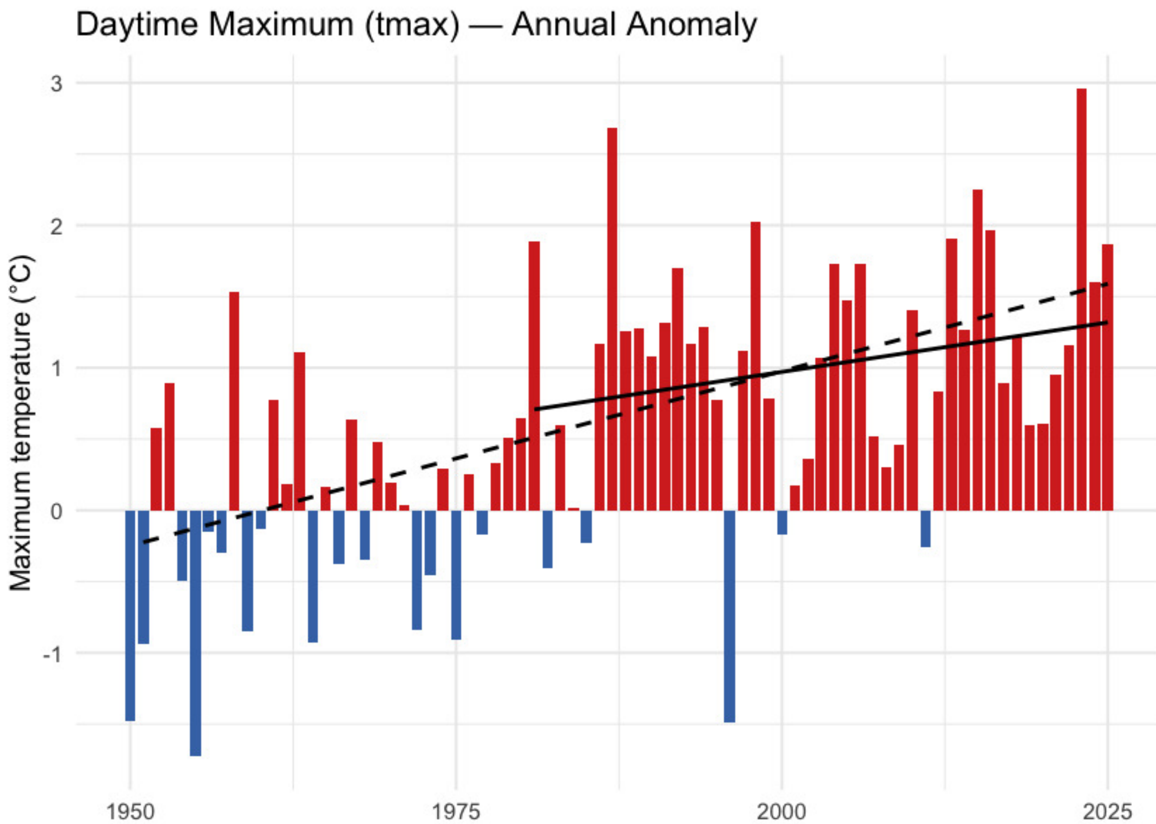


Figure 5.13: Annual daytime maximum temperature (tmax) anomaly relative to the 1951-1980 baseline.

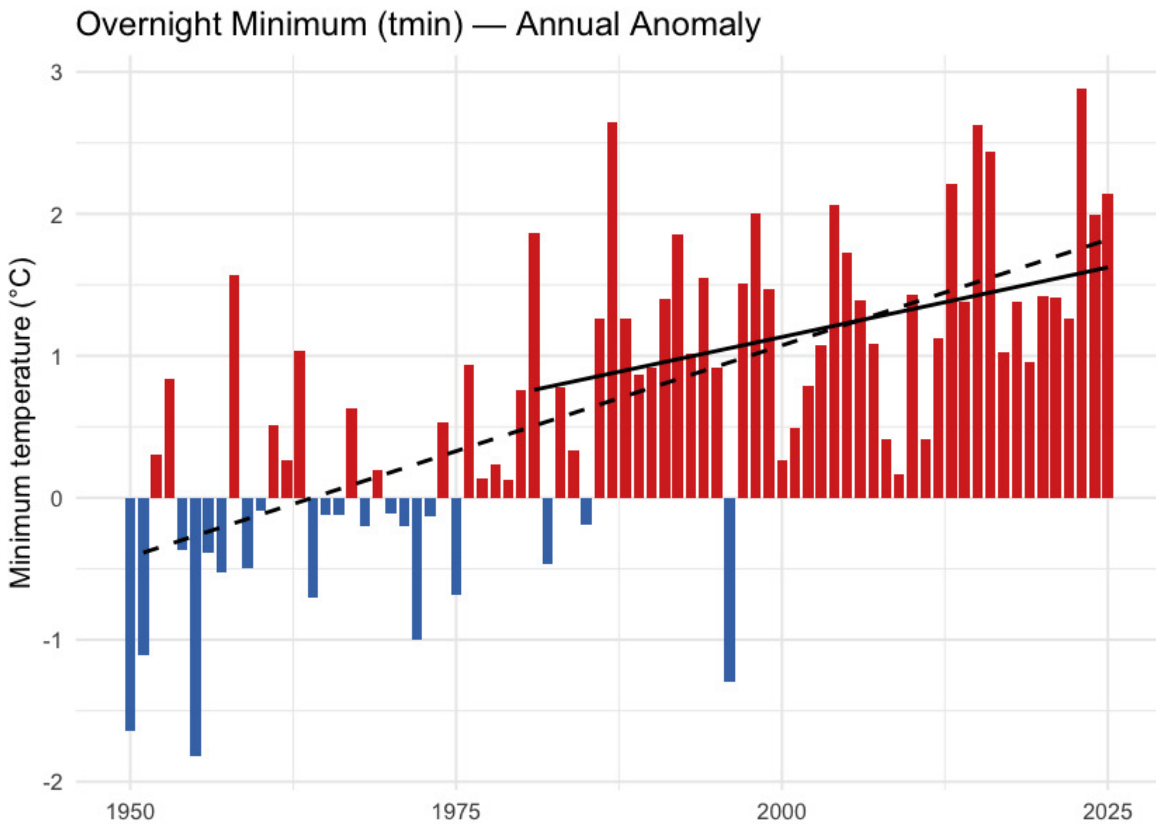


Figure 5.14: Annual overnight minimum temperature (tmin) anomaly relative to the 1951-1980 baseline.

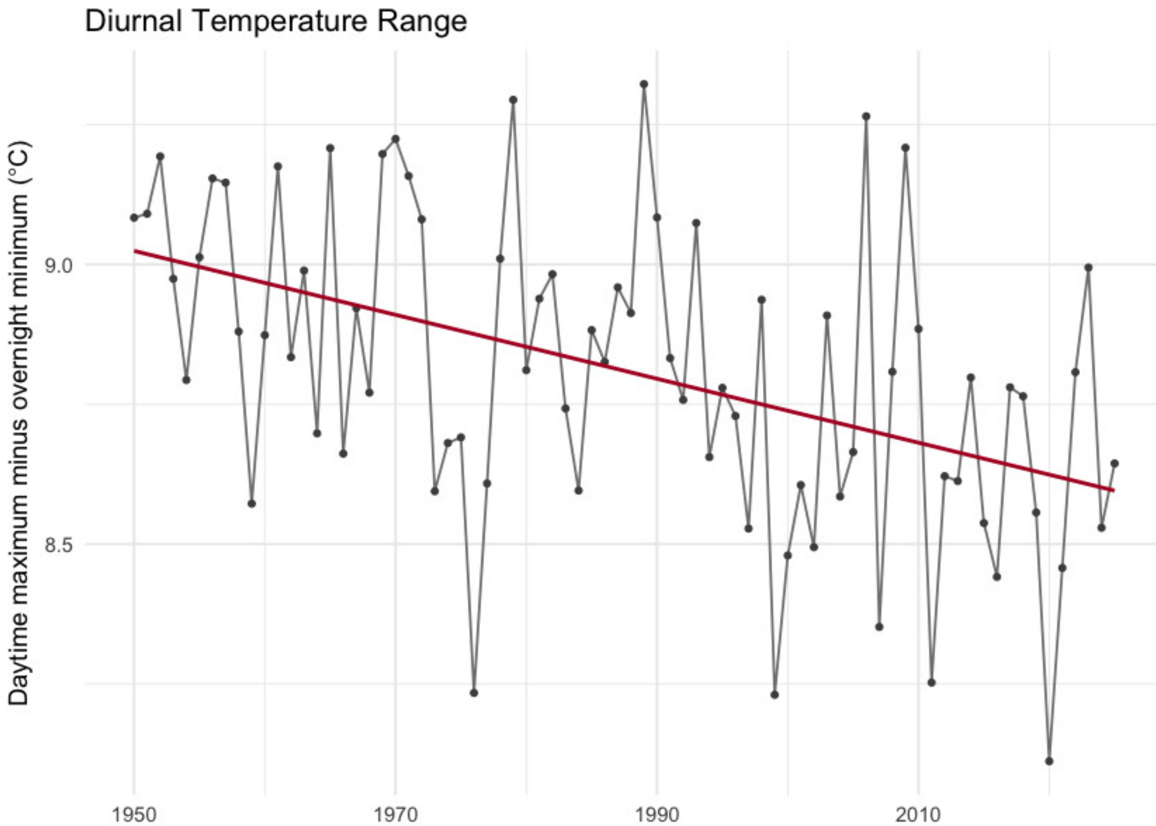


Figure 5.15: Diurnal temperature range (daytime maximum minus overnight minimum) annual mean. The downward slope is the day-night asymmetry — overnight lows are warming faster than daytime highs.

Snowpack

Snowpack drives the seasonal water supply across British Columbia: winter precipitation falls as snow, accumulates on the ground, and releases as meltwater across spring and summer. That seasonal storage is the difference between a late-summer creek that still flows and one that does not. It also sets the timing of the spring freshet — the annual flood pulse that salmonids time their up-river migrations to.

In this study area the snowpack signal is one of **earlier melt, not less snow**. Total annual snowfall fell by about 6 % but neither tests confirm a real change against year-to-year variability. Where the signal is unambiguous is in *when* the snow leaves the ground. The midpoint day of annual snowmelt shifted 12 days earlier in the calendar over the same window ($p < 0.001$), and snow cover dropped 3.8 percentage points ($p = 0.002$).

Seasonal snowpack curve

Aggregated to the four standard meteorological seasons — winter (December–February), spring (March–May), summer (June–August), and fall (September–November) — the table below compares the recent decade (2015–2025) directly against the pre-warming reference (1951–1980) for each of the four monthly-native snow variables. The seasonal redistribution is the headline. Statistically the load-bearing signals are on the summer side — summer SWE has collapsed ($p \approx 0.01$) and summer snowmelt has fallen by a comparable amount ($p \approx 0.02$). The counterbalancing winter and spring snowmelt rises are directionally consistent but noisier season-by-season (Welch $p \approx 0.28$ and 0.06 respectively) — see the *What this means* paragraph below for the full reading.

Table 5.30: Seasonal snowpack: recent decade versus pre-warming reference. Summer SWE collapse (-52 %, $p \approx 0.01$) and summer snowmelt fall (-37 %, $p \approx 0.02$) are the statistically significant signals. Winter (+45 %) and spring (+18 %) snowmelt rises are directionally consistent with the redistribution but do not individually clear statistical significance (Welch $p \approx 0.28$ and 0.06).

Variable	Season	Pre-warming (1951–1980)	Recent (2015–2025)	Δ %
SWE (mm)	winter	219.1	213.8	-2.4
SWE (mm)	spring	323.0	287.9	-10.9
SWE (mm)	summer	40.1	19.3	-51.8
SWE (mm)	fall	24.7	28.0	13.1
Snowfall (mm)	winter	230.8	219.1	-5.0
Snowfall (mm)	spring	116.6	94.9	-18.6
Snowfall (mm)	summer	10.6	6.2	-41.0
Snowfall (mm)	fall	126.4	137.3	8.6
Snowmelt (mm)	winter	6.4	9.3	44.9
Snowmelt (mm)	spring	238.4	282.2	18.4
Snowmelt (mm)	summer	197.5	124.4	-37.0
Snowmelt (mm)	fall	31.1	31.2	0.6

Annual snowpack signals

Four derived annual scalars capture the snowpack signals the literature treats as headline metrics for snow hydrology: peak snowpack (Mote et al. 2018, 2005), the date of melt midpoint (Stewart et al. 2005; Cayan et al. 2001), freshet flashiness, and the fraction of precipitation falling as snow (Knowles et al. 2006).

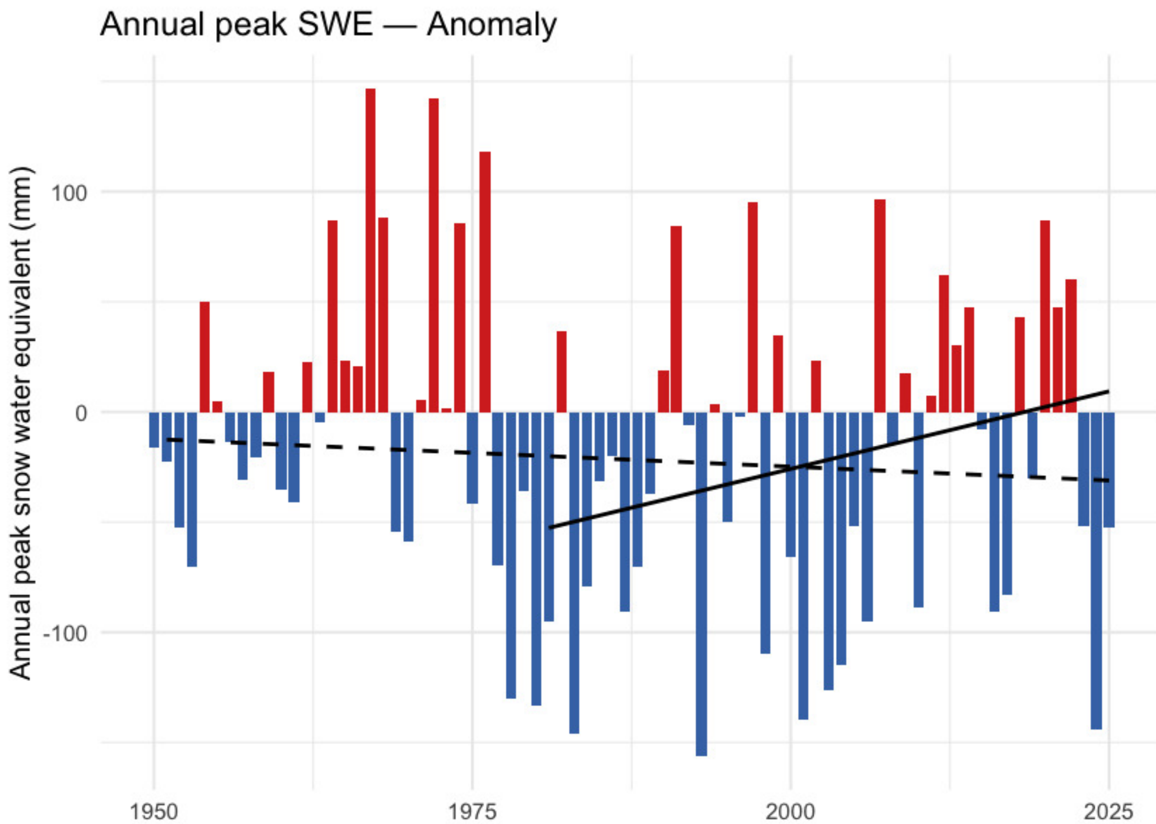


Figure 5.16: Annual peak snow water equivalent (swe_max). ERA5-Land mm SWE, regional spatial mean. The downward trend is present but does not clear statistical significance against year-to-year variability at the regional scale.

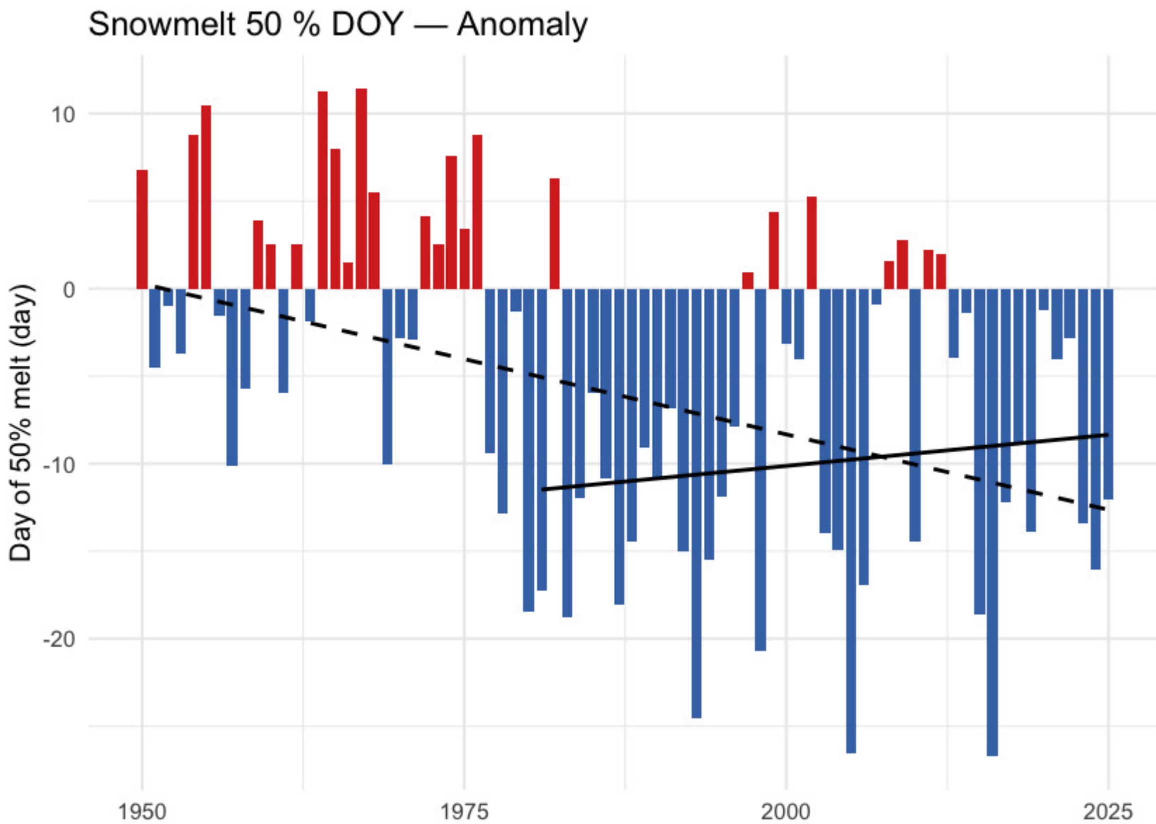


Figure 5.17: Day of year when half the annual snowmelt has accumulated (snowmelt_doy_50). Earlier dates indicate an earlier freshet centroid — directly relevant to crossing-design hydrology and to migration timing. The trend is highly significant ($p < 0.001$ at the regional scale and in every ecoregion).

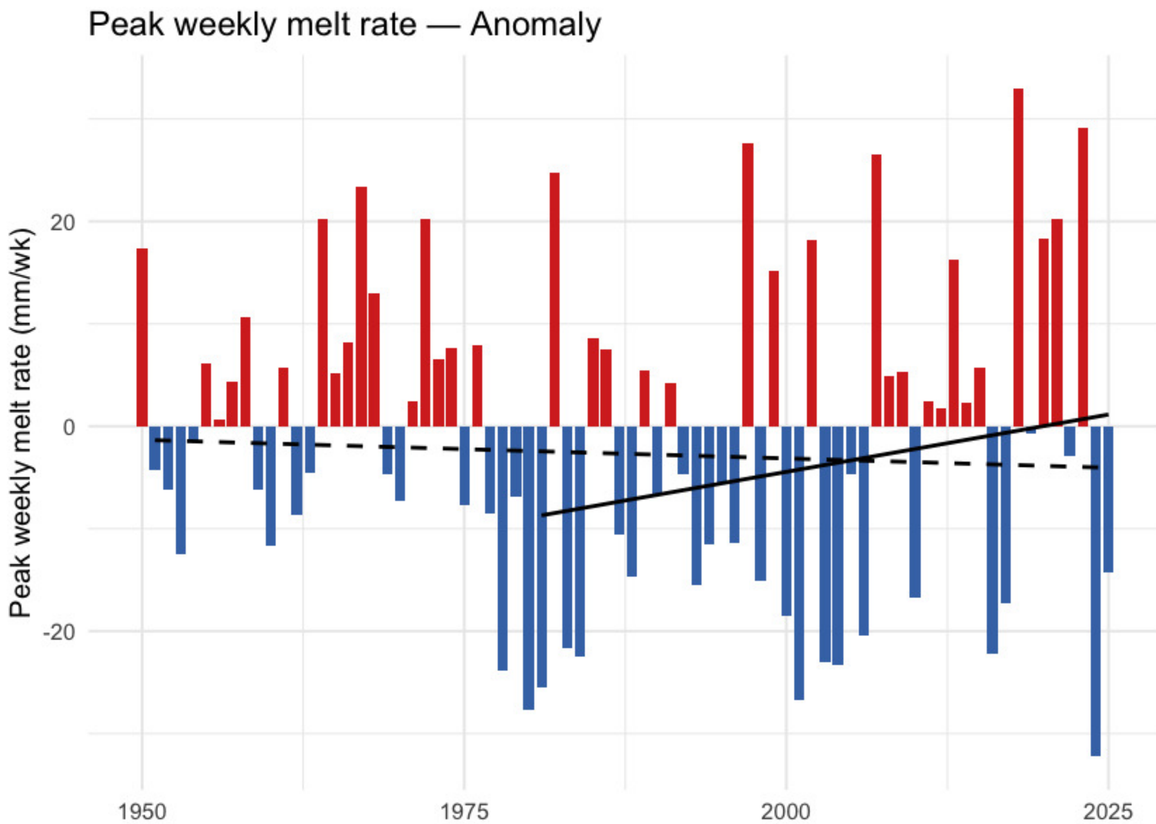


Figure 5.18: Annual maximum of 7-day rolling daily snowmelt (snowmelt_rate_peak). Peak melt intensity has not shifted meaningfully — the freshet is arriving earlier, but its peak rate has not sharpened.

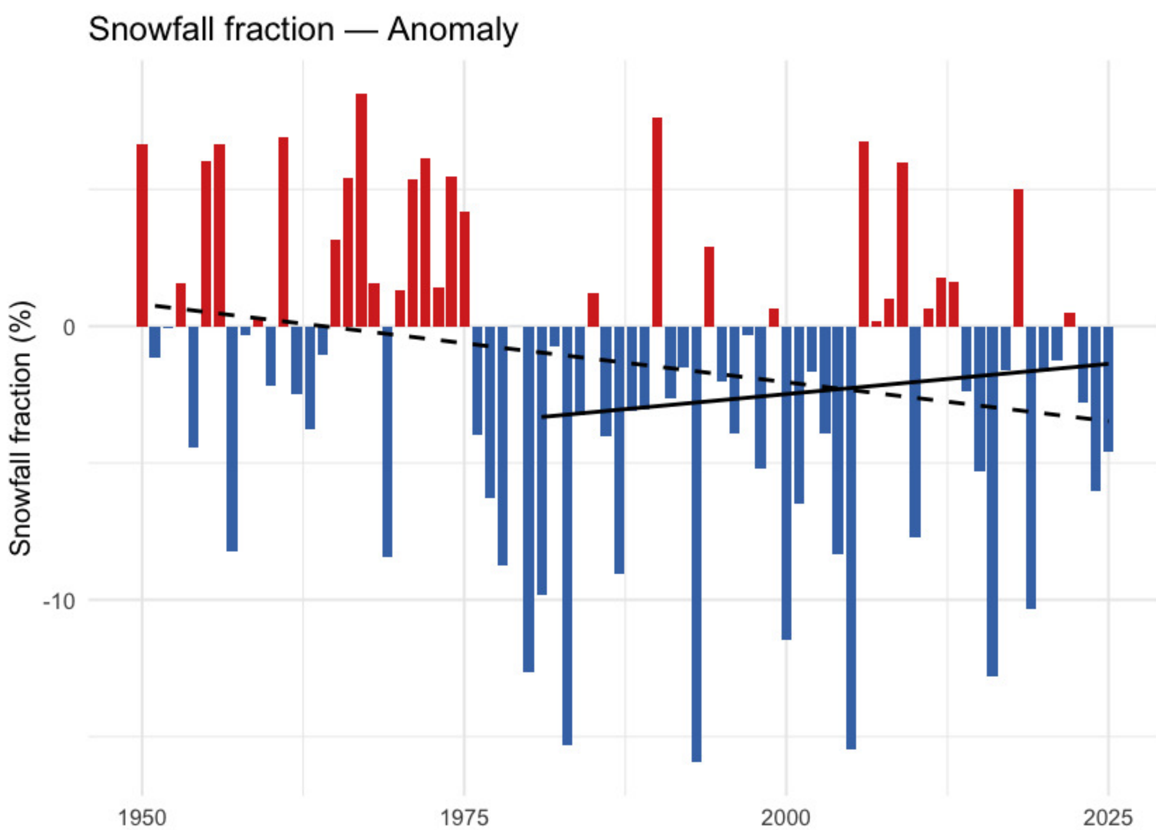


Figure 5.19: Annual snowfall fraction: the percent of annual precipitation that fell as snow. Anomaly is in percentage points relative to the 1951-1980 baseline. The downward trend sits just above statistical significance (Mann-Kendall $p \approx 0.05$) — directionally a shift toward rain-dominant precipitation, but not formally significant on either the trend or the recent-vs-baseline window test.

What this means

Three findings carry the snowpack story.

Snow is leaving the region earlier each year, not falling less. Annual snowfall fell about 6 % and annual SWE about 10 %, but neither change clears statistical significance against year-to-year variability. The clear signals are in *timing*: the snowmelt midpoint shifted nearly two weeks earlier ($p < 0.001$ regionally and in every ecoregion), snow cover declined 3.8 percentage points ($p = 0.002$), and snowfall fraction edged down 3.7 pp (Welch and Mann-Kendall p both just above 0.05 — directionally consistent with a shift toward rain-dominant precipitation, but just shy of statistical significance). This matches in direction the broader Pacific-Northwest signal documented since Mote et al. (2005).

Melt is shifting earlier. The unambiguous signals are on the summer side and at the midpoint: summer snowmelt fell 37 % ($p \approx 0.02$), summer SWE collapsed 52 % ($p \approx 0.01$), and the annual snowmelt midpoint moved ~12 days earlier ($p < 0.001$). Apparent rises in spring snowmelt (+18 %, Welch $p \approx 0.06$) and winter snowmelt (+45 %, Welch $p \approx 0.28$) mirror this redistribution but are noisier season- by-season — winter and spring have smaller absolute snowmelt volumes, so the same shifts produce larger relative-change variance. The total annual melt has not changed; its centre of mass has moved earlier. This is consistent in direction with the 1–4 week earlier streamflow timing across western North America documented by Stewart et al. (2005) and with the ~10-day earlier Fraser freshet at Hope Kang et al. (2016).

Summer SWE has dropped sharply. Summer SWE fell 52 %. The high-elevation snowpack that historically lingered through the warmest weeks of the year no longer carries the same volume — reducing the late-season cold-water input to streams during the window when stream-temperature stress on cold-water salmonids is highest.

Spatial pattern of warming

The regional zonal mean reduces an AOI this size to a single number; the spatial pattern carries the rest of the story.

Warming is not spatially uniform. Individual grid cells range from about +1.1 to +2.0 °C across the AOI, and ecoregion means span +1.4 to +2.0 °C (Figure 5.20). The dominant axis runs **west-warm to east-cool** — the lower-elevation interior plateau ecoregions (Fraser Basin, Fraser Plateau) warmed roughly 0.5 °C more than the high-elevation Western Continental Ranges and adjacent eastern Rocky Mountain ecoregions. A secondary south-to-north component adds about 0.4 °C of additional warming in the northern part of the AOI, consistent with high-latitude amplification. The eastern Rocky Mountain ecoregions show a thermal-buffer effect — likely a combination of slower warming at elevation and a weaker snow-albedo feedback where snow persists longer. The implication for fish passage is geographic: barrier reaches in the interior-plateau WSGs (LCHL, NECR, TABR, FRAN) sit on the warmer side of the gradient; reaches in the eastern WSGs (UFRA, MORK) sit on the cooler side.

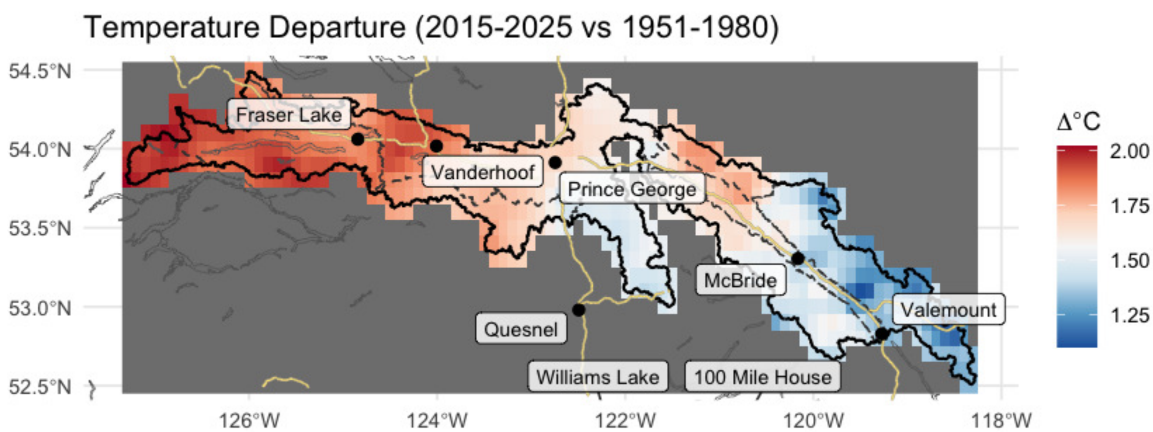


Figure 5.20: Spatial pattern of annual mean temperature departure across the AOI (2015-2025 mean minus 1951-1980 mean, degrees C). The dominant axis runs west-warm (interior plateau) to east-cool (Rocky Mountain ranges), with a secondary south-to-north amplification.

Per-ecoregion variation

The regional zonal mean averages over eight ecoregions with very different elevations and exposures. Running the same pipeline on each ecoregion polygon tests whether the regional story holds within each one, and where it does not.

All eight ecoregions warmed significantly (Mann-Kendall $p < 0.001$ for tmean in every ecoregion). The interior-plateau ecoregions (Fraser Basin, Fraser Plateau) and the northern Eastern Hazelton Mountains lead at +1.7 to +2.0 °C cumulative; the high-elevation Western Continental Ranges trail at +1.4 °C. The day-night asymmetry (tmin > tmax) holds in every ecoregion. Vapour pressure deficit rose significantly ($p < 0.05$) in every ecoregion.

Precipitation is the variable that does not move. Unlike some other BC regions where the long-term precipitation trend is significant in a subset of ecoregions, here annual precipitation shows no significant trend in any of the eight ecoregions (Mann- Kendall p ranges 0.19 to 0.91). The climate signal for this study area is unambiguously about temperature and snowpack timing, not about precipitation magnitude.

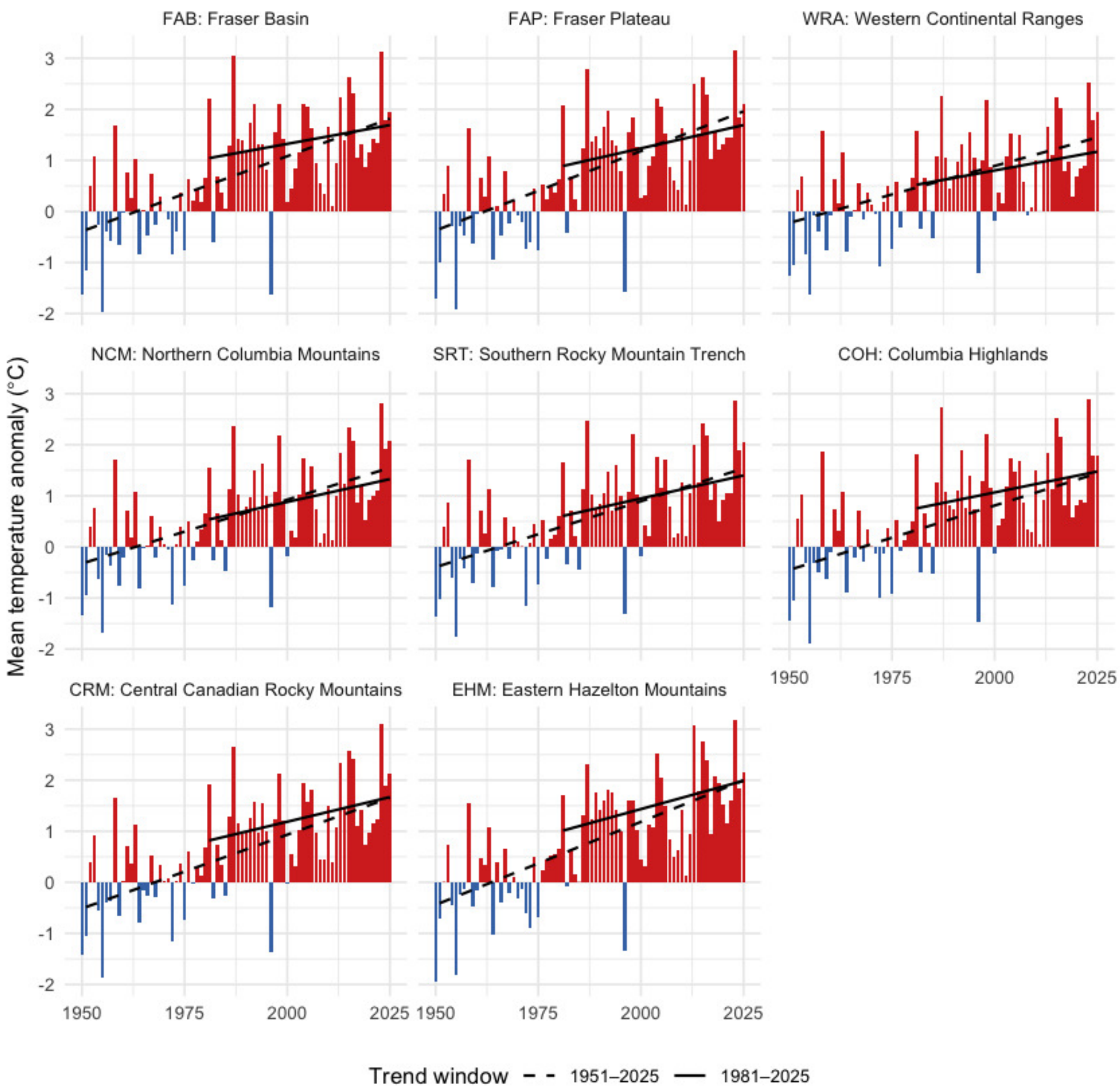


Figure 5.21: Annual mean temperature anomaly relative to the 1951-1980 baseline, by ecoregion. Dashed line is the 75-year Theil-Sen trend (1951-present); solid line is the 45-year trend (1981-present). A solid line steeper than the dashed line indicates accelerating warming — note that across this AOI, the 45-year slope is generally shallower than the 75-year slope, meaning recent warming has not been faster than the long-term rate.

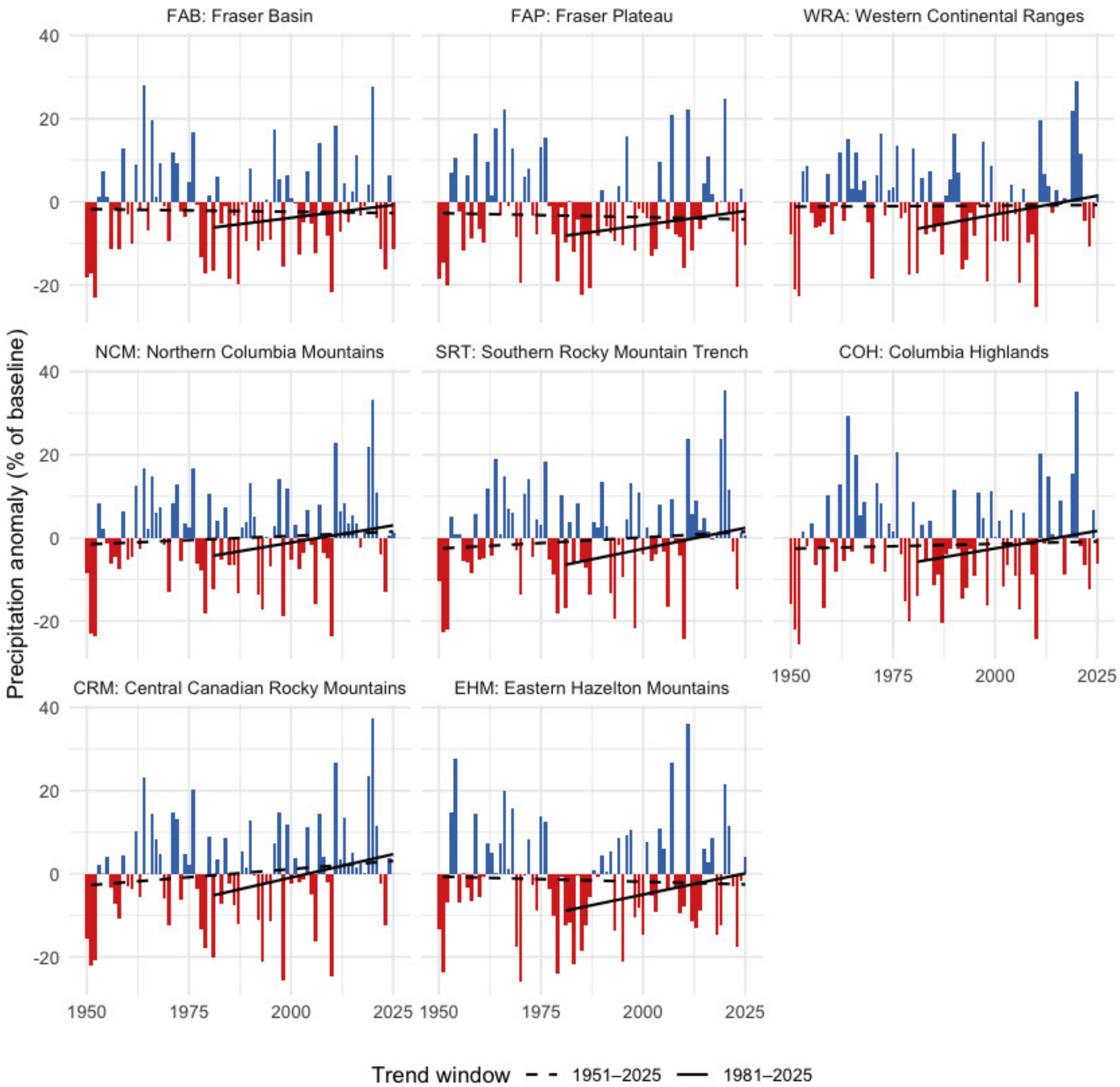


Figure 5.22: Annual precipitation anomaly (% of the 1951-1980 baseline) by ecoregion. None of the eight ecoregions shows a statistically significant precipitation trend over the full record.

Snow by ecoregion

Two snow metrics are worth viewing per-ecoregion: peak SWE (annual snowpack magnitude) and DOY-50 (the day-of-year by which half the year's snowmelt has already happened — earlier values mean an earlier freshet). The peak-SWE signal is noisy at the per- ecoregion scale — none of the eight ecoregions clears statistical significance for the long-term swe_max trend. The DOY-50 signal, in contrast, is highly significant in every ecoregion (Mann- Kendall $p < 0.01$ in each), with shifts ranging from -1.4 to -2.1 days per decade. Earlier melt is universal across the AOI.

Per-ecoregion variation

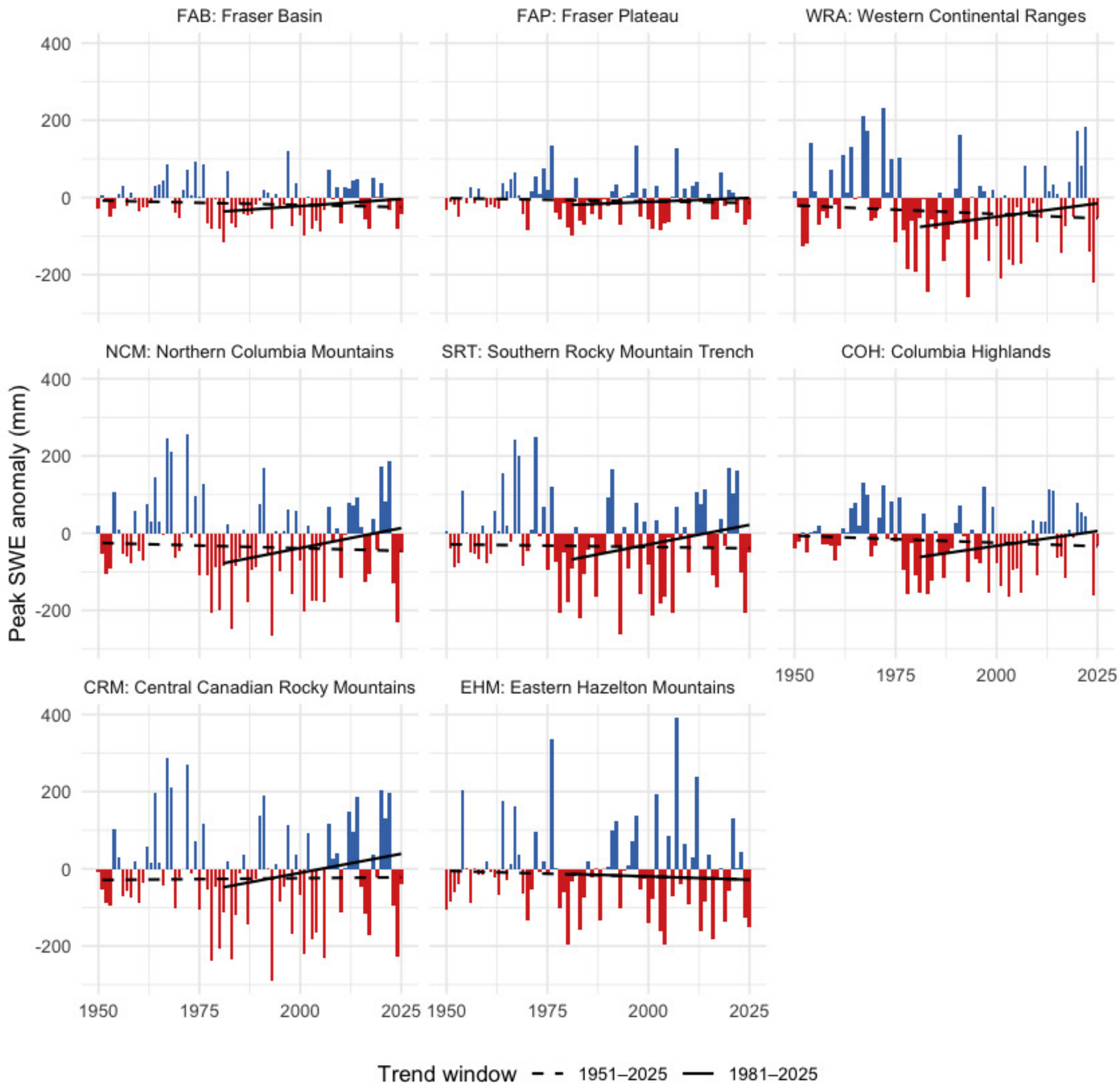


Figure 5.23: Annual peak SWE anomaly by ecoregion, relative to the 1951-1980 baseline. Year-to-year variability is large; the long-term trend is not statistically significant in any single ecoregion.

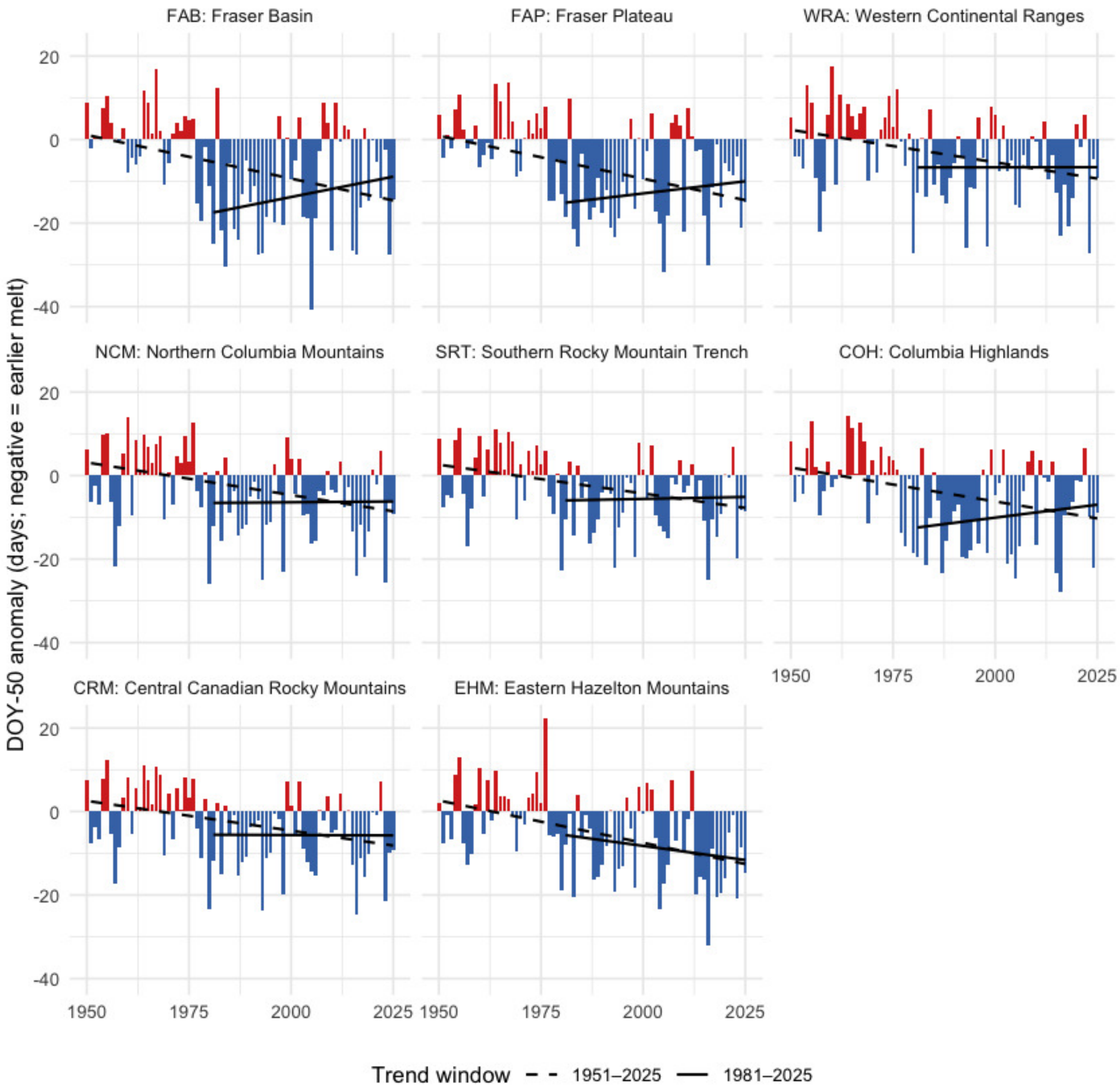


Figure 5.24: Snowmelt 50 % day-of-year (DOY-50) anomaly by ecoregion, relative to the 1951-1980 baseline. Negative values (red) mean the freshet midpoint shifted earlier in the year. The trend is highly significant in every ecoregion.

Table 5.31: Per-ecoregion roll-up over the 75-year window (1951-present): annual mean / daytime maximum / overnight minimum temperature trends (°C per decade); annual precipitation trend (mm per year) and Mann-Kendall p-value; annual vapour pressure deficit trend (hPa per decade) and p-value; and recent (2015-2025) vs pre-warming (1951-1980) percent change for precipitation and soil moisture. All temperature p-values are below 0.001. A tmin slope greater than the tmax slope indicates the day-night asymmetry.

Ecoregion	tmean °C/dec	tmax °C/dec	tmin °C/dec	prcp mm/yr	prcp p	vpd hPa/dec	vpd p	prcp % change	soil moisture % chg
FAB	0.29	0.27	0.32	-0.013	0.812	0.082	0.000	0.4	-1.2
FAP	0.31	0.28	0.33	-0.020	0.770	0.086	0.000	0.1	-1.3
WRA	0.22	0.19	0.26	0.005	0.905	0.031	0.011	4.4	-0.2
NCM	0.25	0.21	0.28	0.041	0.448	0.042	0.004	5.3	-0.2

Ecoregion codes: FAB — Fraser Basin; FAP — Fraser Plateau; WRA — Western Continental Ranges; NCM — Northern Columbia Mountains; SRT — Southern Rocky Mountain Trench; COH — Columbia Highlands; CRM — Central Canadian Rocky Mountains; EHM — Eastern Hazelton Mountains.

Watershed groups across ecoregions

The study area is reported and managed at the watershed-group scale — the British Columbia Freshwater Atlas hydrological reporting unit, and the unit that drives barrier prioritisation in this report. The seven watershed groups (LCHL, NECR, FRAN, MORK, UFRA, TABR, WILL) span the eight ecoregions in different ways: some sit cleanly within one or two, others split across several (Figure 5.25). The table that follows gives, for each watershed group, the share of its area falling in each ecoregion — the cross-reference that maps the regional and ecoregion climate signals onto the per-watershed prioritisation unit.

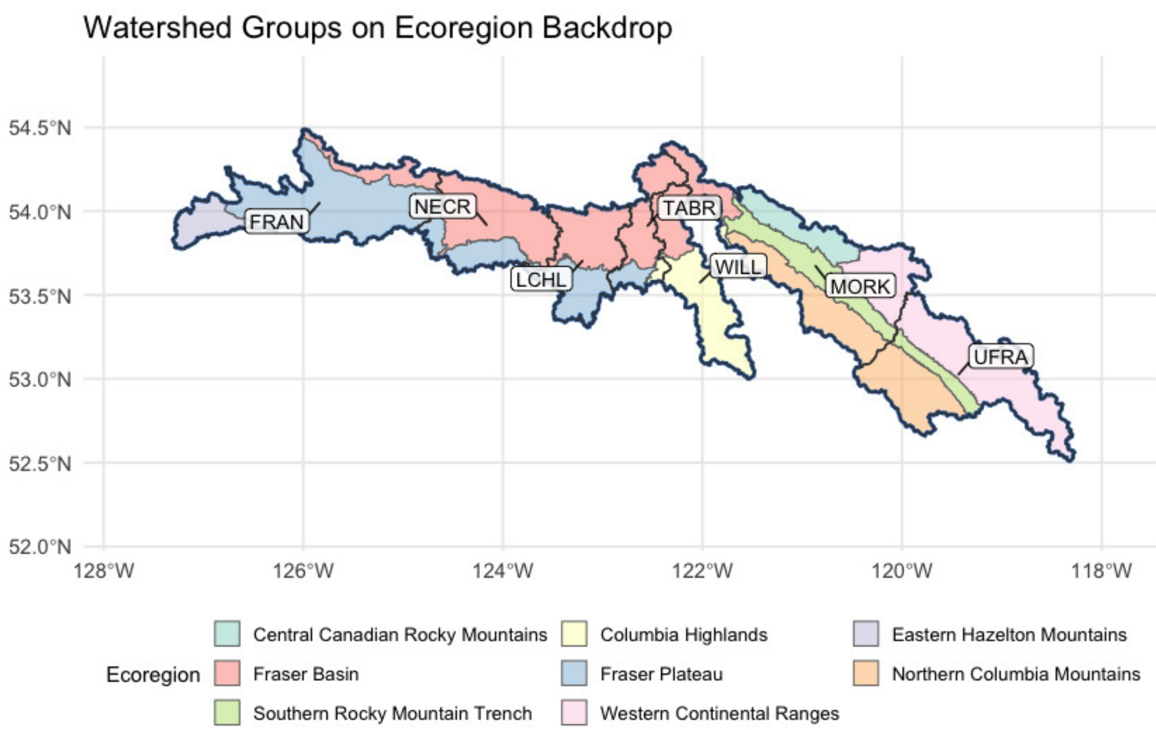


Figure 5.25: Watershed groups in the study area, labelled with their codes, on top of ecoregion fills. LCHL / NECR / TABR sit dominantly in the interior plateau ecoregions (FAB, FAP); FRAN extends into the Eastern Hazelton Mountains; WILL sits dominantly in the Columbia Highlands; UFRA and MORK reach into the Western Continental Ranges and adjacent Rocky Mountain ecoregions.

Table 5.32: Watershed groups in the study area with the percent of each group's area falling in each of the eight ecoregions. Rows where one ecoregion is at or near 100 % indicate watershed groups contained within a single climate-physiography zone; rows split across two or more ecoregions indicate watershed groups whose climate departure can pull from more than one regional pattern. MORK in particular spans six ecoregions, reaching from the Fraser Basin into the Western Continental Ranges.

Code	Name	km²	FAB %	FAP %	WRA %	NCM %	SRT %	COH %	CRM %	EHM %
FRAN	Francois Lake	6574	12.9	72.5	0.0	0.0	0.0	0.0	0.0	14.5
LCHL	Lower Chilako River	3139	57.2	42.8	0.0	0.0	0.0	0.0	0.0	0.0
MORK	Morkill River	8203	9.9	0.0	20.9	28.9	22.7	0.8	16.8	0.0
NECR	Nechako River	4146	70.6	29.4	0.0	0.0	0.0	0.0	0.0	0.0

Interpretation for fish passage

Four findings tie the regional climate-departure analysis to per-watershed barrier prioritisation.

Warming is broad and significant; the gradient is real but small. All eight ecoregions warmed +1.4 to +2.0 °C since 1951. Temperature is not the variable that *separates* watershed groups within the AOI — what it does set is the regional context: every reach in the study area is operating in air ~1.6 °C warmer than the conditions the channel, floodplain, and fish community evolved under. The within-AOI thermal differentiation comes from elevation, aspect, riparian cover, and groundwater contribution — not from climate-departure magnitude.

Precipitation has not moved. Across every ecoregion, the long-term precipitation trend is not statistically distinguishable from natural year-to-year variability. The climate-driven hydrology story for these seven watershed groups is dominated by *atmospheric drying* (rising VPD, significant in every ecoregion) and *snowpack timing*, not by precipitation magnitude. This is the place where the local result differs from some other BC regions: there is no precipitation-trend gradient within the AOI for barrier prioritisation to lean on.

Snow is leaving the AOI earlier, not falling less. The snowmelt midpoint shifted nearly two weeks earlier ($p < 0.001$ regionally and in every ecoregion), summer SWE collapsed by 52 % ($p \approx 0.01$) and summer snowmelt fell 37 % ($p \approx 0.02$); the counterbalancing winter (+45 %) and spring (+18 %) snowmelt rises follow in direction but do not individually clear statistical significance season-by-season. The total annual melt has not changed; its centre of mass has moved earlier. The freshet — the dominant high-flow event that mobilises spawning gravels, refills off-channel rearing habitat, and that crossing structures are sized against — is arriving weeks earlier across all seven watershed groups, regardless of which ecoregion they sit in. Crossing-structure design assumptions and peak-flow-timing baselines baked into the assessment scoring need to track this shift. The late-summer cold-water input that high-elevation snowpack historically provided during the warmest, most thermally stressful weeks of summer is dropping in parallel.

The geographic gradient runs west-warm to east-cool, following the elevation profile. Warming has been roughly 0.5 °C greater in the lower-elevation interior plateau ecoregions (Fraser Basin, Fraser Plateau) and the Eastern Hazelton Mountains than in the higher-elevation Western Continental Ranges and adjacent Rocky Mountain ecoregions. Mapping that onto the watershed-group reporting unit: LCHL, NECR, TABR and FRAN sit dominantly on the interior-plateau side of the gradient and have seen the larger warming + drying signal; UFRA and MORK reach into the Rocky Mountain ecoregions and have experienced the smaller shifts; WILL sits in Columbia Highlands with an intermediate signal.

For the cold-water resident salmonids the study area supports — bull trout, chinook, mountain whitefish, rainbow trout — the implication for barrier prioritisation is that two fundamentally different climate-departure stories play out across the seven watershed groups, and that the priority weight on barrier removal should respond to both directions. Headwater reaches that were cold-limited (most likely in the mountain-dominated WSGs at elevation) can gain growing-degree-days from warming, raising the intrinsic potential of habitat above the barrier and *increasing* the recovery value of restoring passage. Reaches sitting closer to the upper edge of the cold-water thermal niche (most likely in the lower-elevation, interior-plateau WSGs) face the opposite direction — the same warming reduces usable habitat and lowers the payoff. The watershed-group ecoregion mapping in the table above is the bridge between the regional climate-departure summary and the per-watershed weight applied to each barrier.

Appendix - Environmental DNA Results

This appendix carries the site-level detail behind the headline counts in the Results chapter. Field blanks are reported separately because they test the sampling protocol rather than a location. Office blanks are excluded entirely — they are distilled-water controls filtered at the end of a session at accommodation, so although they carry real GPS coordinates, those coordinates record where the filtering happened rather than a stream site. The interactive map is linked at the end.

Per-site detection results

Format: `CODE(max_droplets)`, with $*$ flagging samples that UNBC reran. Detected = ≥4 positive droplets per the standard ddPCR call threshold; Sub-threshold = 1-3 droplets, not interpretable as species presence under ddPCR convention; Not detected = 0 droplets across all runs.

Table 5.33: Site-level eDNA detection results across real environmental samples in the Fraser project area. UNBC ID is the lab sample identifier; Pos. control = yes indicates the sample was collected at a site where the target species was also confirmed by electrofishing.

Site ID	Stream	UNBC ID	Pos. control	Detected	Sub-threshold	Not detected
126158_ds_ed1	Tributary To Stony Lake	F61121		RAIN(77)	—	BULT(0), CHIN(0), COHO(0)
126158_us_ed1	Tributary To Stony Lake	F61122	yes	RAIN(21)	—	BULT(0), CHIN(0), COHO(0)
126349_ds_ed1	Tributary To Cameron Lake	F21236		RAIN(39)	—	BULT(0)
196076_ds_ed2a	Tributary To Fraser River	F41016		RAIN(23)	BURB(1)	BULT(0), CHIN(0)
196076_ds_ed2b	Tributary To Fraser River	F41017		RAIN(5)	—	BULT(0), BURB(0), CHIN(0)
196076_us_ed1	Tributary To Fraser River	F61125		RAIN(44)	—	BULT(0), CHIN(0), COHO(0)
196085_ds_ed1	Tabor Creek	F40915		BURB(8), CHIN(181)	BULT(1)	SOCK(0)
196085_us_ed1a	Tabor Creek	F41019		CHIN(8), RAIN(6)	—	BULT(0), BURB(0)
196085_us_ed1b	Tabor Creek	F41020		CHIN(144), RAIN(128)	—	BULT(0), BURB(0)
196209_us_ed1	Bittner Creek	F60808		—	—	BULT(0), CHIN(0), COHO(0), SOCK(0)
196332_ds_ed1	S. Yuzkli Creek	F21237		—	—	BULT(0), RAIN2(0)
196332_us_ed1	S. Yuzkli Creek	F21238		—	BULT(2)*	RAIN(0)*, RAIN2(0)
199171_ds_ed1	Tributary To Fraser Lake	F60503		RAIN(4)	—	BULT(0), CHIN(0), SOCK(0)

Site ID	Stream	UNBC ID	Pos. control	Detected	Sub-threshold	Not detected
199171_us_ed1	Tributary To Fraser Lake	F60504		—	CHIN(1), RAIN(3)	BULT(0), SOCK(0)
199173_ds_ed1	Tributary To Nechako River	F61126		CHIN(16), RAIN(255)	COHO(1)	BULT(0)
199173_us_ed1	Tributary To Nechako River	F61127	yes	RAIN(512)	—	BULT(0), CHIN(0), COHO(0)
199174_us_ed1	Tributary To Nechako River	F61128		RAIN(38)	—	BULT(0), CHIN(0), COHO(0)
199190_ds_ed1	Creek Creek	F60505		RAIN(17)	—	BULT(0), CHIN(0)*, COHO(0), SOCK(0)
199190_us_ed1	Creek Creek	F61129		—	—	BULT(0), CHIN(0), COHO(0), RAIN(0)
199237_ds_ed1	Snowshoe Creek	F60809		—	—	BULT(0), CHIN(0), COHO(0), SOCK(0)
199237_us_ed1	Snowshoe Creek	F60810		—	—	BULT(0), CHIN(0), COHO(0), SOCK(0)
199267_ds_ed1	Driscoll Creek	F60811		CHIN(4)	BULT(1)	COHO(0), SOCK(0)
199267_us_ed1b	Driscoll Creek	F20401		—	BULT(1)	CHIN(0)
199273_ds_ed1	Hankins Creek	F60812		BULT(10)	—	CHIN(0), COHO(0), SOCK(0)
199278_ds_ed1	Teepee Creek	F61131		BULT(127), RAIN(47)	—	CHIN(0), COHO(0)
199278_us_ed1	Teepee Creek	F61132		BULT(138), RAIN(20)	—	CHIN(0), COHO(0)
203578_ds_ed1	Tributary To Willow River	F61135		RAIN(67)	—	BULT(0), CHIN(0), COHO(0)
203581_us_ed1a	Tributary To Fraser River	F61123		RAIN(54)	BULT(2), *CHIN(2)*	COHO(0)
203581_us_ed1b	Tributary To Fraser River	F61124		RAIN(51)	CHIN(3)*	BULT(0), COHO(0)
203582_us_ed1a	Tabor Creek	F60506		CHIN(56)*, RAIN(1281)	SOCK(1)	BULT(0), COHO(0)
203582_us_ed1b	Tabor Creek	F60507		CHIN(37)*, RAIN(1871)	—	BULT(0), COHO(0), SOCK(0)
23919_ds_ed1a	Swift Creek	F60813	yes	BULT(62), CHIN(138), SOCK(332)	—	COHO(0)
24715757_ds_ed1	Snowshoe Creek	F20402		BULT(4), CHIN(49)	—	—

Field blanks

Field blanks filter distilled water at the field site to test for protocol contamination during collection and processing. **Detections in blanks indicate potential cross-contamination, not species**

presence at the location — these rows are presented for QA transparency only, not as eDNA detections.

Table 5.34: Field blanks collected during the 2025 eDNA program in the Fraser project area. Detections in blanks reflect protocol-side contamination, not site eDNA.

Site ID	Stream	UNBC ID	Detected	Sub-threshold	Clean (0 droplets)
199267_us_ed1a	Driscoll Creek	F61130	—	—	BULT(0), CHIN(0), COHO(0), RAIN(0)
23919_ds_ed1b	Swift Creek	F60814	—	—	BULT(0), CHIN(0), COHO(0), SOCK(0)

Retests

UNBC reran a subset of samples to confirm borderline results. The table below lists every site × species combination that was rerun, including reruns of both real environmental samples and field blanks.

Table 5.35: UNBC retests for the 2025 Fraser eDNA samples. Runs is the total number of times the sample x species combination was assayed; Max droplets is the highest positive-droplet count observed across all runs.

Site ID	Stream	Species	Code	Sample type	Runs	Max droplets	Result
196332_us_ed1	S. Yuzkli Creek	Bull Trout	BULT	environmental	6	2	Sub-threshold (1-3 droplets, NOT a detection)
196332_us_ed1	S. Yuzkli Creek	Rainbow Trout	RAIN	environmental	4	0	No detection (0 droplets)
199190_ds_ed1	Creek Creek	Chinook Salmon	CHIN	environmental	2	0	No detection (0 droplets)
203581_us_ed1a	Tributary To Fraser River	Bull Trout	BULT	environmental	4	2	Sub-threshold (1-3 droplets, NOT a detection)
203581_us_ed1a	Tributary To Fraser River	Chinook Salmon	CHIN	environmental	4	2	Sub-threshold (1-3 droplets, NOT a detection)
203581_us_ed1b	Tributary To Fraser River	Chinook Salmon	CHIN	environmental	4	3	Sub-threshold (1-3 droplets, NOT a detection)
203582_us_ed1a	Tabor Creek	Chinook Salmon	CHIN	environmental	2	56	Detected (≥4 droplets)
203582_us_ed1b	Tabor Creek	Chinook Salmon	CHIN	environmental	2	37	Detected (≥4 droplets)

Interactive map

The interactive Fraser eDNA map shows site-level detection results with toggleable per-species layers, an opt-in sub-threshold-signals layer (off by default), and an opt-in Controls layer for field blanks. See the interactive Fraser eDNA map.

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Note that this reference was not actually cited in the body of the report but rather in the `index.Rmd` file `yaml` headerj under the `no_cite` key.

Session Info

Information about the computing environment is important for reproducibility. A summary of the computing environment is saved to `session_info.csv`, which can be viewed and downloaded from here.

```
## ─ Session info ───────────────────────────────────────────────────────────────
##  setting  value
##  version  R version 4.5.2 (2025-10-31)
##  os       macOS Tahoe 26.2
##  system   aarch64, darwin20
##  ui       X11
##  language (EN)
##  collate  en_US.UTF-8
##  ctype    en_US.UTF-8
##  tz       America/Vancouver
##  date     2026-07-31
##  pandoc   3.9.0.2 @ /opt/homebrew/bin/ (via rmarkdown)
##  quarto   1.4.553 @ /usr/local/bin/quarto
##
## ─ Packages ───────────────────────────────────────────────────────────────────
##  package         * version    date (UTC) lib source
##  arrow             22.0.0.1   2025-12-23 [2] CRAN (R 4.5.2)
##  askpass           1.2.1      2024-10-04 [1] CRAN (R 4.5.0)
##  assertthat        0.2.1      2019-03-21 [2] CRAN (R 4.5.0)
##  base64enc         0.1-6      2026-02-02 [1] CRAN (R 4.5.2)
##  bit               4.6.0      2025-03-06 [1] CRAN (R 4.5.0)
##  bit64             4.8.0      2026-04-21 [1] CRAN (R 4.5.2)
##  blob              1.3.0      2026-01-14 [1] CRAN (R 4.5.2)
##  bookdown        * 0.46       2025-12-05 [2] CRAN (R 4.5.2)
##  boot              1.3-32     2025-08-29 [2] CRAN (R 4.5.2)
##  bslib             0.10.0     2026-01-26 [1] CRAN (R 4.5.2)
##  cachem            1.1.0      2024-05-16 [1] CRAN (R 4.5.0)
##  cd              * 0.3.0      2026-05-12 [1] Github
(NewGraphEnvironment/cd@a3e4733)
##  chk               0.10.0     2025-01-24 [1] CRAN (R 4.5.0)
##  class             7.3-23     2025-01-01 [2] CRAN (R 4.5.2)
##  classInt          0.4-11     2025-01-08 [1] CRAN (R 4.5.0)
##  cli               3.6.6      2026-04-09 [1] CRAN (R 4.5.2)
##  codetools         0.2-20     2024-03-31 [2] CRAN (R 4.5.2)
##  crayon            1.5.3      2024-06-20 [1] CRAN (R 4.5.0)
##  crosstalk         1.2.2      2025-08-26 [1] CRAN (R 4.5.0)
##  curl              7.1.0      2026-04-22 [1] CRAN (R 4.5.2)
##  data.table        1.18.2.1   2026-01-27 [1] CRAN (R 4.5.2)
```

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## DBI              1.3.0      2026-02-25 [1] CRAN (R 4.5.2)
## dbplyr           2.5.2      2026-02-13 [1] CRAN (R 4.5.2)
## devtools         2.4.6      2025-10-03 [2] CRAN (R 4.5.0)
## digest           0.6.39     2025-11-19 [1] CRAN (R 4.5.2)
## doParallel       1.0.17     2022-02-07 [2] CRAN (R 4.5.0)
## dplyr          * 1.2.1      2026-04-03 [1] CRAN (R 4.5.2)
## DT               0.34.0     2025-09-02 [2] CRAN (R 4.5.0)
## e1071            1.7-17     2025-12-18 [1] CRAN (R 4.5.2)
## elevatr          0.99.1     2025-09-10 [2] CRAN (R 4.5.0)
## ellipsis         0.3.2      2021-04-29 [2] CRAN (R 4.5.0)
## english          1.2-6      2021-08-21 [2] CRAN (R 4.5.0)
## evaluate         1.0.5      2025-08-27 [1] CRAN (R 4.5.0)
## farver           2.1.2      2024-05-13 [1] CRAN (R 4.5.0)
## fasstr           0.5.3      2024-09-27 [2] CRAN (R 4.5.0)
## fastmap          1.2.0      2024-05-15 [1] CRAN (R 4.5.0)
## fishbc         * 0.2.1.9000 2026-01-15 [2] Github (lucy-
schick/fishbc@0607ffd)
## forcats        * 1.0.1      2025-09-25 [2] CRAN (R 4.5.0)
## foreach          1.5.2      2022-02-02 [2] CRAN (R 4.5.0)
## fpr            * 1.2.0      2026-02-03 [2] Github
(NewGraphEnvironment/fpr@2817533)
## fs               2.1.0      2026-04-18 [1] CRAN (R 4.5.2)
## generics         0.1.4      2025-05-09 [1] CRAN (R 4.5.0)
## ggplot2        * 4.0.3      2026-04-22 [1] CRAN (R 4.5.2)
## ggrepel          0.9.7      2026-02-25 [2] CRAN (R 4.5.2)
## gh               1.5.0      2025-05-26 [2] CRAN (R 4.5.0)
## glue             1.8.1      2026-04-17 [1] CRAN (R 4.5.2)
## gtable           0.3.6      2024-10-25 [1] CRAN (R 4.5.0)
## here             1.0.2      2025-09-15 [2] CRAN (R 4.5.0)
## hms              1.1.4      2025-10-17 [1] CRAN (R 4.5.0)
## htmltools        0.5.9      2025-12-04 [1] CRAN (R 4.5.2)
## htmlwidgets      1.6.4      2023-12-06 [1] CRAN (R 4.5.0)
## httr             1.4.8      2026-02-13 [1] CRAN (R 4.5.2)
## iterators        1.0.14     2022-02-05 [2] CRAN (R 4.5.0)
## janitor          2.2.1      2024-12-22 [1] CRAN (R 4.5.0)
## jquerylib        0.1.4      2021-04-26 [1] CRAN (R 4.5.0)
## jsonlite         2.0.0      2025-03-27 [1] CRAN (R 4.5.0)
## kableExtra     * 1.4.0      2024-01-24 [2] CRAN (R 4.5.0)
## Kendall          2.2.2      2025-12-22 [2] CRAN (R 4.5.2)
## KernSmooth       2.23-26    2025-01-01 [2] CRAN (R 4.5.2)
## knitr          * 1.51       2025-12-20 [1] CRAN (R 4.5.2)
## labeling         0.4.3      2023-08-29 [1] CRAN (R 4.5.0)
## lattice          0.22-9     2026-02-09 [2] CRAN (R 4.5.2)
## leafem         * 0.2.5      2025-08-28 [2] CRAN (R 4.5.0)
## leaflet        * 2.2.3      2025-09-04 [1] CRAN (R 4.5.0)
## lifecycle        1.0.5      2026-01-08 [1] CRAN (R 4.5.2)
## lubridate      * 1.9.5      2026-02-04 [1] CRAN (R 4.5.2)
```

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##  magick           2.9.0      2025-09-08 [2] CRAN (R 4.5.0)
##  magrittr         2.0.5      2026-04-04 [1] CRAN (R 4.5.2)
##  Matrix           1.7-4      2025-08-28 [2] CRAN (R 4.5.2)
##  memoise          2.0.1      2021-11-26 [1] CRAN (R 4.5.0)
##  mgcv             1.9-3      2025-04-04 [2] CRAN (R 4.5.2)
##  ngr            * 0.0.1      2026-01-22 [2] Github
(newgraphenvironment/ngr@22c9714)
##  nlme             3.1-168    2025-03-31 [2] CRAN (R 4.5.2)
##  otel             0.2.0      2025-08-29 [1] CRAN (R 4.5.0)
##  pagedown       * 0.23       2025-08-20 [2] CRAN (R 4.5.0)
##  pak              0.11.1     2026-07-22 [1] CRAN (R 4.5.2)
##  pdftools       * 3.7.0      2026-01-30 [2] CRAN (R 4.5.2)
##  pillar           1.11.1     2025-09-17 [1] CRAN (R 4.5.0)
##  pkgbuild         1.4.8      2025-05-26 [2] CRAN (R 4.5.0)
##  pkgconfig        2.0.3      2019-09-22 [1] CRAN (R 4.5.0)
##  pkgload          1.5.0      2026-02-03 [2] CRAN (R 4.5.2)
##  plyr             1.8.9      2023-10-02 [1] CRAN (R 4.5.0)
##  png              0.1-9      2026-03-15 [1] CRAN (R 4.5.2)
##  prettyunits      1.2.0      2023-09-24 [1] CRAN (R 4.5.0)
##  processx         3.9.0      2026-04-22 [1] CRAN (R 4.5.2)
##  progress         1.2.3      2023-12-06 [2] CRAN (R 4.5.0)
##  progressr        0.18.0     2025-11-06 [2] CRAN (R 4.5.0)
##  proxy            0.4-29     2025-12-29 [1] CRAN (R 4.5.2)
##  purrr          * 1.2.2      2026-04-10 [1] CRAN (R 4.5.2)
##  qpdf             1.4.1      2025-07-02 [2] CRAN (R 4.5.0)
##  R6               2.6.1      2025-02-15 [1] CRAN (R 4.5.0)
##  rappdirs         0.3.4      2026-01-17 [1] CRAN (R 4.5.2)
##  raster           3.6-32     2025-03-28 [1] CRAN (R 4.5.0)
##  rayshader        0.37.3     2024-02-21 [2] CRAN (R 4.5.0)
##  rbbt           * 0.0.0.9000 2026-01-15 [2] Github
(paleolimbot/rbbt@212680c)
##  RColorBrewer     1.1-3      2022-04-03 [1] CRAN (R 4.5.0)
##  Rcpp             1.1.1-1.1  2026-04-24 [1] CRAN (R 4.5.2)
##  RcppRoll         0.3.1      2024-07-07 [2] CRAN (R 4.5.0)
##  readr          * 2.2.0      2026-02-19 [1] CRAN (R 4.5.2)
##  readwritesqlite * 0.2.0.9021 2026-01-15 [2] Github
(poissonconsulting/readwritesqlite@ccdb297)
##  remotes          2.5.0      2024-03-17 [2] CRAN (R 4.5.0)
##  rgl              1.3.31     2025-11-14 [2] CRAN (R 4.5.2)
##  rlang            1.2.0      2026-04-06 [1] CRAN (R 4.5.2)
##  rmarkdown      * 2.31       2026-03-26 [1] CRAN (R 4.5.2)
##  RPostgres      * 1.4.10     2026-02-16 [1] CRAN (R 4.5.2)
##  rprojroot        2.1.1      2025-08-26 [2] CRAN (R 4.5.0)
##  RSQLite          3.52.0     2026-05-10 [1] CRAN (R 4.5.2)
##  rstudioapi       0.18.0     2026-01-16 [2] CRAN (R 4.5.2)
##  rvest            1.0.5      2025-08-29 [2] CRAN (R 4.5.0)
##  s2               1.1.9      2025-05-23 [1] CRAN (R 4.5.0)
```

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##  S7                 0.2.2      2026-04-22 [1] CRAN (R 4.5.2)
##  sass               0.4.10     2025-04-11 [1] CRAN (R 4.5.0)
##  scales             1.4.0      2025-04-24 [1] CRAN (R 4.5.0)
##  sessioninfo        1.2.3      2025-02-05 [2] CRAN (R 4.5.0)
##  sf               * 1.1-0      2026-02-24 [1] CRAN (R 4.5.2)
##  snakecase          0.11.1     2023-08-27 [1] CRAN (R 4.5.0)
##  sp                 2.2-1      2026-02-13 [1] CRAN (R 4.5.2)
##  staticimports    * 0.0.0.9001 2026-01-15 [2] Github
(newgraphenvironment/staticimports@19cd73c)
##  stringi            1.8.7      2025-03-27 [1] CRAN (R 4.5.0)
##  stringr          * 1.6.0      2025-11-04 [1] CRAN (R 4.5.0)
##  svglite            2.2.2      2025-10-21 [2] CRAN (R 4.5.0)
##  systemfonts        1.3.1      2025-10-01 [2] CRAN (R 4.5.0)
##  terra            * 1.9-11     2026-03-26 [1] CRAN (R 4.5.2)
##  textshaping        1.0.4      2025-10-10 [2] CRAN (R 4.5.0)
##  tibble           * 3.3.1      2026-01-11 [1] CRAN (R 4.5.2)
##  tidyhydat          0.7.2      2025-10-23 [2] CRAN (R 4.5.0)
##  tidyr            * 1.3.2      2025-12-19 [1] CRAN (R 4.5.2)
##  tidyselect         1.2.1      2024-03-11 [1] CRAN (R 4.5.0)
##  tidyterra        * 1.1.0      2026-03-11 [2] CRAN (R 4.5.2)
##  tidyverse        * 2.0.0      2023-02-22 [2] CRAN (R 4.5.0)
##  tidyxl             1.0.10     2026-01-15 [2] Github
(nacnudus/tidyxl@7e2fbe7)
##  timechange         0.4.0      2026-01-29 [1] CRAN (R 4.5.2)
##  tzdb               0.5.0      2025-03-15 [1] CRAN (R 4.5.0)
##  units              1.0-1      2026-03-11 [1] CRAN (R 4.5.2)
##  usethis            3.2.1      2025-09-06 [2] CRAN (R 4.5.0)
##  vctrs              0.7.3      2026-04-11 [1] CRAN (R 4.5.2)
##  viridisLite        0.4.3      2026-02-04 [1] CRAN (R 4.5.2)
##  vroom              1.7.1      2026-03-31 [1] CRAN (R 4.5.2)
##  withr              3.0.2      2024-10-28 [1] CRAN (R 4.5.0)
##  wk                 0.9.5      2025-12-18 [1] CRAN (R 4.5.2)
##  xfun               0.57       2026-03-20 [1] CRAN (R 4.5.2)
##  xml2               1.5.2      2026-01-17 [1] CRAN (R 4.5.2)
##  yaml               2.3.12     2025-12-10 [1] CRAN (R 4.5.2)
##  zyp                0.11-1     2023-03-22 [2] CRAN (R 4.5.0)
##
##  [1] /Users/airvine/Library/R/arm64/4.5/library
##  [2] /Library/Frameworks/R.framework/Versions/4.5-
arm64/Resources/library
##  * ── Packages attached to the search path.
##
##
```

Attachment - Maps

All georeferenced field maps are presented at:

- https://hillcrestgeo.ca/outgoing/fishpassage/projects/parsnip/archive/2022-05-27/

Maps are also available zipped for bulk download at:

- https://hillcrestgeo.ca/outgoing/fishpassage/projects/parsnip/archive/2022-05-27/2022-05-27.zip

Attachment - Phase 1 Data and Photos

Data and photos for all Phase 1 - fish passage assessments are provided online at https://www.newgraphenvironment.com/fish_passage_peace_2024_reporting/appendix---phase-1-fish-passage-assessment-data-and-photos.html - with a pdf version available at https://github.com/NewGraphEnvironment/fish_passage_peace_2024_reporting/raw/main/docs/Appendix_1.pdf

Attachment - Habitat Assessment and Fish Sampling Data

All field data collected is available here.

Habitat assessment data is available for download here https://github.com/NewGraphEnvironment/fish_passage_fraser_2025_reporting/blob/main/data/habitat_confirmations.xls.

Attachment - Bayesian analysis to map stream discharge and temperature causal effects pathways

Details of this analysis and subsequent outputs can be reviewed in the report Spatial Stream Network Analysis of Nechako Watershed Stream Temperatures 2022b (Hill et al. 2024). At the time of reporting, ongoing work regarding the project was tracked https://github.com/poissonconsulting/fish-passage-22/issues and https://github.com/poissonconsulting/fish-passage-22b/issues.